Summary Report, v4: Nine-Component Composite, Three Sampled Designs, Honest Selection

2026

1 Setup

1.1 Shared setup (identical across versions)

- **Suite.** 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Official test or dev splits only; 150 items per dataset by seeded stratified sampling, 100 flagged for answering (OfficeQA Pro V2: 90; Alaska schema: 93); 3,083 answered items per model. Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % ratio); schema matching is one question per source column with a candidate list; BIRD is graded on the executed result of the gold SQL.
- **Answers.** Ten models (Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b, Llama 3.3 70B); one prompt per item ending in `FINAL ANSWER: <answer>`; temperature 0 where accepted (Claude 5 and GPT-5.5 ran at provider defaults, recorded per row); output budget 700 tokens with mechanical re-asks at 2,500 and 8,000 for replies cut off before a final line; every answer cached with tokens, cost and finish reason. The same cache feeds every version, so leaderboards are directly comparable.
- **Scoring primitives.** Extraction of the final line, Unicode / case / punctuation normalisation, number parsing with sign, percent and scale words, alias lists per gold (best alias wins), list golds, a hedge flag. Native metric per dataset: numeric tolerance (1 %) for numeric golds, exact match for text and boolean, ROUGE-L for free-form.
- **Coverage rule.** A model is in the study only if it covered ≥ 90 % of the suite; a dataset counts only items every model answered and only with ≥ 20 such items.
- **Leaderboard machinery.** Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, SE from the observed information) as the headline pooling; Borda, Kemeny (with cost), Copeland, mean-z, reciprocal-rank fusion and family-balanced z as checks; 1,000 item bootstraps of the Borda and Bradley–Terry ranks; Friedman with Iman–Davenport, Nemenyi CD, Holm-corrected pairwise Wilcoxon across datasets; paired permutation and Wilcoxon per item within datasets (Holm); Kendall's W among datasets; majority-class baselines for boolean sets; post-stratified official-split estimates with a weighted bootstrap and positive-class F1 at the official match ratio; floor / ceiling / self-stability diagnostics per dataset.
- **Sampled designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples on the weight simplex at step 0.05 (20 seeds each) with centred L_2 discrepancy and nearest-neighbour distances, compared with the exhaustive optimum where one exists.
- **Reporting.** Each version's `run_all.sh` regenerates its `REPORT.md` from the cached answers; every number in this document is read from those output files.

1.2 What v4 searches

- **Per-answer rule.** A weighted sum of nine lexical components per answer: exact match, numeric tolerance (1 %), numeric decay $e^{-2.5 \cdot \text{rel. err}}$, token precision, token recall, token F1, edit similarity, ROUGE-L, Jaccard. The v1 rubric is the point (F1 0.35, decay 0.35, precision 0.15, recall 0.15) embedded in nine dimensions.
- **Candidates.** The step-0.05 lattice on the nine-weight simplex has 3,108,105 points, too many to enumerate, so three sampled designs of 100 points each (Dirichlet-snapped, Latin hypercube on the simplex, uniform lattice) plus five baselines (exact match only, F1 only, numeric tolerance only, uniform, v1 rubric): 305 candidates. Design comparison over 20 seeds with coverage diagnostics; regret is relative to the best over all seeds and designs since no exhaustive optimum exists.
- **Objective.** The same as v2: J = mean Kendall τ between each dataset's model ranking under the candidate and the Borda pool of the 31 rankings (also under Kemeny, Copeland and Bradley–Terry pooling; τ to native metrics beside). J measures consistency between datasets, not correctness.
- **Correctness.** 31,380 synthetic anchors of known utility (same generator as v2 and v5); Spearman ρ, AUC correct vs wrong, mean score of wrong answers, hedge gap; (J, ρ) Pareto front and knee.
- **Honesty and transfer.** Nested item bootstrap (B = 1,000: select in-bag, score out-of-bag; optimism gap; plateau), leave-one-dataset-out and leave-one-family-out, per-family optima, one-step sensitivity (neighbours one lattice step away within the candidate set).
- **Leaderboards.** Exact match, v1 rubric (headline for comparability with v2), the five best by J, the best by ρ and the Pareto knee, each pooled by Bradley–Terry with the full significance battery.

2 Results

2.1 A single typed metric matches the nine-component composite

The best candidate is numeric tolerance alone ($J = 0.442$), with exact match alone second (0.439); every mixed candidate is below both (Figures 1, 2, 3). The v1 rubric sits at rank 295 of 305, but the whole candidate range is 0.042 wide. The three designs find the same region (Table 11); the uniform lattice has the best coverage and the highest best-J.

Table 1: v4 audit summary.

Candidates	305 (3 designs × 100 + 5 baselines) on a lattice of 3,108,105 points; J range 0.400–0.442
Best	numeric tolerance 1.00 (num_tol_only), $J = 0.442$, τ native 0.97; exact match alone $J = 0.439$ (rank 2); v1 rubric $J = 0.404$ (rank 295 of 305)
Best per design	Dirichlet 0.439, LHS 0.437, uniform lattice 0.442; regret over seeds $\leq$ 0.014
Nested selection ($B = 1{,}000$)	in-bag J of selected 0.387, out-of-bag 0.337 (optimism gap 0.050); v1 rubric out-of-bag 0.339
Selected vs v1 rubric, out of bag	-0.002, 95 % CI $[-0.033, +0.024]$; 204 distinct candidates selected
Plateau	305 of 305 not significantly below the best; practical plateau (within 0.02): 303; v1 rubric inside: yes
Transfer (LODO)	mean held-out τ: re-fitted weights +0.41, full best +0.41, v1 rubric +0.37
Anchors	v1 rubric ρ 0.878; best-J 0.796; best-ρ 0.884; knee 0.884; exact match nan; Spearman(J, ρ) across candidates -0.19
Sensitivity	one step moves J by at most 0.008 around the v1 rubric and 0.021 around the best
Leaderboard (v1 rubric, BT)	first Gemini 3.8 Flash, then Claude Opus 5, GPT-5.5; last Llama 3.3 70B; Friedman p = 1.0e-12; $W = 0.26$
Leaderboard (best J = numeric tolerance, BT)	first Claude Opus 5; identical ranking to exact match ($\tau = 1.00$)

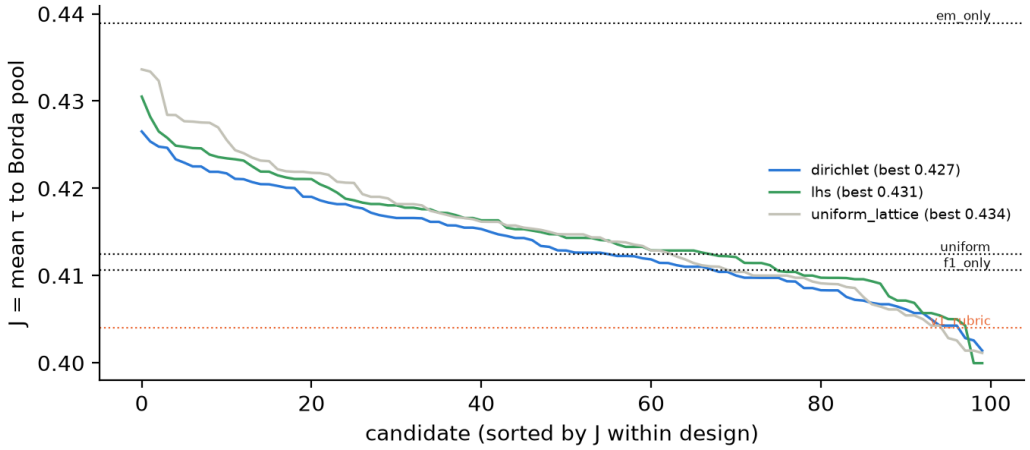


Figure 1: All 305 candidates sorted by J, with the baselines marked.

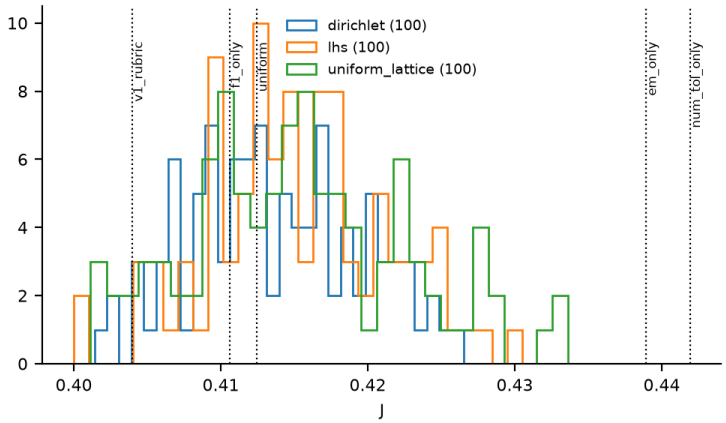


Figure 2: Distribution of J over the 100 candidates of each design, with the five baselines marked.

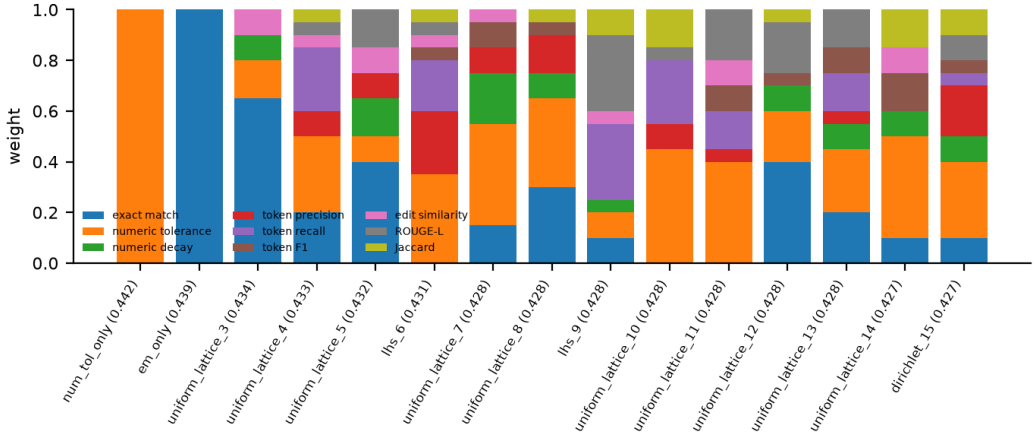


Figure 3: Component weights of the fifteen best candidates.

Table 2: Sampled designs: summary over 20 seeds.

design	best_J_mean	best_J_sd	best_J_max	mean_J	regret_mean	regret_max	min_pairwise_L1	mean_nearest_L1	discrepancy	zero_share
dirichlet_lattice	0.4320	0.0030	0.4391	0.4153	0.0100	0.0138	0.1950	0.4027	2.6738	0.8820
lhs_simplex	0.4313	0.0022	0.4373	0.4151	0.0107	0.0141	0.2150	0.4096	2.6718	0.8780
uniform_lattice	0.4343	0.0033	0.4420	0.4155	0.0077	0.0129	0.2350	0.4650	2.5904	0.9705

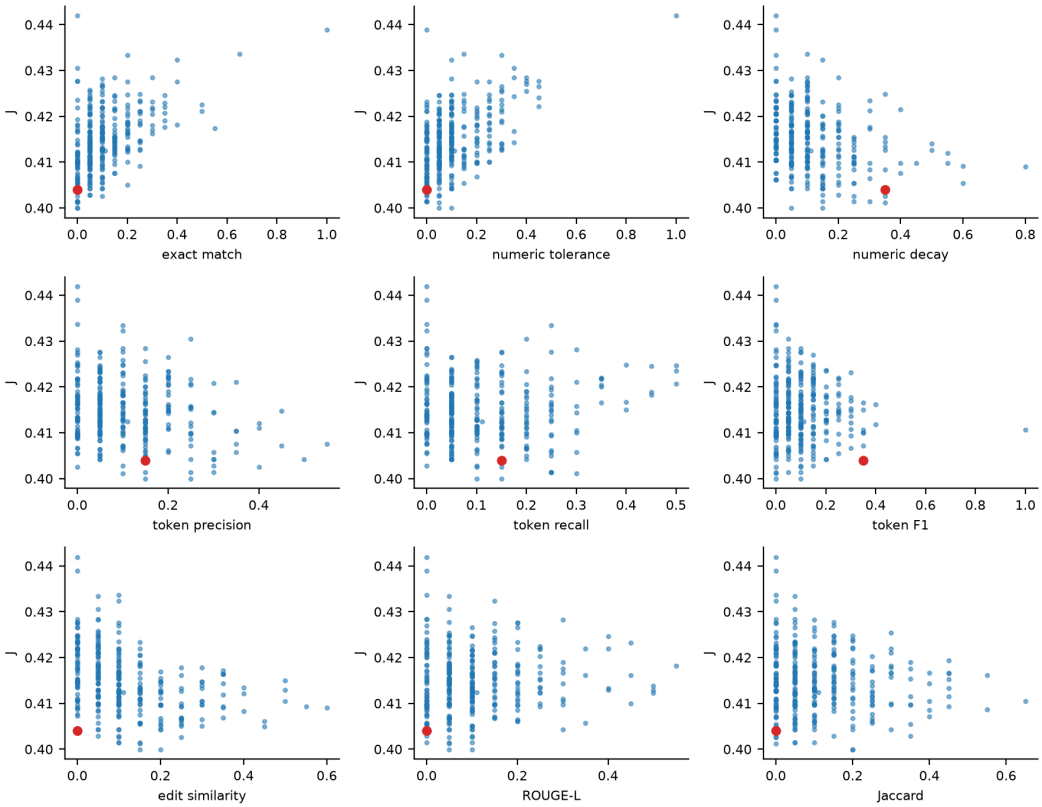


Figure 4: J against each component weight over all candidates; red = v1 rubric.

2.2 Honest selection

As in v2, the in-bag winner loses 0.050 out of bag and ends -0.002 from the v1 rubric with an interval that includes zero; 204 different candidates are selected across resamples and the plateau covers 305 of 305 (Figures 5, 6, 7).

2.3 Transfer

Re-fitted weights predict a held-out dataset no better than the v1 rubric (Figure 8); the per-family optima are numeric tolerance for EM and SM but recall- and precision-heavy mixes for TQA and FIN (Figure 9, Table 41).

Table 3: Per-family optimum.

family	best weights	J own best	J global best
EM	numeric tolerance 1.00	0.361	0.361
SM	numeric tolerance 1.00	0.580	0.580
TQA	exact match 0.50, numeric decay 0.05, token precision 0.15, token recall 0.05, token F1 0.10, edit similarity 0.05, ROUGE-L 0.10	0.513	0.512
FIN	exact match 0.10, numeric tolerance 0.10, token precision 0.20, token recall 0.35, token F1 0.10, ROUGE-L 0.15	0.834	0.619

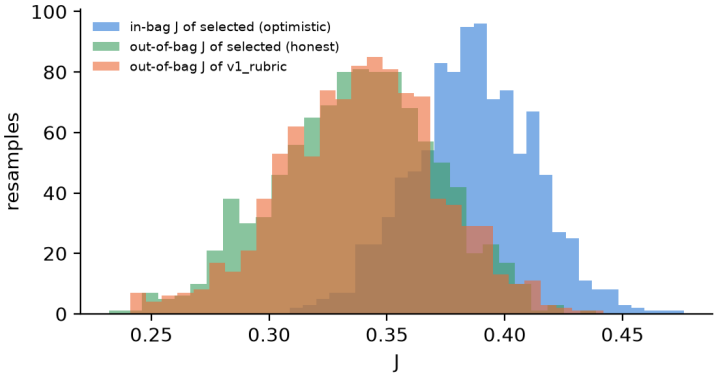


Figure 5: In-bag vs out-of-bag J of the selected candidate against the v1 rubric.

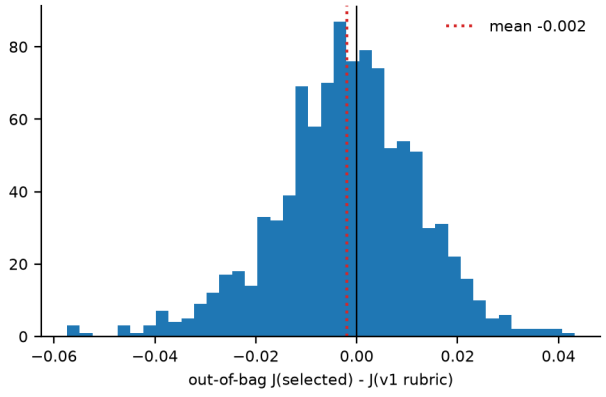


Figure 6: Paired out-of-bag difference between the selected candidate and the v1 rubric over 1,000 resamples.

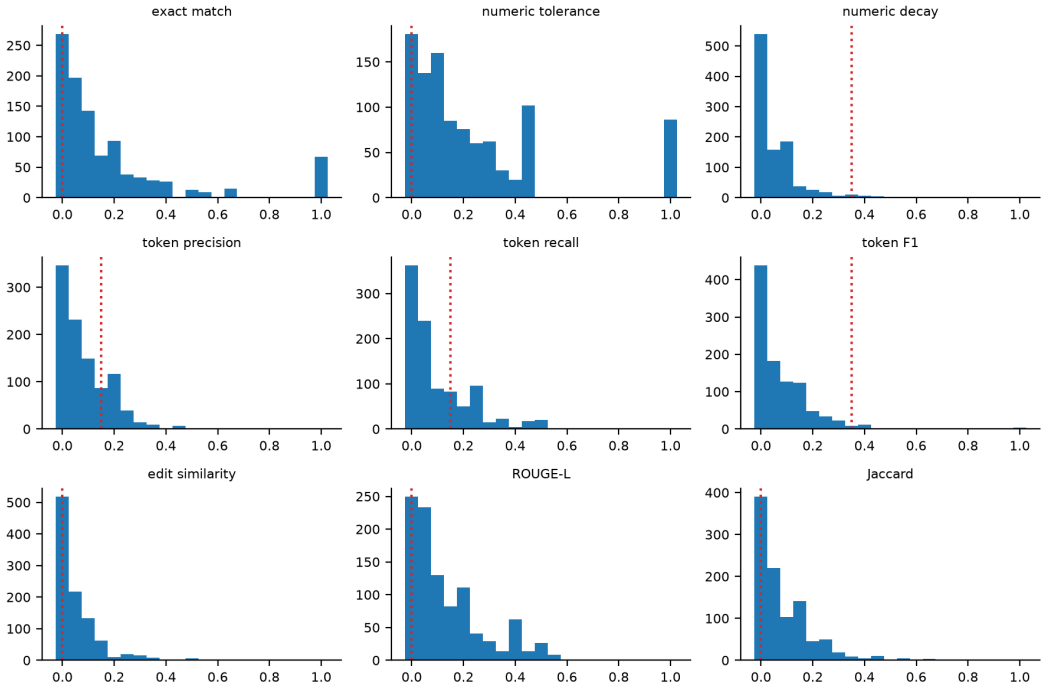


Figure 7: Weights selected as in-bag best across 1,000 bootstrap resamples; dotted = v1 rubric.

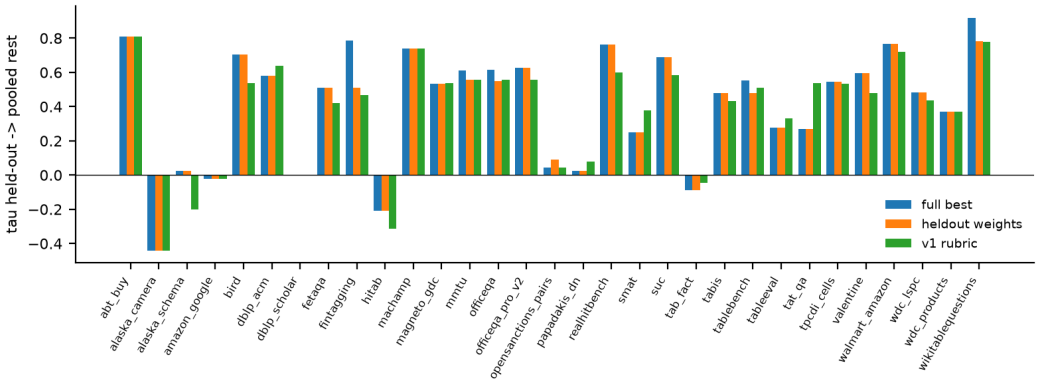


Figure 8: Leave-one-dataset-out: agreement of the held-out dataset with the pooled rest under re-fitted weights, the full best and the v1 rubric.

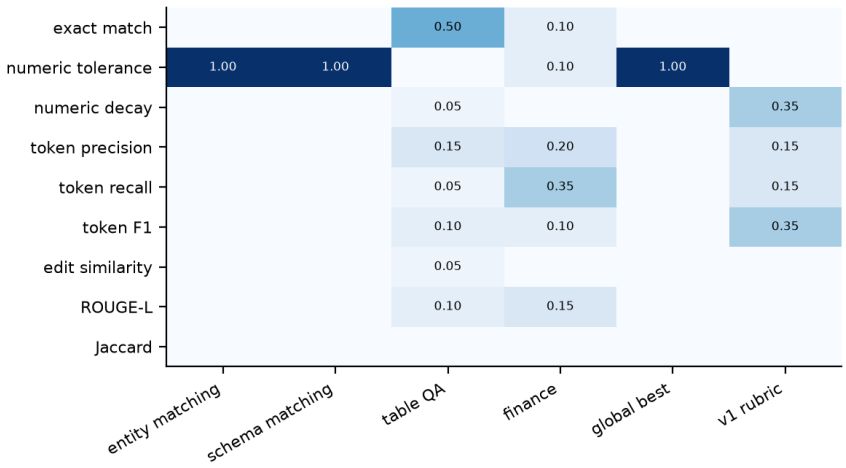


Figure 9: Per-family optimum, global optimum and the v1 rubric over the nine components.

2.4 Correctness

Numeric tolerance alone is the best by J but the worst labelled rule on the anchors ($\rho = 0.796$ vs 0.878 for the v1 rubric): it gives no partial credit to near misses, which the anchors reward (Figure 11). The Pareto knee (em 0.40, num_tol 0.10, num_decay 0.15, tok_prec 0.10, edit_sim 0.10, rouge_l 0.15) keeps $J = 0.432$ and $\rho = 0.884$. Across candidates J and ρ correlate -0.19: agreement and correctness pull apart here as in v2.

Table 4: Anchor correctness for the labelled candidates.

label	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J	rho	rho_ds	auc	wrong_mean	hedge_gap
num_tol_only	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.442	0.796	0.808	0.989	0.022	-0.595
em_only	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.828	0.830	0.989	0.011	-0.807
uniform	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.412	0.878	0.861	0.994	0.046	-0.555
f1_only	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.411	0.856	0.846	0.989	0.030	-0.496
v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.404	0.878	0.868	0.994	0.034	-0.506

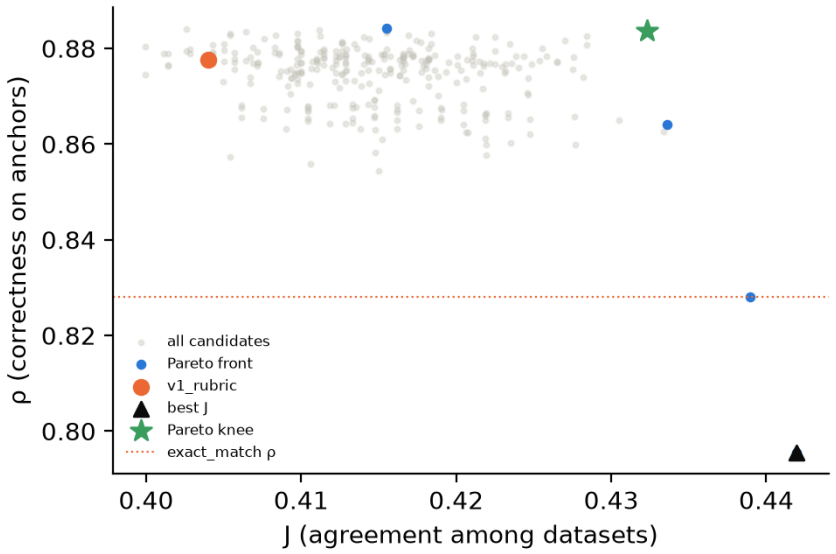


Figure 10: Agreement J against anchor correctness ρ for every candidate.

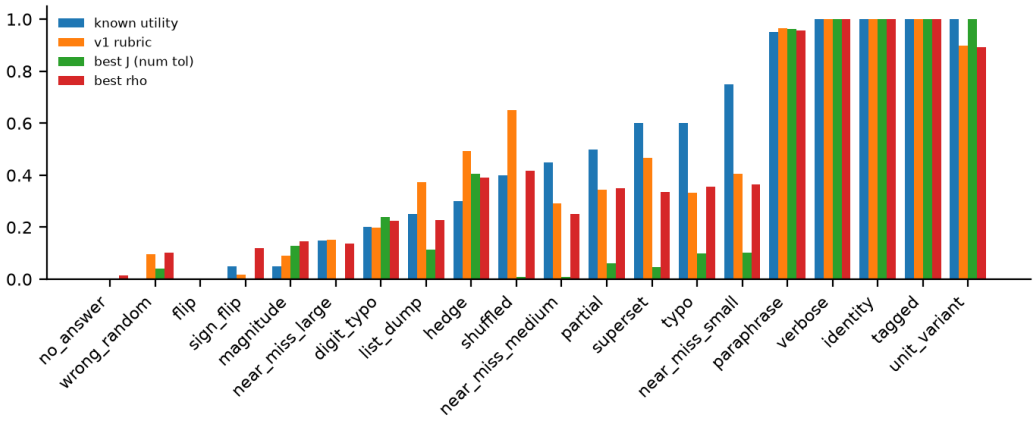


Figure 11: Anchors: known utility of each operator against the mean score it receives under the v1 rubric, the best-J and the best-ρ candidates.

2.5 Leaderboard

Under the v1 rubric the ranking is the one v2 reports (same answers, same rule): Gemini 3.8 Flash first by Bradley–Terry, GPT-5.5 first by Borda, Llama 3.3 70B last. Under the best-J rule (numeric tolerance) and under exact match Claude Opus 5 is first. Rule choice moves the top three and nothing else.

Table 5: Leaderboard under the headline rule (v1 rubric).

rank	model	BT strength	SE	Borda rank	Borda 95%	P(first) BT	mean score
1	Gemini 3.8 Flash	0.121	0.016	3	[1, 3]	0.990	0.812
2	Claude Opus 5	0.095	0.000	2	[1, 3]	0.010	0.811
3	GPT-5.5	0.052	0.016	1	[1, 3]	0.000	0.808
4	Claude Sonnet 5	0.029	0.016	4	[4, 6]	0.000	0.795
5	Grok 4	0.027	0.016	5	[4, 6]	0.000	0.797
6	GPT-5.4-mini	-0.010	0.016	8	[5, 8]	0.000	0.790
7	DeepSeek-chat	-0.018	0.016	7	[5, 8]	0.000	0.784
8	DeepSeek-R1	-0.025	0.016	6	[5, 8]	0.000	0.788
9	gpt-oss-120b	-0.100	0.016	9	[9, 10]	0.000	0.767
10	Llama 3.3 70B	-0.170	0.016	10	[9, 10]	0.000	0.738

Table 6: Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
exact match	3 (3)	1 (1)	2 (2)	5 (4)	4 (5)	6 (8)	8 (7)	7 (6)	9 (9)	10 (10)	5.3e-14	0.28
v1 rubric	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (8)	7 (7)	8 (6)	9 (9)	10 (10)	1.0e-12	0.26
top1	3 (3)	1 (1)	2 (2)	5 (4)	4 (5)	6 (7)	8 (8)	7 (6)	9 (9)	10 (10)	7.2e-14	0.28
top2	3 (3)	1 (1)	2 (2)	5 (4)	4 (5)	6 (8)	8 (7)	7 (6)	9 (9)	10 (10)	5.3e-14	0.28
top3	1 (3)	2 (1)	3 (2)	4 (4)	5 (5)	6 (7)	7 (8)	8 (6)	9 (9)	10 (10)	9.2e-14	0.28
top4	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	7 (8)	6 (7)	8 (6)	9 (9)	10 (10)	8.5e-14	0.28
top5	1 (3)	2 (1)	3 (1)	4 (4)	5 (5)	6 (7)	7 (8)	8 (6)	9 (9)	10 (10)	1.1e-13	0.28
best-rho	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (7)	7 (8)	8 (6)	9 (9)	10 (10)	3.6e-13	0.27
knee	1 (3)	2 (1)	3 (1)	4 (4)	5 (5)	6 (7)	7 (8)	8 (6)	9 (9)	10 (10)	1.1e-13	0.28

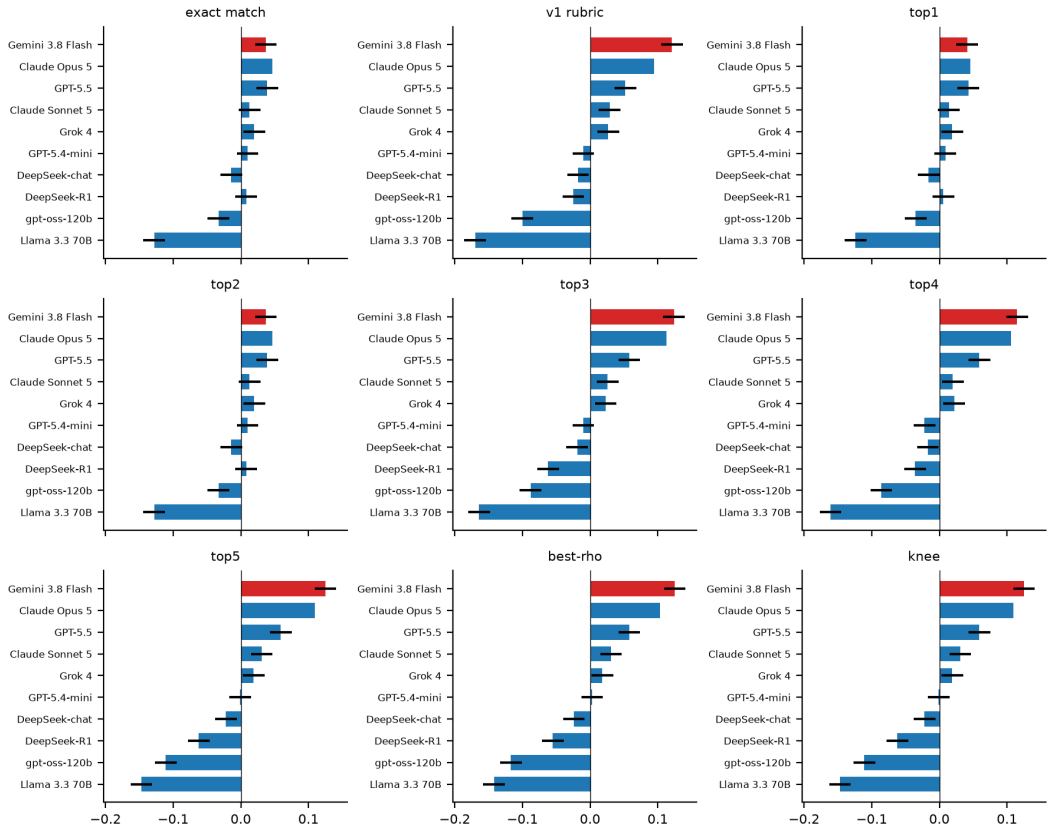


Figure 12: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

2.6 Comparability and diagnostics

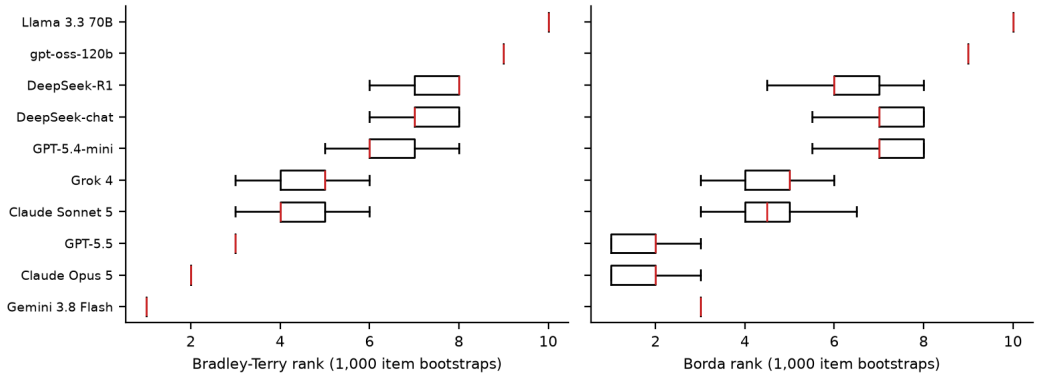


Figure 13: Bootstrap distribution of each model's rank under the headline rule (v1 rubric).

Table 7: Official-split estimates under the headline rule averaged over the ten models.

dataset	suite_mean	official_ estimate	suite_f1_ pos	official_precision_ pos	official_recall_ pos	official_f1_ pos	official_positive_ share
abt_buy	0.970	0.986	0.968	0.926	0.949	0.936	0.108
alaska_camera	0.958	0.958	0.955	1.000	0.921	0.955	0.500
alaska_schema	0.928	0.928	–	–	–	–	–
amazon_google	0.936	0.932	0.929	0.611	0.942	0.739	0.102
bird	0.187	0.213	–	–	–	–	–
dblp_acm	0.986	0.991	0.985	0.979	0.978	0.976	0.180
dblp_scholar	0.949	0.966	0.948	0.900	0.924	0.910	0.186
fetaqa	0.382	0.382	–	–	–	–	–
fintagging	0.880	0.876	–	–	–	–	–
hitab	0.770	0.773	–	–	–	–	–
machamp	0.887	0.890	0.892	0.644	0.853	0.733	0.175
magneto_gdc	0.934	0.938	–	–	–	–	–
mmtu	0.867	0.829	–	–	–	–	–
officeqa	0.129	0.129	–	–	–	–	–
officeqa_pro_v2	0.139	0.139	–	–	–	–	–
opensanctions_ pairs	0.821	0.727	0.773	0.993	0.649	0.777	0.769
papadakis_dn	0.941	0.988	0.833	0.550	0.963	0.683	0.018
realhitbench	0.673	0.673	–	–	–	–	–
smat	0.918	0.916	–	–	–	–	–
suc	0.914	0.912	–	–	–	–	–
tab_fact	0.798	0.789	0.758	0.896	0.674	0.764	0.500
tabis	0.915	0.918	–	–	–	–	–
tablebench	0.818	0.816	–	–	–	–	–
tableeval	0.749	0.754	–	–	–	–	–
tat_qa	0.803	0.758	–	–	–	–	–
tpcdi_cells	0.456	0.461	–	–	–	–	–
valentine	0.984	0.979	–	–	–	–	–
walmart_amazon	0.958	0.975	0.947	0.832	0.927	0.875	0.094
wdc_lspc	0.982	0.983	0.983	0.965	0.979	0.971	0.264
wdc_products	0.978	0.985	0.979	0.980	0.968	0.974	0.280
wikitablequestions	0.846	0.844	–	–	–	–	–

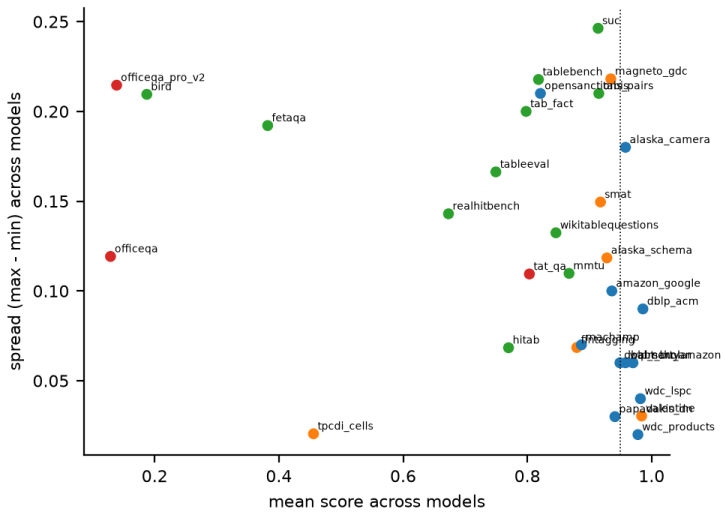


Figure 14: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

A All figures

Every figure that can be drawn from the saved outputs; source data in `data/`.

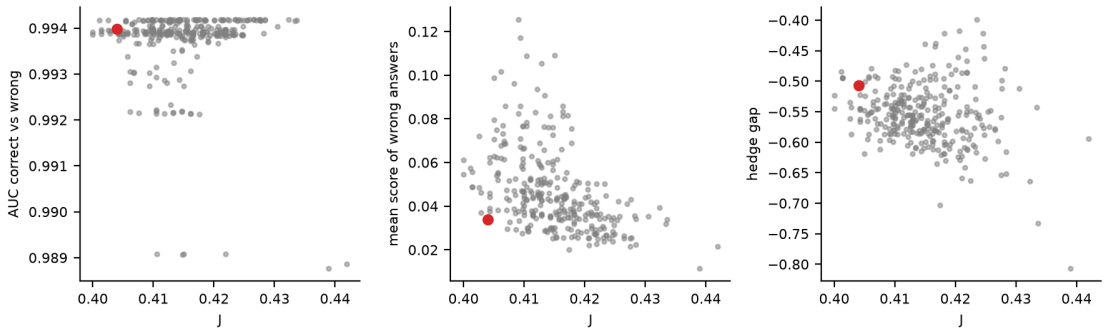


Figure 15: Three further anchor diagnostics against J for every candidate (red = v1 rubric).

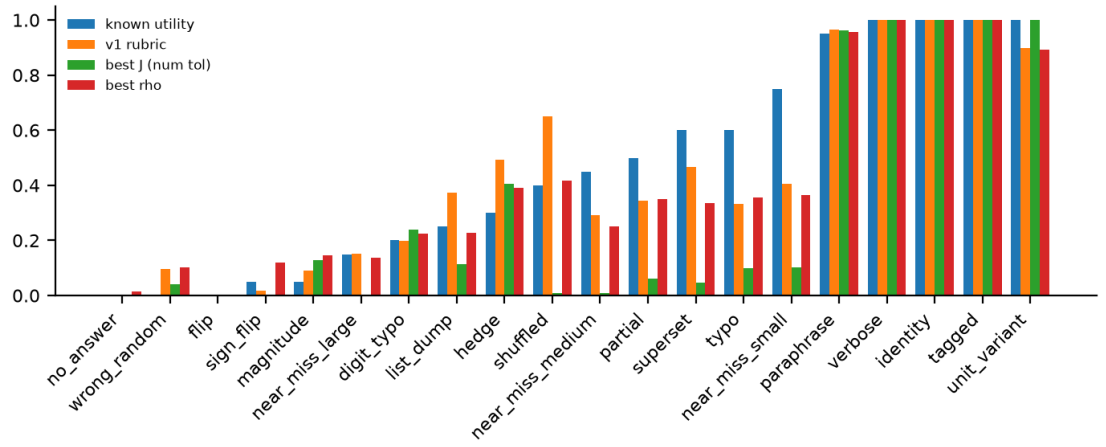


Figure 16: Anchors: known utility of each operator against the mean score it receives under the v1 rubric, the best-J and the best-ρ candidates.

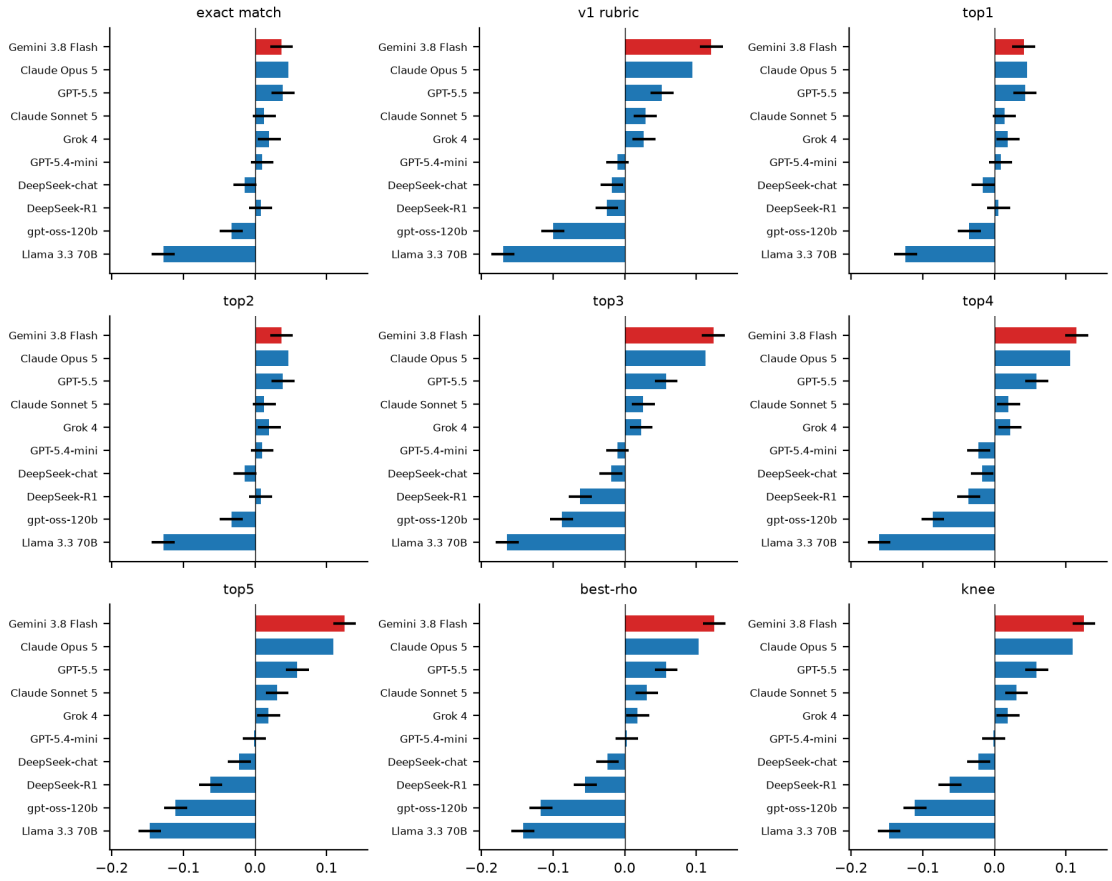


Figure 17: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

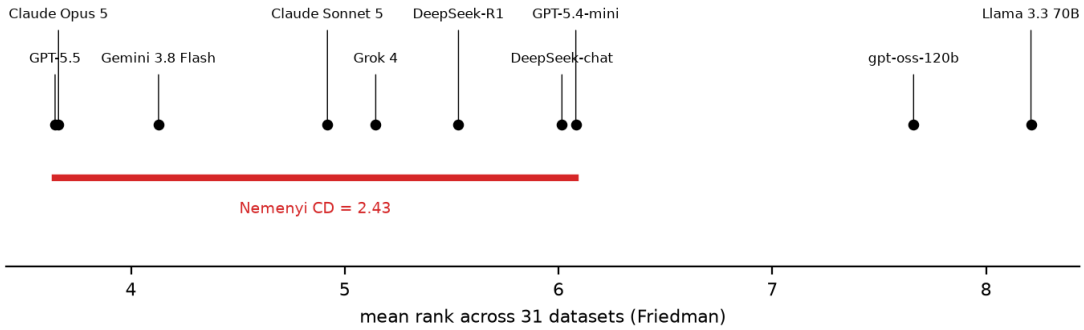


Figure 18: Critical-difference view of the Friedman test under the headline rule: mean rank per model and the Nemenyi critical difference at $\alpha = .05$.

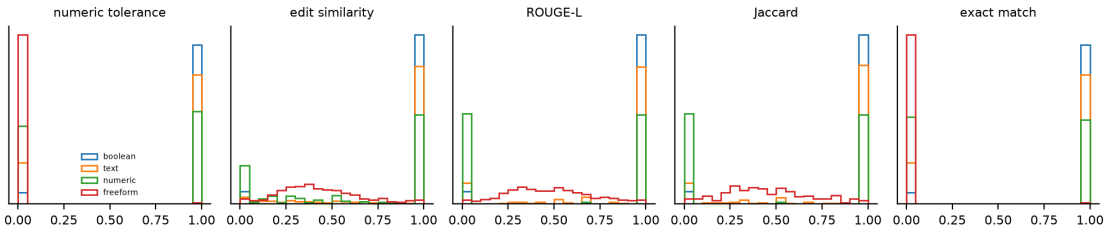


Figure 19: Distributions of the five components not in v1, by answer type.

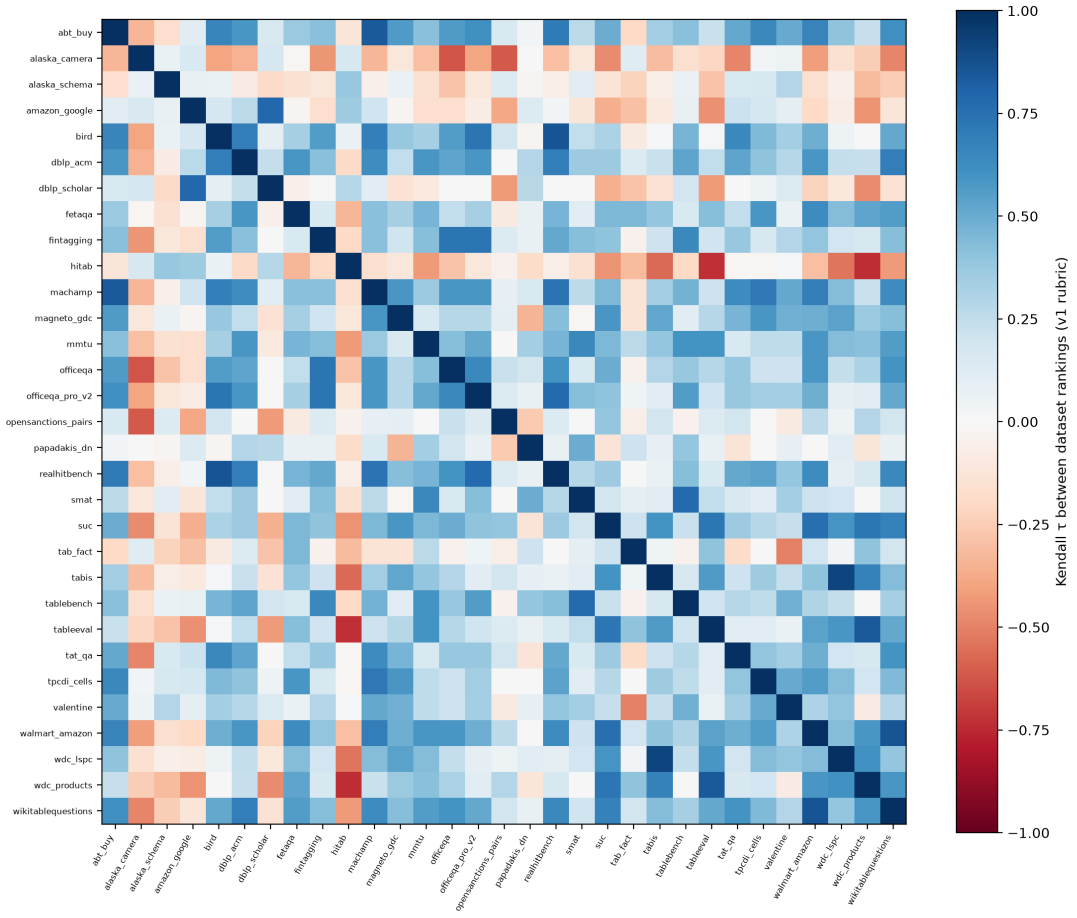


Figure 20: Kendall τ between the model rankings of each pair of datasets under the headline rule.

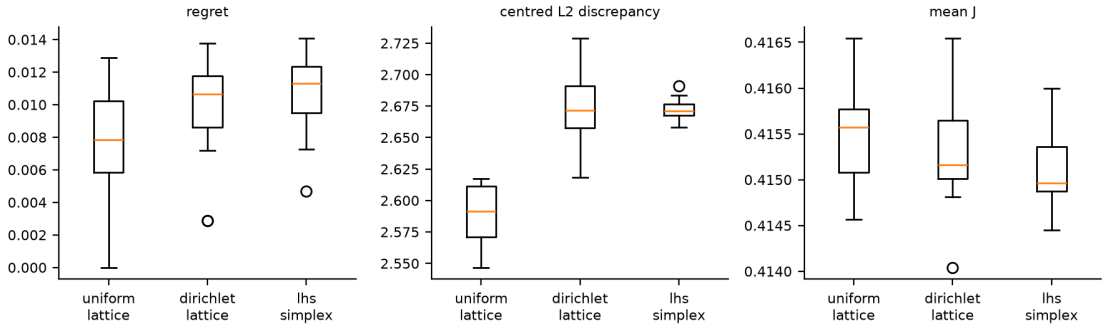


Figure 21: Sampled designs over 20 seeds: regret against the best over all seeds and designs, centred L_2 discrepancy and mean J of the design.

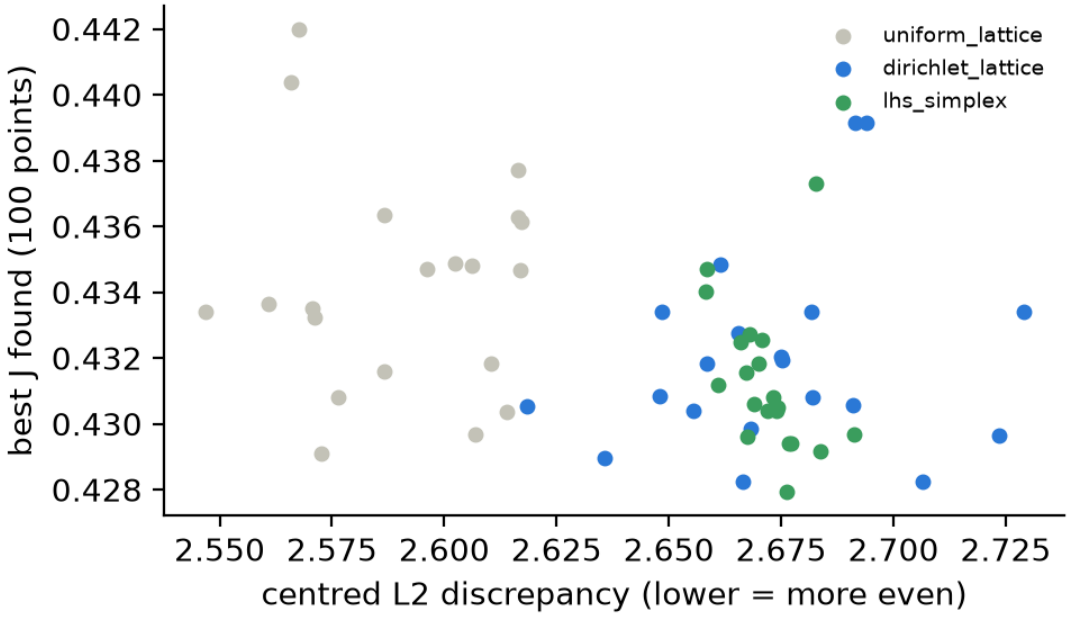


Figure 22: Sampled designs: best J found per seed.

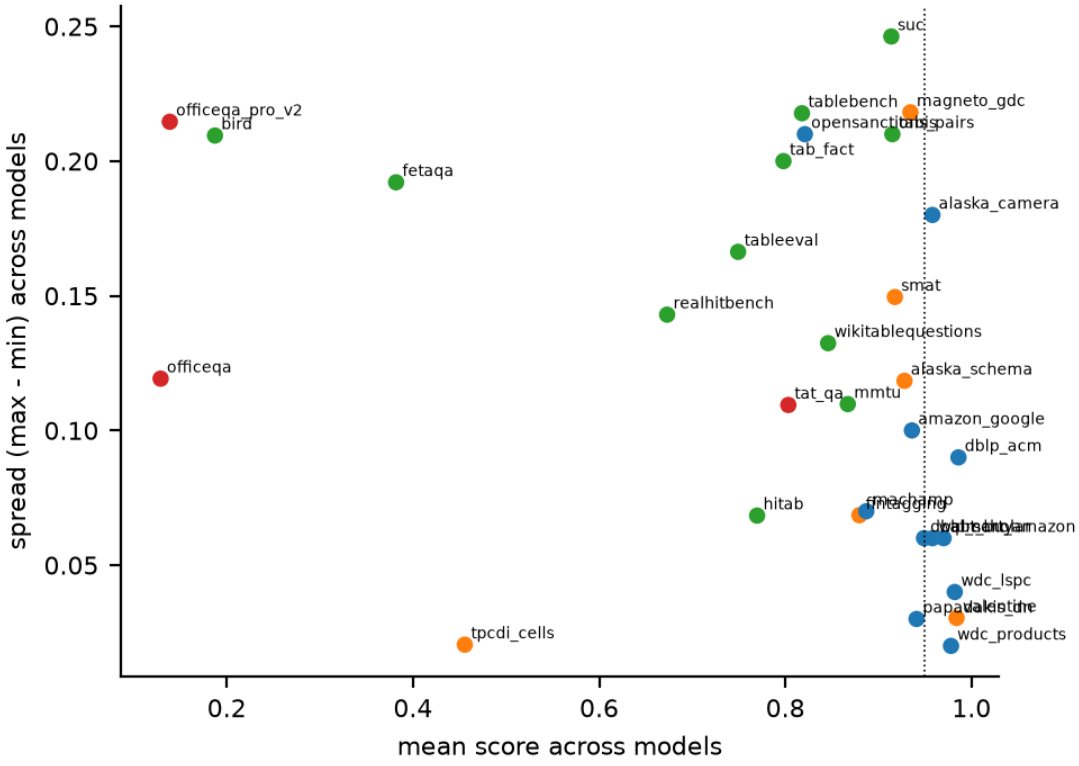


Figure 23: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

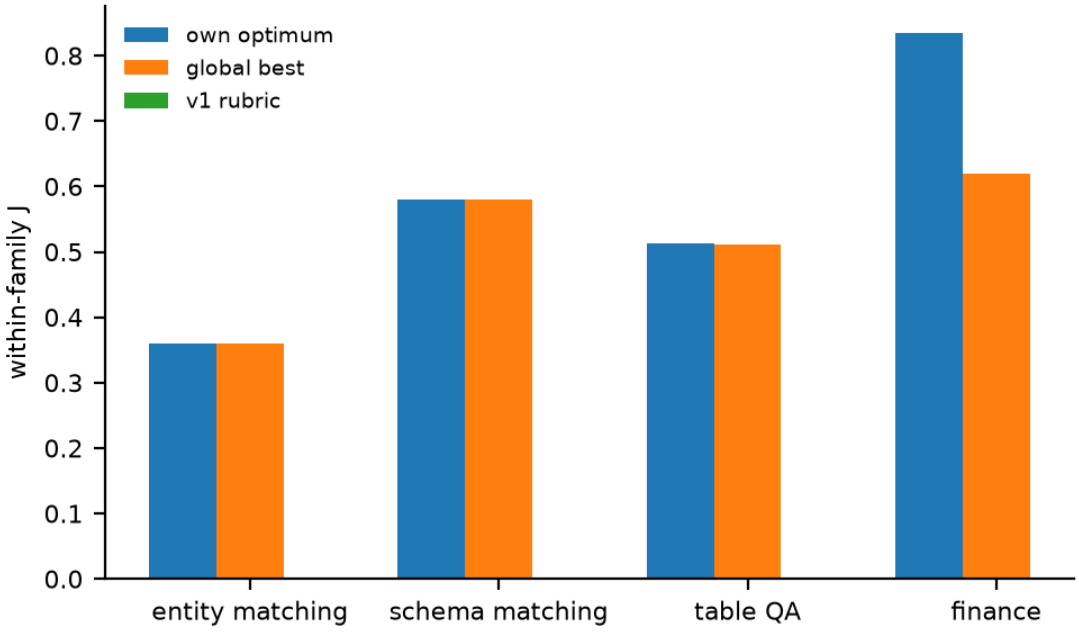


Figure 24: Within-family J under each family's own optimum, the global best and the v1 rubric.

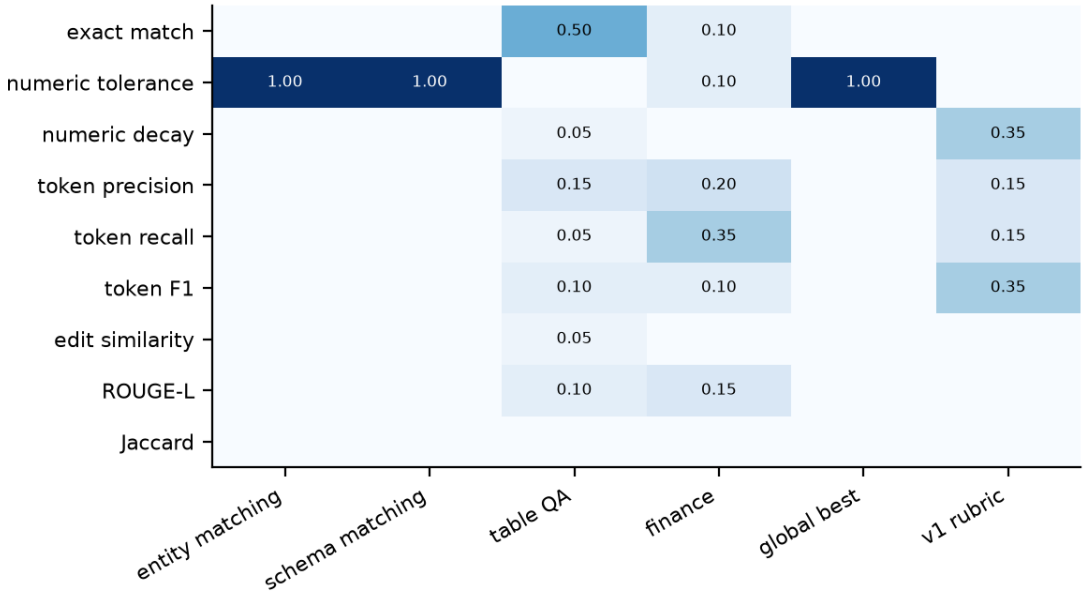


Figure 25: Per-family optimum, global optimum and the v1 rubric over the nine components.

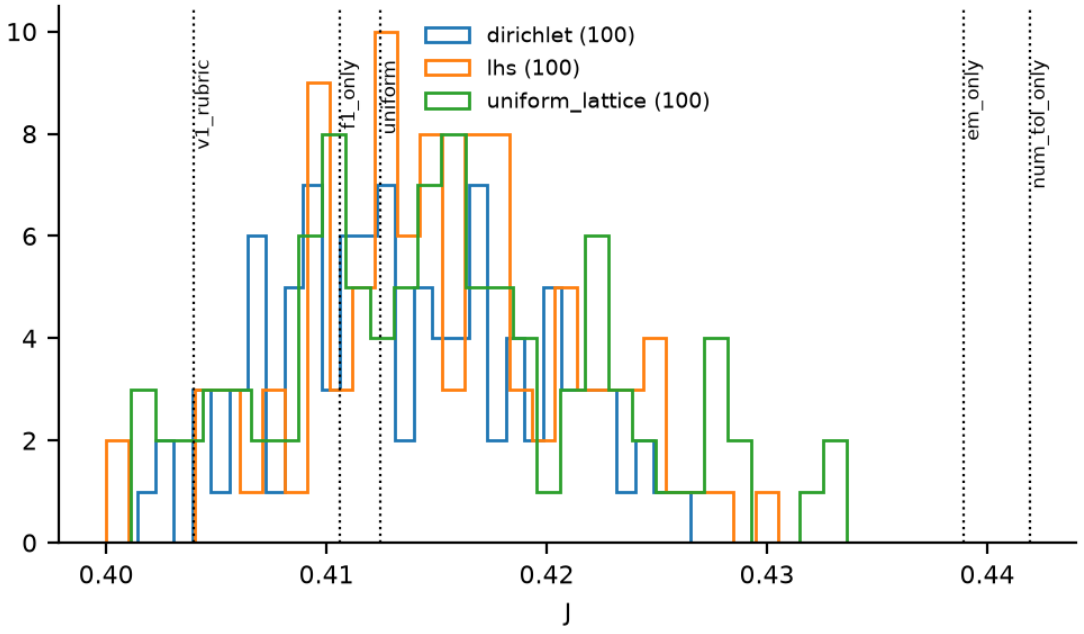


Figure 26: Distribution of J over the 100 candidates of each design, with the five baselines marked.

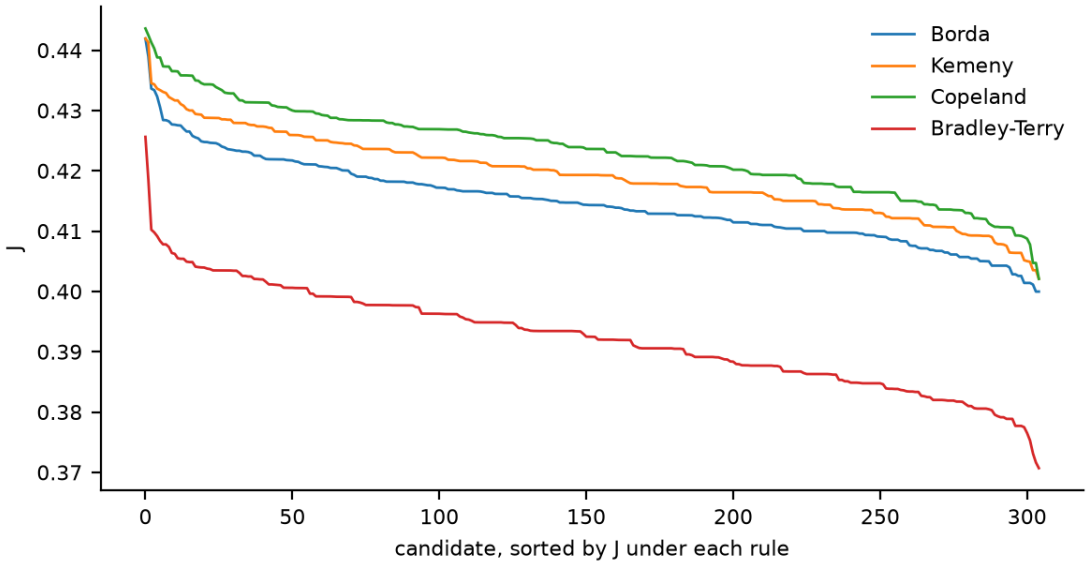


Figure 27: Objective over all candidates under four pooling rules.

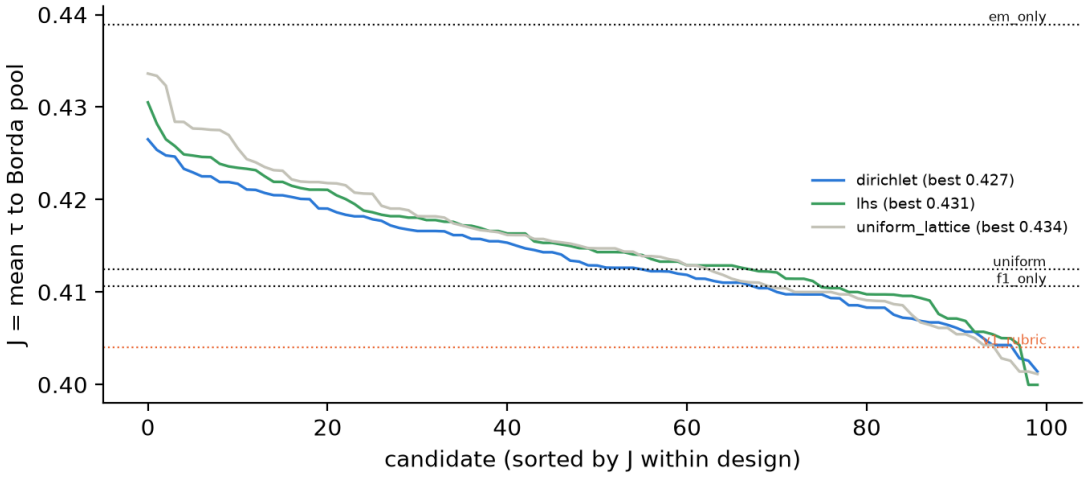


Figure 28: All 305 candidates sorted by J, with the baselines marked.

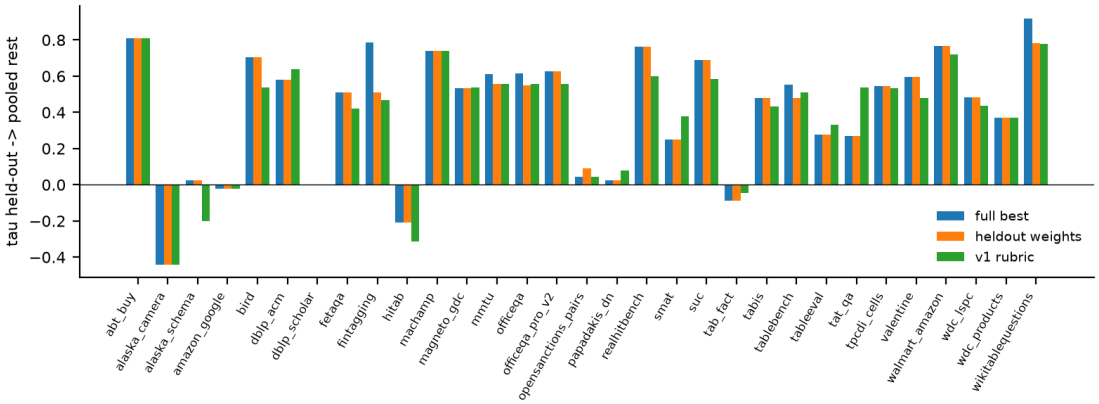


Figure 29: Leave-one-dataset-out: agreement of the held-out dataset with the pooled rest under re-fitted weights, the full best and the v1 rubric.

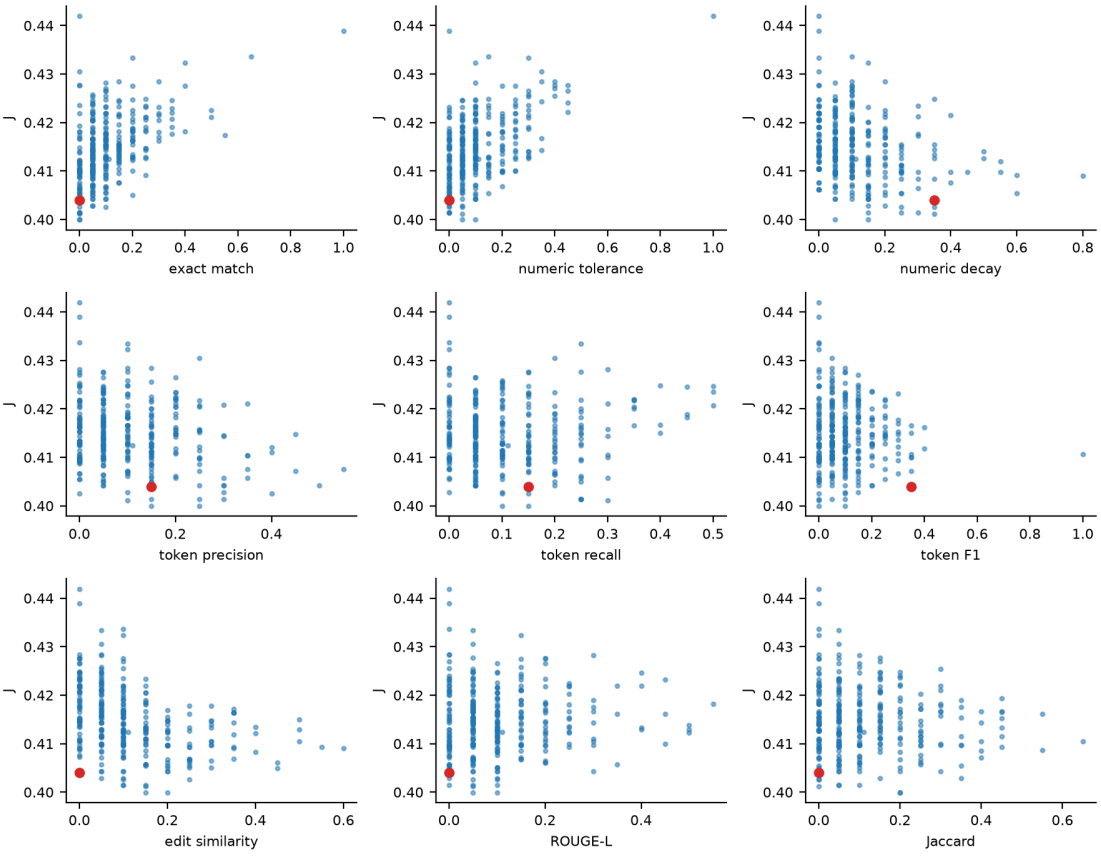


Figure 30: J against each component weight over all candidates; red = v1 rubric.

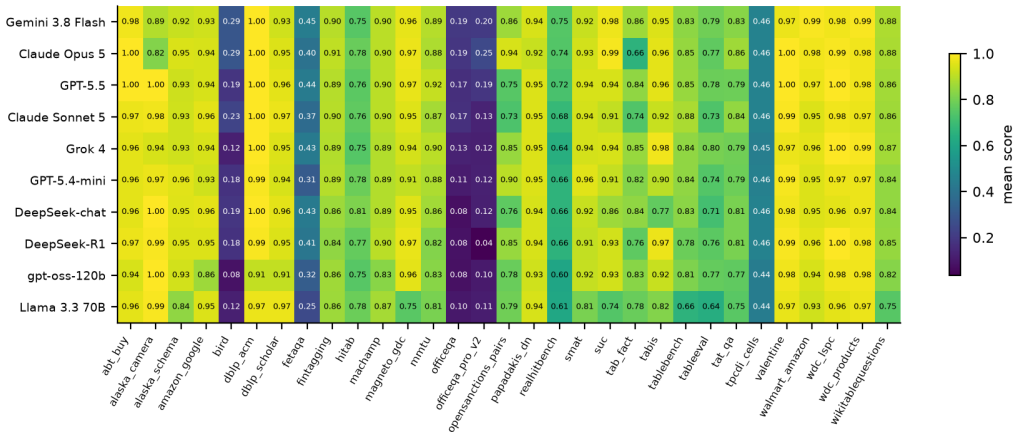


Figure 31: Mean score per model and dataset under the headline rule.

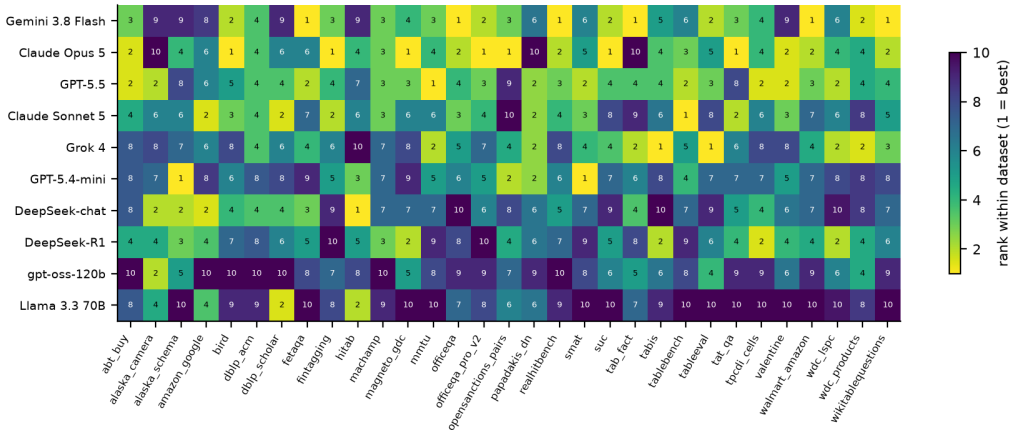


Figure 32: Rank per model and dataset under the headline rule.

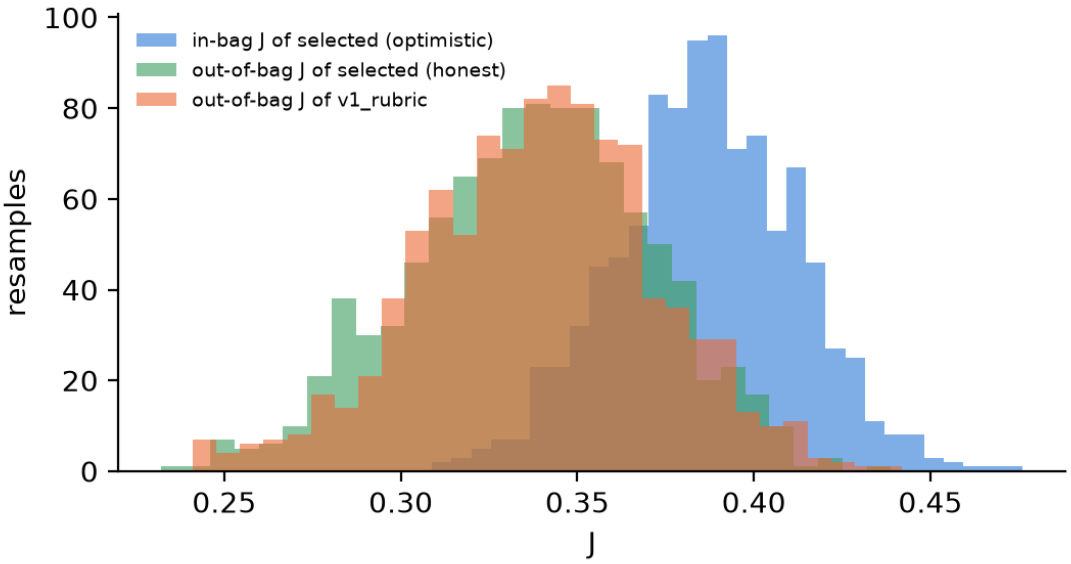


Figure 33: In-bag vs out-of-bag J of the selected candidate against the v1 rubric.

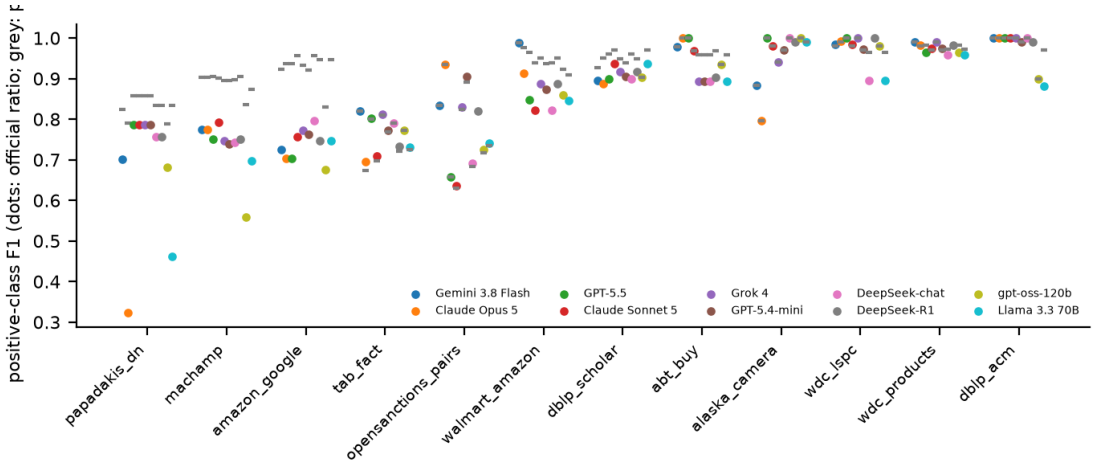


Figure 34: Positive-class F1 on the balanced pool (grey) and at the official match ratio (coloured) per model and matching dataset.

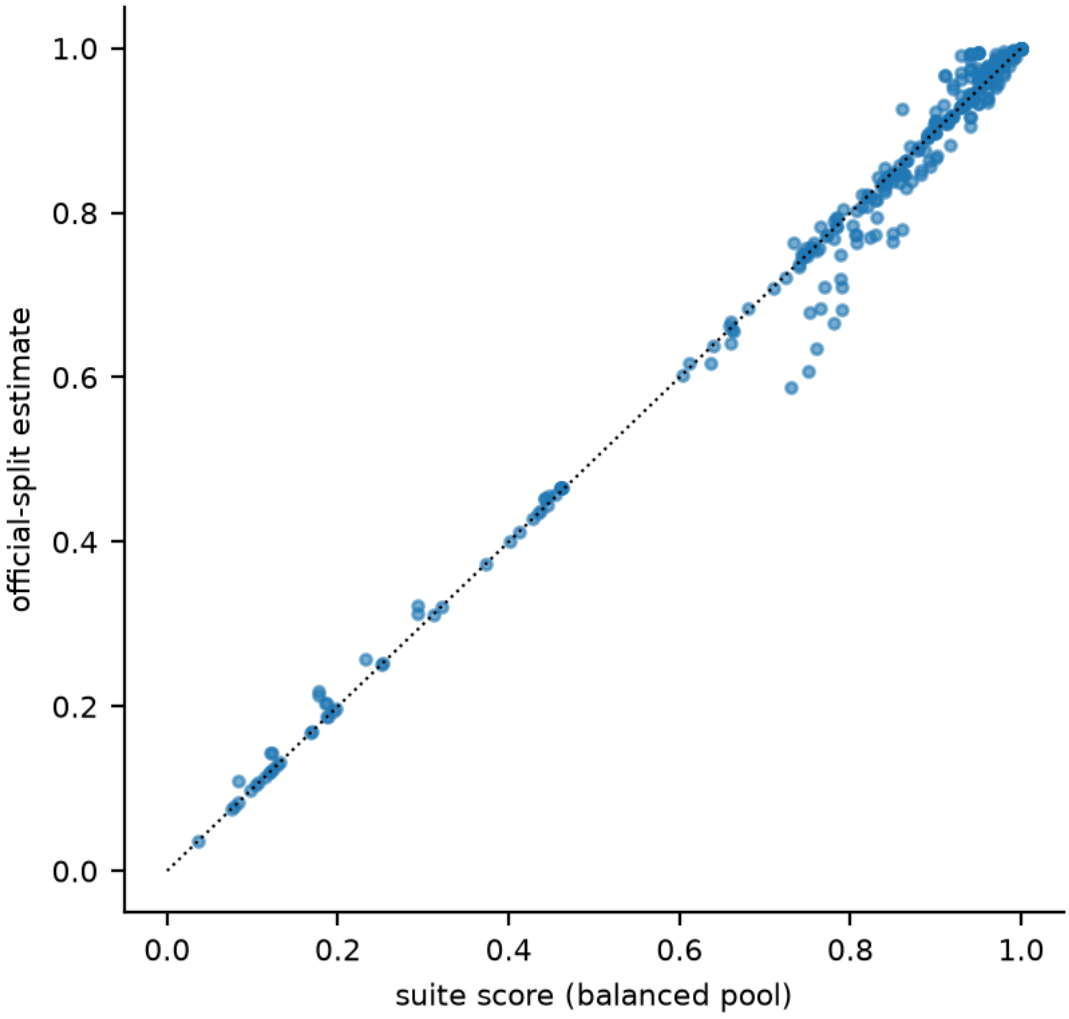


Figure 35: Suite score against post-stratified official-split estimate for every (dataset, model) pair under the headline rule.

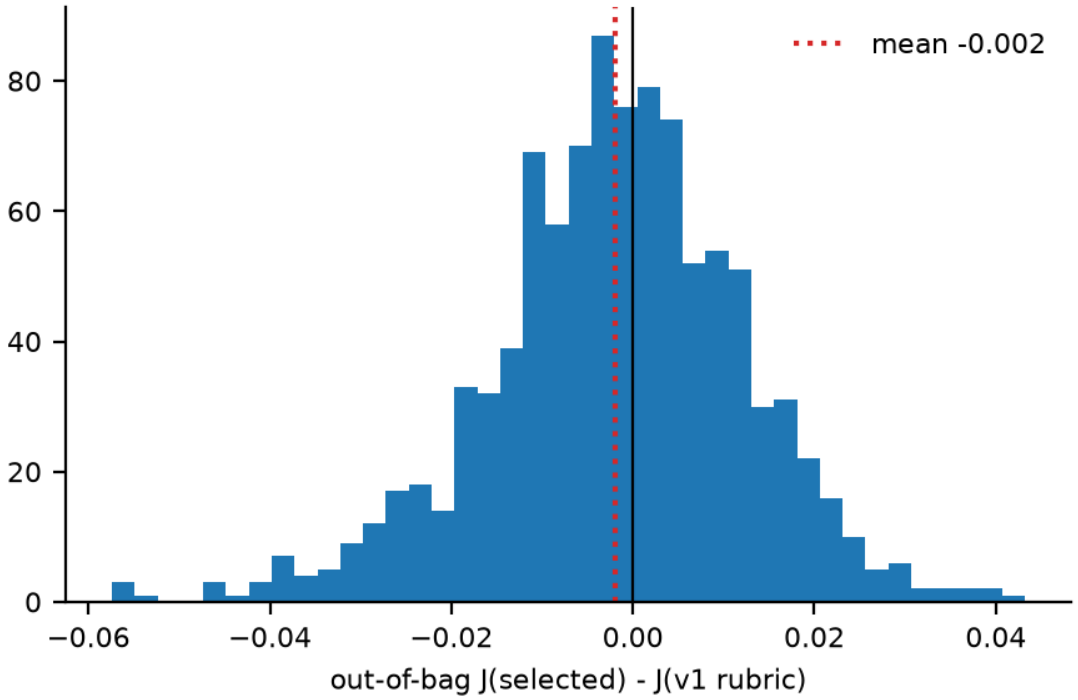


Figure 36: Paired out-of-bag difference between the selected candidate and the v1 rubric over 1,000 resamples.

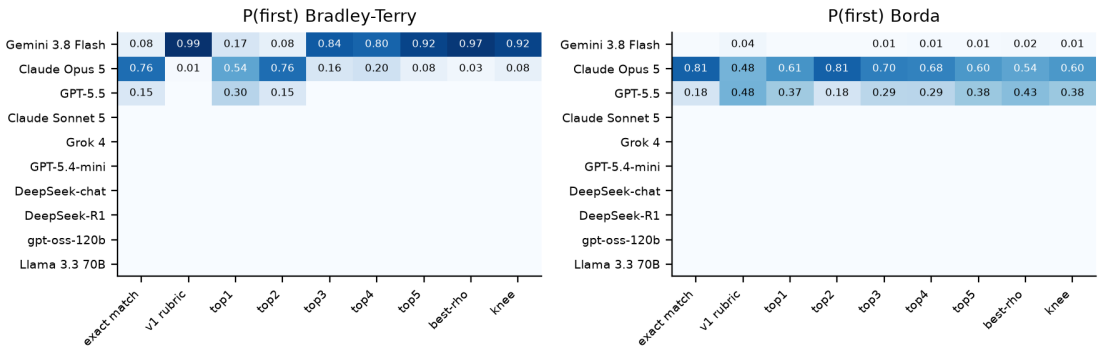


Figure 37: Bootstrap probability of first place under every rule, by pooling method.

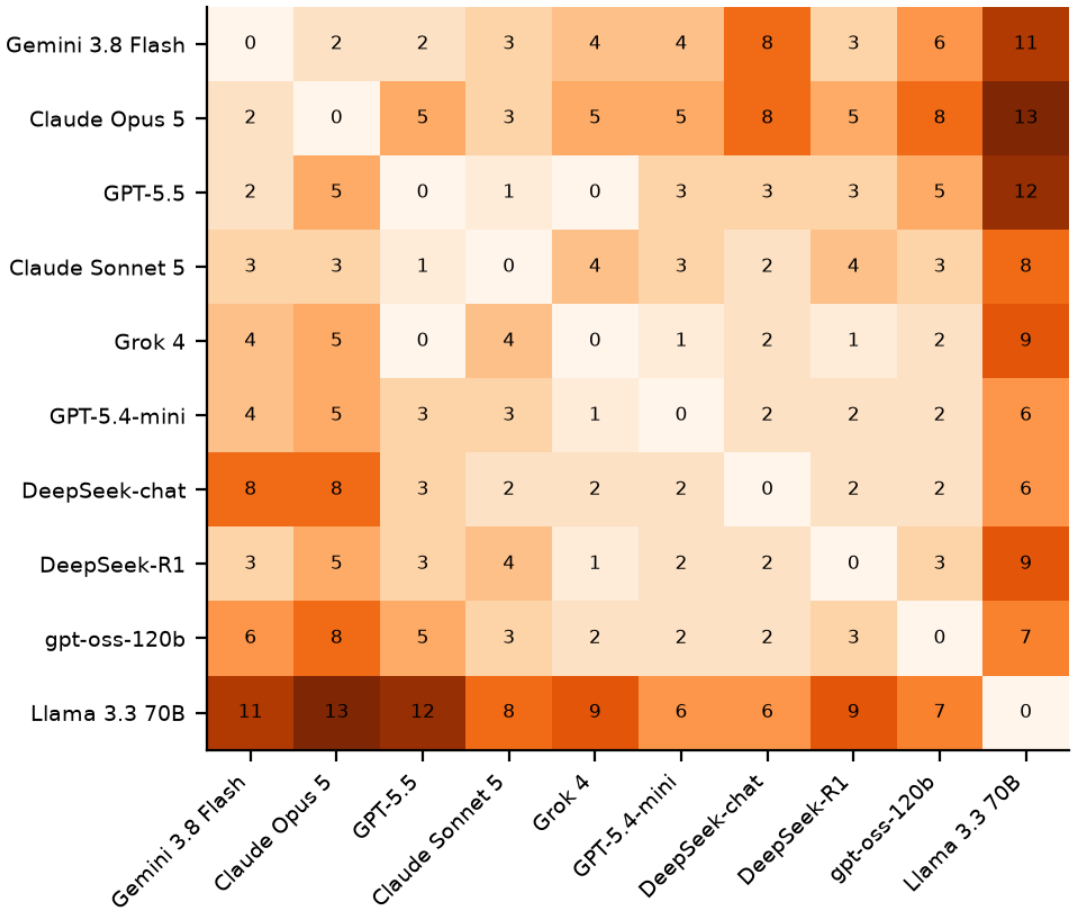


Figure 38: Number of datasets (of 31) in which the per-item paired test separates each pair of models under the headline rule (Holm within dataset).

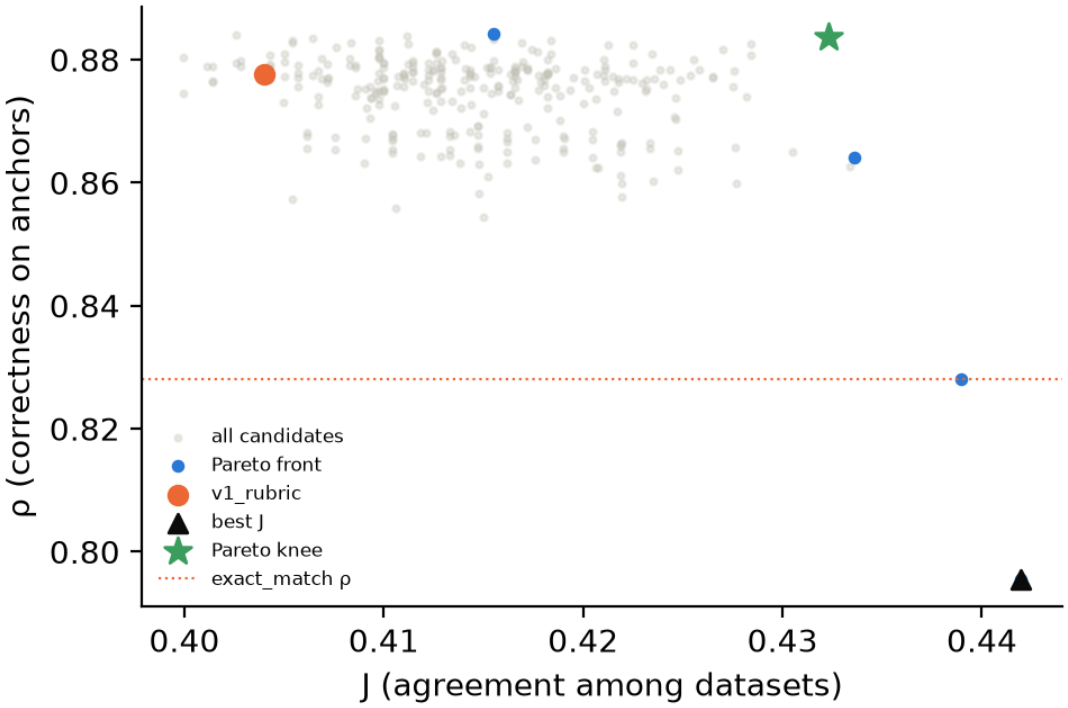


Figure 39: Agreement J against anchor correctness ρ for every candidate.

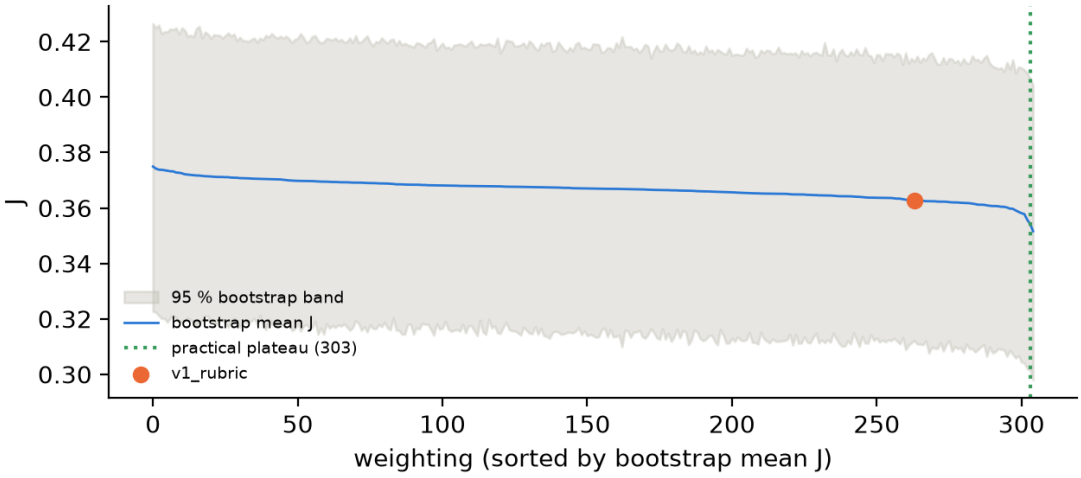


Figure 40: Bootstrap band over sorted candidates and the plateau boundary.

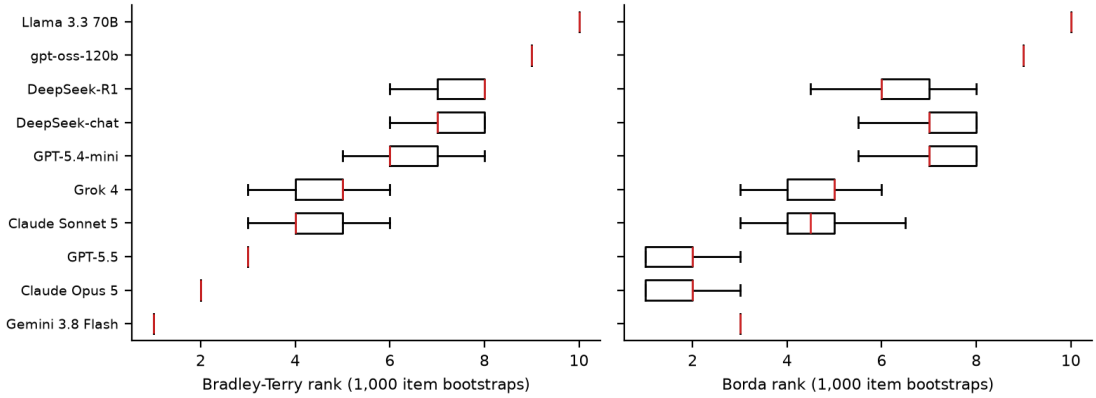


Figure 41: Bootstrap distribution of each model's rank under the headline rule (v1 rubric).

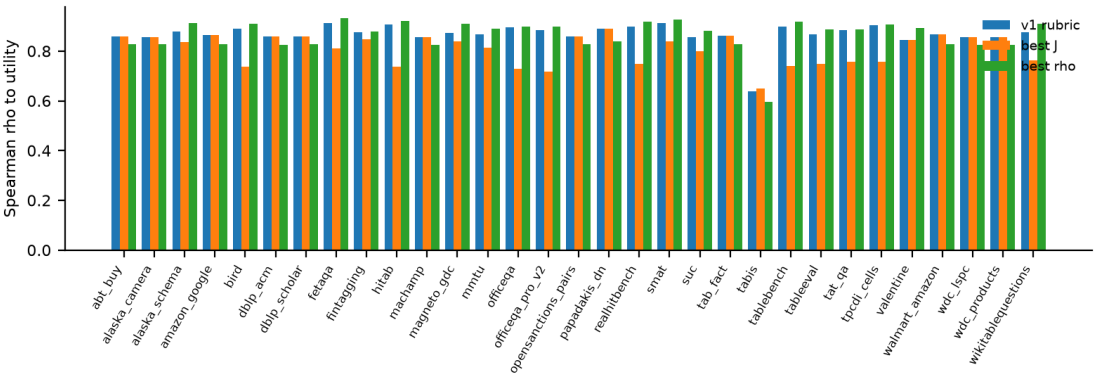


Figure 42: Anchor correctness per dataset under the v1 rubric, the best-J and the best-ρ candidates.

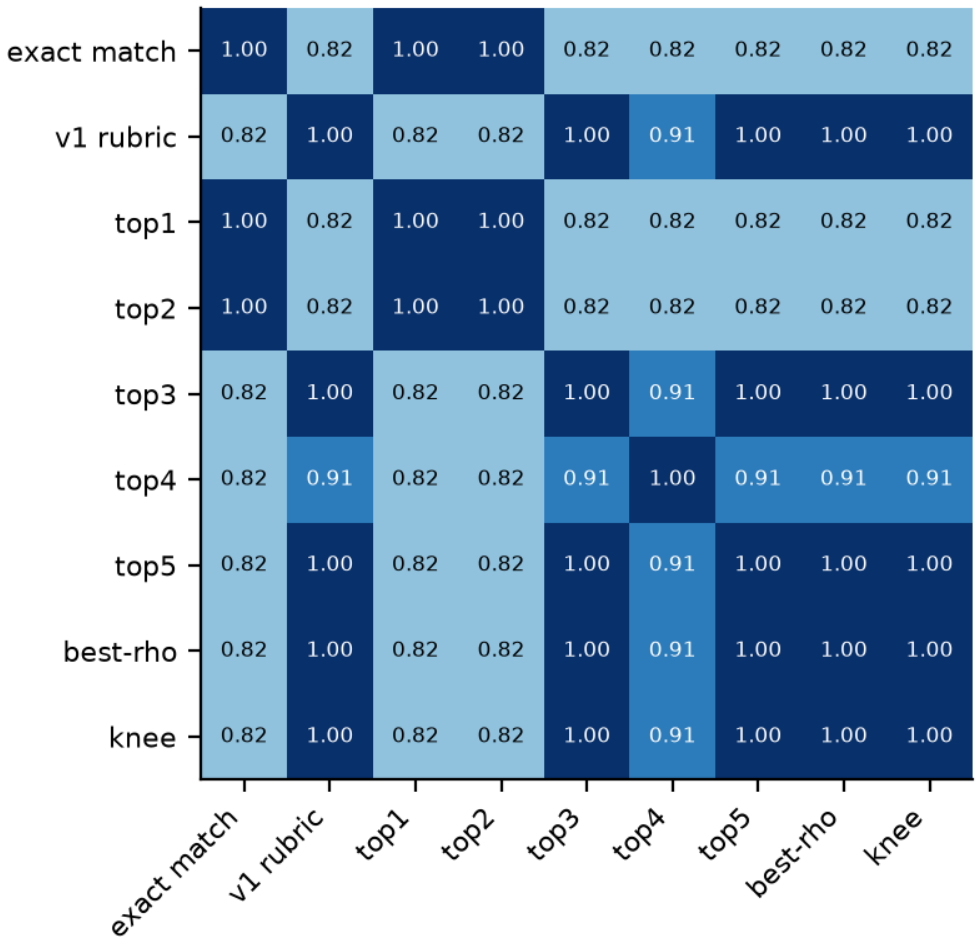


Figure 43: Kendall τ between the Bradley–Terry rankings of each pair of rules.

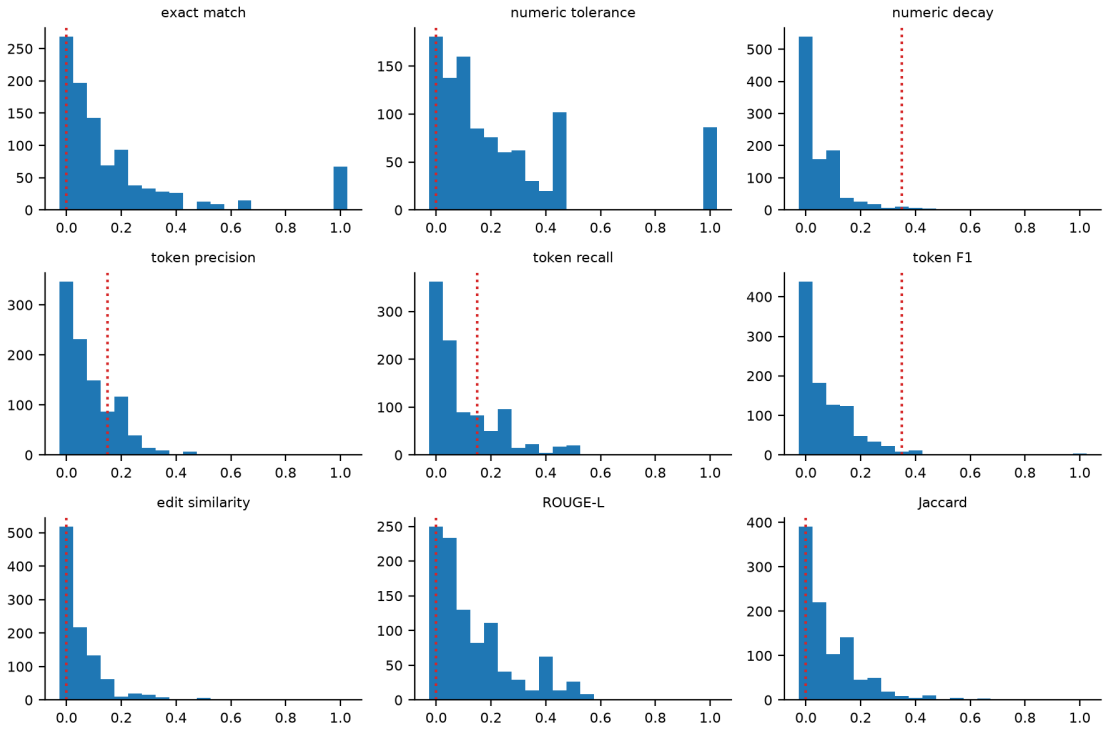


Figure 44: Weights selected as in-bag best across 1,000 bootstrap resamples; dotted = v1 rubric.

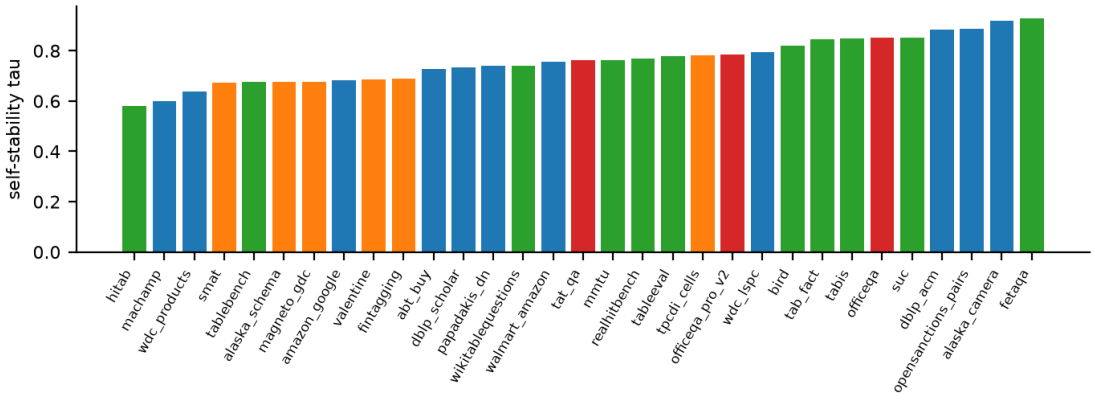


Figure 45: Self-stability of each dataset's model ranking (Kendall τ between bootstrap halves).



Figure 46: One-step sensitivity around the v1 rubric and the best candidate.

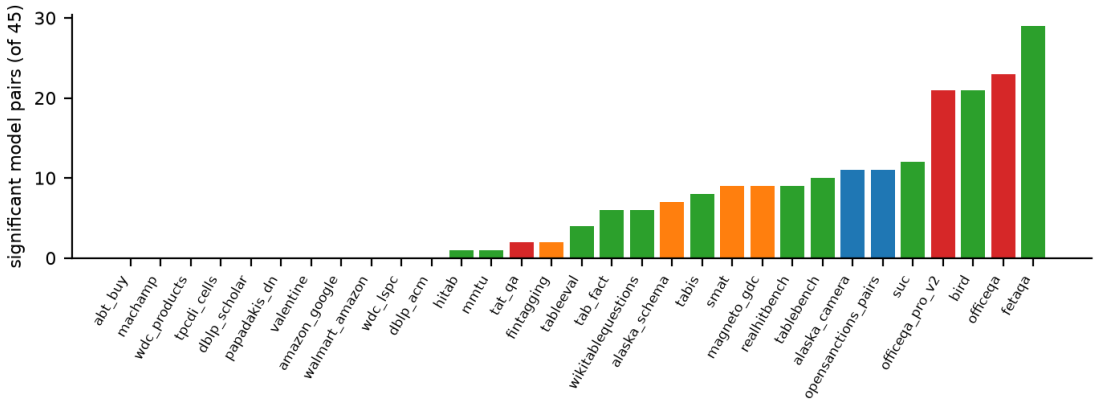


Figure 47: How many of the 45 model pairs each dataset separates at the item level under the headline rule (colour = family).

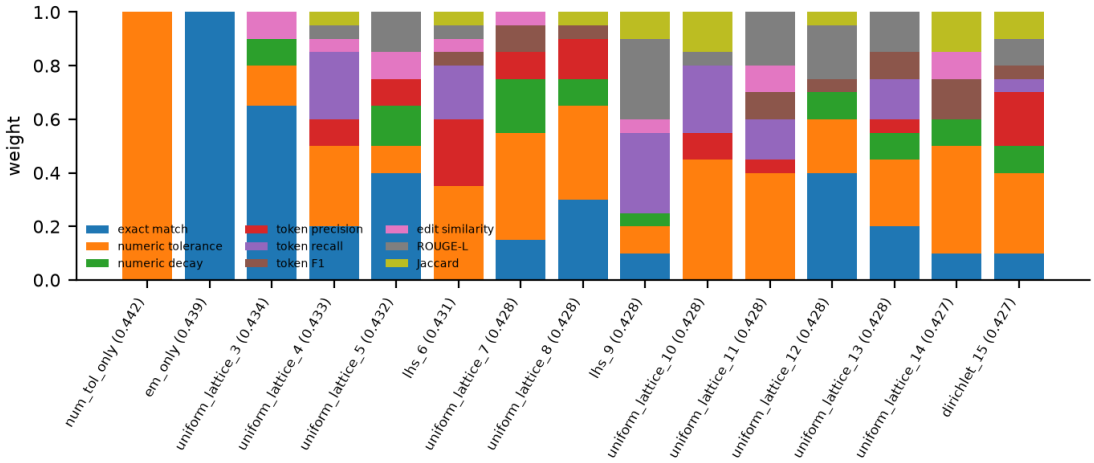


Figure 48: Component weights of the fifteen best candidates.

B All tables

Every saved table, one file each under `tables/`; include any in the main text with `\input`.

Table 8: Anchor operators: utility and mean score under three rules.

op	utility	v1	bestJ	bestrho	em	n
no_answer	0.000	0.001	0.000	0.014	0.000	3083
wrong_random	0.000	0.097	0.042	0.101	0.041	1845
flip	0.000	0.000	0.000	0.000	0.000	1238
sign_flip	0.050	0.018	0.003	0.119	0.003	632
magnitude	0.050	0.091	0.128	0.146	0.013	632
near_miss_large	0.150	0.150	0.000	0.137	0.000	632
digit_typo	0.200	0.199	0.241	0.224	0.000	632
list_dump	0.250	0.373	0.113	0.226	0.013	1845
hedge	0.300	0.493	0.405	0.392	0.192	2974
shuffled	0.400	0.652	0.009	0.418	0.009	109
near_miss_medium	0.450	0.293	0.008	0.252	0.008	632
partial	0.500	0.345	0.061	0.350	0.061	1213
superset	0.600	0.467	0.045	0.335	0.045	1213
typo	0.600	0.334	0.101	0.355	0.101	1104
near_miss_small	0.750	0.405	0.101	0.366	0.097	632
paraphrase	0.950	0.965	0.964	0.958	0.930	3083
verbose	1.000	1.000	0.999	1.000	0.999	3083
identity	1.000	1.000	0.999	1.000	0.999	3083
tagged	1.000	1.000	0.999	1.000	0.999	3083
unit_variant	1.000	0.898	1.000	0.891	0.837	632

Table 9: Anchor correctness for the labelled candidates.

label	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J	rho	rho_ds	auc	wrong_mean	hedge_gap
num_tol_only	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.442	0.796	0.808	0.989	0.022	-0.595
em_only	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.828	0.830	0.989	0.011	-0.807
uniform	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.412	0.878	0.861	0.994	0.046	-0.555
f1_only	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.411	0.856	0.846	0.989	0.030	-0.496
v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.404	0.878	0.868	0.994	0.034	-0.506

Table 10: The five fixed baselines on the candidate set.

rank	label	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J	J_kemeny	J_copeland	J_bt	tau_native
1	num_tol_only	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.442	0.442	0.444	0.426	0.974
2	em_only	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.441	0.441	0.419	0.922
189	uniform	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.412	0.418	0.427	0.395	0.916
217	f1_only	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.411	0.418	0.422	0.375	0.893
295	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.404	0.408	0.409	0.390	0.912

Table 11: Sampled designs: summary over 20 seeds.

design	best_J_mean	best_J_sd	best_J_max	mean_J	regret_mean	regret_max	min_pairwise_L1	mean_nearest_L1	discrepancy	zero_share
dirichlet_lattice	0.4320	0.0030	0.4391	0.4153	0.0100	0.0138	0.1950	0.4027	2.6738	0.8820
lhs_simplex	0.4313	0.0022	0.4373	0.4151	0.0107	0.0141	0.2150	0.4096	2.6718	0.8780
uniform_lattice	0.4343	0.0033	0.4420	0.4155	0.0077	0.0129	0.2350	0.4650	2.5904	0.9705

Table 12: Dataset diagnostics under the headline rule.

dataset	family	mean_score	spread_ across_models	floor	ceiling	share_ constant_ bootstrap_ rankings	ranking_self_ stability_tau	tau_to_ others_mean	n_items	exclude_ from_ objective
abt_buy	EM	0.970	0.060	no	no	0.000	0.726	0.356	100	no
alaska_camera	EM	0.958	0.180	no	no	0.000	0.919	-0.218	100	no
alaska_schema	SM	0.928	0.118	no	no	0.000	0.675	-0.044	93	no
amazon_google	EM	0.936	0.100	no	no	0.000	0.684	-0.012	100	no
bird	TQA	0.187	0.209	no	no	0.000	0.820	0.313	100	no
dblp_acm	EM	0.986	0.090	no	no	0.000	0.884	0.350	100	no
dblp_scholar	EM	0.949	0.060	no	no	0.000	0.733	-0.019	100	no
fetaqa	TQA	0.382	0.192	no	no	0.000	0.930	0.271	100	no
fintagging	SM	0.880	0.069	no	no	0.000	0.689	0.256	100	no
hitab	TQA	0.770	0.068	no	no	0.000	0.581	-0.181	100	no
machamp	EM	0.887	0.070	no	no	0.000	0.600	0.377	100	no
magneto_gdc	SM	0.934	0.218	no	no	0.000	0.677	0.252	100	no
mmtu	TQA	0.867	0.110	no	no	0.000	0.763	0.298	100	no
officeqa	FIN	0.129	0.119	no	no	0.000	0.853	0.276	100	no
officeqa_pro_v2	FIN	0.139	0.215	no	no	0.000	0.785	0.311	90	no
opensanctions_pairs	EM	0.821	0.210	no	no	0.000	0.888	0.030	100	no
papadakis_dn	EM	0.941	0.030	no	no	0.002	0.739	0.067	100	no
realhitbench	TQA	0.673	0.143	no	no	0.000	0.769	0.344	100	no
smat	SM	0.918	0.150	no	no	0.000	0.674	0.200	100	no
suc	TQA	0.914	0.246	no	no	0.000	0.853	0.293	100	no
tab_fact	TQA	0.798	0.200	no	no	0.000	0.845	0.007	100	no
tabis	TQA	0.915	0.210	no	no	0.000	0.849	0.224	100	no
tablebench	TQA	0.818	0.218	no	no	0.000	0.675	0.276	100	no
tableeval	TQA	0.749	0.166	no	no	0.000	0.778	0.195	100	no
tat_qa	FIN	0.803	0.109	no	no	0.000	0.762	0.262	100	no
tpcdi_cells	SM	0.456	0.020	no	no	0.000	0.782	0.298	100	no
valentine	SM	0.984	0.030	no	yes	0.000	0.685	0.229	100	yes
walmart_amazon	EM	0.958	0.060	no	no	0.000	0.755	0.360	100	no
wdc_lspc	EM	0.982	0.040	no	yes	0.004	0.796	0.246	100	yes
wdc_products	EM	0.978	0.020	no	yes	0.006	0.637	0.178	100	yes
wikitablequestions	TQA	0.846	0.132	no	no	0.000	0.739	0.350	100	no

Table 13: Mean v1-rubric score per model within each family.

model	entity matching	finance	schema matching	table QA
Gemini 3.8 Flash	0.945	0.414	0.854	0.765
Claude Opus 5	0.947	0.439	0.869	0.745
GPT-5.5	0.950	0.388	0.864	0.751
Claude Sonnet 5	0.942	0.391	0.862	0.724
Grok 4	0.948	0.354	0.851	0.738
GPT-5.4-mini	0.947	0.347	0.859	0.716
DeepSeek-chat	0.941	0.342	0.852	0.711
DeepSeek-R1	0.953	0.318	0.853	0.717
gpt-oss-120b	0.915	0.323	0.848	0.698
Llama 3.3 70B	0.936	0.328	0.777	0.633

Table 14: The 25 best candidates with the objective under the other pooling rules.

rank	label	w_em	w_num_ tol	w_num_ decay	w_tok_ prec	w_tok_ rec	w_tok_ f1	w_edit_ sim	w_ rouge_l	w_ jaccard	J	J_ kemeny	J_ copeland	J_bt	tau_ native
1	num_ tol_only	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.442	0.442	0.444	0.426	0.974
2	em_only	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.441	0.441	0.419	0.922
3	uniform_ lattice_3	0.650	0.150	0.100	0.000	0.000	0.000	0.100	0.000	0.000	0.434	0.434	0.434	0.398	0.907
4	uniform_ lattice_4	0.200	0.300	0.000	0.100	0.250	0.000	0.050	0.050	0.050	0.433	0.433	0.433	0.387	0.933
5	uniform_ lattice_5	0.400	0.100	0.150	0.100	0.000	0.000	0.100	0.150	0.000	0.432	0.431	0.440	0.408	0.916
6	lhs_6	0.000	0.350	0.000	0.250	0.200	0.050	0.050	0.050	0.050	0.431	0.435	0.435	0.401	0.927
7	uniform_ lattice_7	0.150	0.400	0.200	0.100	0.000	0.100	0.050	0.000	0.000	0.428	0.429	0.431	0.405	0.932
8	uniform_ lattice_8	0.300	0.350	0.100	0.150	0.000	0.050	0.000	0.000	0.050	0.428	0.432	0.431	0.399	0.930
9	lhs_9	0.100	0.100	0.050	0.000	0.300	0.000	0.050	0.300	0.100	0.428	0.431	0.431	0.398	0.907
10	uniform_ lattice_ 10	0.000	0.450	0.000	0.100	0.250	0.000	0.000	0.050	0.150	0.428	0.428	0.428	0.404	0.932
11	uniform_ lattice_ 11	0.000	0.400	0.000	0.050	0.150	0.100	0.100	0.200	0.000	0.428	0.432	0.432	0.392	0.933
12	uniform_ lattice_ 12	0.400	0.200	0.100	0.000	0.000	0.050	0.000	0.200	0.050	0.428	0.433	0.442	0.410	0.930
13	uniform_ lattice_ 13	0.200	0.250	0.100	0.050	0.150	0.100	0.000	0.150	0.000	0.428	0.429	0.427	0.401	0.935
14	uniform_ lattice_ 14	0.100	0.400	0.100	0.000	0.000	0.150	0.100	0.000	0.150	0.427	0.428	0.434	0.409	0.938
15	dirichlet_ 15	0.100	0.300	0.100	0.200	0.050	0.050	0.000	0.100	0.100	0.427	0.434	0.434	0.405	0.932
16	lhs_16	0.100	0.450	0.100	0.050	0.050	0.000	0.000	0.200	0.050	0.427	0.433	0.432	0.408	0.944
17	lhs_17	0.050	0.300	0.100	0.100	0.100	0.050	0.050	0.150	0.100	0.426	0.432	0.432	0.403	0.930
18	uniform_ lattice_ 18	0.100	0.400	0.100	0.150	0.100	0.050	0.100	0.000	0.000	0.426	0.426	0.427	0.405	0.935
19	dirichlet_ 19	0.050	0.300	0.050	0.000	0.100	0.100	0.050	0.050	0.300	0.425	0.428	0.428	0.393	0.929
20	lhs_20	0.250	0.150	0.350	0.000	0.000	0.150	0.000	0.050	0.050	0.425	0.427	0.427	0.410	0.926
21	dirichlet_ 21	0.200	0.250	0.000	0.100	0.100	0.000	0.050	0.100	0.200	0.425	0.432	0.437	0.391	0.929
22	lhs_22	0.100	0.100	0.000	0.100	0.400	0.050	0.050	0.050	0.150	0.425	0.426	0.426	0.384	0.912
23	dirichlet_ 23	0.350	0.050	0.050	0.000	0.150	0.000	0.000	0.400	0.000	0.425	0.430	0.437	0.401	0.920
24	lhs_24	0.150	0.050	0.050	0.050	0.500	0.000	0.000	0.050	0.150	0.425	0.425	0.425	0.404	0.909
25	lhs_25	0.050	0.100	0.050	0.000	0.450	0.150	0.000	0.100	0.100	0.425	0.425	0.429	0.404	0.908

Table 15: Full leaderboard under rule best-rho; weights: em 0.20, num_decay 0.25, tok_prec 0.25, tok_rec 0.05, tok_f1 0.05, edit_sim 0.15, rouge_l 0.05.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.125	0.016	1.000	1.000	2.000	0.968	3.000	2.000	3.012	0.017	3.000	3.000	3.000	3.000	2.000	0.806
Claude Opus 5	0.103	0.000	2.000	1.000	2.000	0.032	2.000	1.000	3.000	0.541	1.000	1.000	1.000	2.000	1.000	0.806
GPT-5.5	0.058	0.016	3.000	3.000	3.000	0.000	1.000	1.000	3.000	0.433	2.000	2.000	2.000	1.000	3.000	0.803
Claude Sonnet 5	0.031	0.016	4.000	4.000	5.000	0.000	4.000	4.000	6.025	0.000	4.000	4.000	4.000	4.000	4.000	0.790
Grok 4	0.018	0.016	5.000	4.000	6.000	0.000	5.000	4.000	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.792
GPT-5.4-mini	0.003	0.016	6.000	5.000	7.000	0.000	7.000	5.000	8.000	0.000	6.000	7.000	6.000	7.000	6.000	0.787
DeepSeek-chat	-0.024	0.016	7.000	6.000	7.000	0.000	8.000	5.000	8.000	0.000	7.000	7.000	8.000	8.000	7.000	0.778
DeepSeek-R1	-0.055	0.016	8.000	8.000	8.000	0.000	6.000	4.500	8.000	0.000	8.000	7.000	7.000	6.000	8.000	0.781
gpt-oss-120b	-0.117	0.016	9.000	9.000	10.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.762
Llama 3.3 70B	-0.142	0.016	10.000	9.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.733

Table 16: Full leaderboard under rule exact match; weights: em 1.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.047	0.000	1.000	1.000	3.000	0.764	1.000	1.000	2.000	0.812	1.000	1.000	1.000	1.000	1.000	0.768
GPT-5.5	0.039	0.016	2.000	1.000	3.000	0.152	2.000	1.000	3.000	0.177	2.000	2.000	2.000	2.000	2.000	0.764
Gemini 3.8 Flash	0.037	0.016	3.000	1.000	3.000	0.084	3.000	2.000	5.000	0.004	3.000	3.000	3.000	3.000	3.000	0.763
Grok 4	0.020	0.016	4.000	4.000	6.000	0.000	5.000	3.000	6.012	0.000	5.000	5.000	5.000	5.000	5.000	0.754
Claude Sonnet 5	0.013	0.016	5.000	4.000	7.000	0.000	4.000	3.000	6.000	0.000	4.000	4.000	4.000	4.000	4.000	0.751
GPT-5.4-mini	0.010	0.016	6.000	4.000	7.000	0.000	8.000	6.000	8.000	0.000	7.000	7.000	7.000	8.000	8.000	0.749
DeepSeek-R1	0.008	0.016	7.000	5.000	7.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	6.000	0.748
DeepSeek-chat	-0.014	0.016	8.000	8.000	9.000	0.000	7.000	6.000	8.000	0.000	8.000	8.000	8.000	7.000	7.000	0.737
gpt-oss-120b	-0.033	0.016	9.000	8.000	9.000	0.000	9.000	9.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.728
Llama 3.3 70B	-0.128	0.016	10.000	10.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.680

Table 17: Leaderboard under the headline rule (v1 rubric).

rank	model	BT strength	SE	Borda rank	Borda 95%	P(first) BT	mean score
1	Gemini 3.8 Flash	0.121	0.016	3	[1, 3]	0.990	0.812
2	Claude Opus 5	0.095	0.000	2	[1, 3]	0.010	0.811
3	GPT-5.5	0.052	0.016	1	[1, 3]	0.000	0.808
4	Claude Sonnet 5	0.029	0.016	4	[4, 6]	0.000	0.795
5	Grok 4	0.027	0.016	5	[4, 6]	0.000	0.797
6	GPT-5.4-mini	-0.010	0.016	8	[5, 8]	0.000	0.790
7	DeepSeek-chat	-0.018	0.016	7	[5, 8]	0.000	0.784
8	DeepSeek-R1	-0.025	0.016	6	[5, 8]	0.000	0.788
9	gpt-oss-120b	-0.100	0.016	9	[9, 10]	0.000	0.767
10	Llama 3.3 70B	-0.170	0.016	10	[9, 10]	0.000	0.738

Table 18: Full leaderboard under rule knee; weights: em 0.40, num_tol 0.10, num_decay 0.15, tok_prec 0.10, edit_sim 0.10, rouge_l 0.15.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.125	0.016	1.000	1.000	2.000	0.920	3.000	2.000	4.000	0.014	3.000	3.000	3.000	3.000	3.000	0.790
Claude Opus 5	0.109	0.000	2.000	1.000	2.000	0.080	1.500	1.000	3.000	0.597	1.000	1.000	1.000	1.000	1.000	0.792
GPT-5.5	0.059	0.016	3.000	3.000	3.000	0.000	1.500	1.000	3.000	0.381	2.000	2.000	2.000	2.000	2.000	0.788
Claude Sonnet 5	0.031	0.016	4.000	4.000	5.000	0.000	4.000	4.000	6.500	0.000	4.000	4.000	4.000	4.000	4.000	0.776
Grok 4	0.019	0.016	5.000	4.000	6.000	0.000	5.000	4.000	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.778
GPT-5.4-mini	-0.001	0.016	6.000	5.000	7.000	0.000	7.000	5.000	8.000	0.000	6.000	7.000	7.000	7.000	6.000	0.773
DeepSeek-chat	-0.022	0.016	7.000	6.000	7.000	0.000	8.000	5.000	8.000	0.000	7.000	7.000	8.000	8.000	8.000	0.763
DeepSeek-R1	-0.062	0.016	8.000	8.000	8.000	0.000	6.000	4.975	8.000	0.000	8.000	7.000	6.000	6.000	7.000	0.768
gpt-oss-120b	-0.111	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.750
Llama 3.3 70B	-0.147	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.713

Table 19: Full leaderboard under rule top1; weights: num_tol 1.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.046	0.000	1.000	1.000	3.000	0.538	1.000	1.000	2.000	0.613	1.000	1.000	1.000	1.000	1.000	0.778
GPT-5.5	0.043	0.016	2.000	1.000	3.000	0.296	2.000	1.000	3.000	0.369	2.000	2.000	2.000	2.000	2.000	0.776
Gemini 3.8 Flash	0.041	0.016	3.000	1.000	3.000	0.165	3.000	2.000	4.012	0.004	3.000	3.000	3.000	3.000	3.000	0.775
Grok 4	0.019	0.016	4.000	4.000	6.000	0.000	5.000	3.000	6.000	0.000	5.000	4.500	5.000	5.000	5.000	0.764
Claude Sonnet 5	0.014	0.016	5.000	4.000	7.000	0.000	4.000	3.000	6.000	0.000	4.000	4.500	4.000	4.000	4.000	0.762
GPT-5.4-mini	0.009	0.016	6.000	4.000	7.000	0.000	7.000	6.000	8.000	0.000	7.000	7.000	7.000	7.000	7.000	0.759
DeepSeek-R1	0.006	0.016	7.000	5.000	7.000	0.000	6.000	4.000	7.000	0.000	6.000	6.000	6.000	6.000	6.000	0.758
DeepSeek-chat	-0.016	0.016	8.000	8.000	9.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.747
gpt-oss-120b	-0.035	0.016	9.000	8.000	9.000	0.000	9.000	9.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.737
Llama 3.3 70B	-0.124	0.016	10.000	10.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.693

Table 20: Full leaderboard under rule top2; weights: em 1.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.047	0.000	1.000	1.000	3.000	0.764	1.000	1.000	2.000	0.812	1.000	1.000	1.000	1.000	1.000	0.768
GPT-5.5	0.039	0.016	2.000	1.000	3.000	0.152	2.000	1.000	3.000	0.177	2.000	2.000	2.000	2.000	2.000	0.764
Gemini 3.8 Flash	0.037	0.016	3.000	1.000	3.000	0.084	3.000	2.000	5.000	0.004	3.000	3.000	3.000	3.000	3.000	0.763
Grok 4	0.020	0.016	4.000	4.000	6.000	0.000	5.000	3.000	6.012	0.000	5.000	5.000	5.000	5.000	5.000	0.754
Claude Sonnet 5	0.013	0.016	5.000	4.000	7.000	0.000	4.000	3.000	6.000	0.000	4.000	4.000	4.000	4.000	4.000	0.751
GPT-5.4-mini	0.010	0.016	6.000	4.000	7.000	0.000	8.000	6.000	8.000	0.000	7.000	7.000	7.000	8.000	8.000	0.749
DeepSeek-R1	0.008	0.016	7.000	5.000	7.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	6.000	0.748
DeepSeek-chat	-0.014	0.016	8.000	8.000	9.000	0.000	7.000	6.000	8.000	0.000	8.000	8.000	8.000	7.000	7.000	0.737
gpt-oss-120b	-0.033	0.016	9.000	8.000	9.000	0.000	9.000	9.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.728
Llama 3.3 70B	-0.128	0.016	10.000	10.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.680

Table 21: Full leaderboard under rule top3; weights: em 0.65, num_tol 0.15, num_decay 0.10, edit_sim 0.10.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.124	0.016	1.000	1.000	2.000	0.837	3.000	2.000	4.000	0.006	3.000	3.000	3.000	3.000	3.000	0.777
Claude Opus 5	0.113	0.000	2.000	1.000	2.000	0.163	1.000	1.000	2.000	0.696	1.000	1.000	1.000	1.000	1.000	0.780
GPT-5.5	0.058	0.016	3.000	3.000	3.000	0.000	2.000	1.000	3.000	0.287	2.000	2.000	2.000	2.000	2.000	0.776
Claude Sonnet 5	0.026	0.016	4.000	4.000	5.000	0.000	4.000	3.488	7.000	0.000	4.000	4.000	4.000	4.000	4.000	0.763
Grok 4	0.023	0.016	5.000	4.000	5.000	0.000	5.000	3.000	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.766
GPT-5.4-mini	-0.010	0.016	6.000	6.000	7.000	0.000	7.000	5.000	8.000	0.000	7.000	7.000	7.000	7.000	6.000	0.761
DeepSeek-chat	-0.019	0.016	7.000	6.000	7.000	0.000	8.000	5.500	8.000	0.000	8.000	8.000	8.000	8.000	8.000	0.749
DeepSeek-R1	-0.062	0.016	8.000	8.000	9.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	7.000	0.757
gpt-oss-120b	-0.088	0.016	9.000	8.000	9.000	0.000	9.000	9.000	9.512	0.000	9.000	9.000	9.000	9.000	9.000	0.738
Llama 3.3 70B	-0.164	0.016	10.000	10.000	10.000	0.000	10.000	9.488	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.695

Table 22: Full leaderboard under rule top4; weights: em 0.20, num_tol 0.30, tok_prec 0.10, tok_rec 0.25, edit_sim 0.05, rouge_l 0.05, jaccard 0.05.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.016	1.000	1.000	2.000	0.797	3.000	2.000	4.000	0.015	3.000	3.000	3.000	3.000	3.000	0.793
Claude Opus 5	0.106	0.000	2.000	1.000	2.000	0.203	1.000	1.000	3.000	0.681	1.000	1.000	1.000	1.000	1.000	0.794
GPT-5.5	0.059	0.016	3.000	3.000	3.000	0.000	2.000	1.000	3.000	0.290	2.000	2.000	2.000	2.000	2.000	0.792
Grok 4	0.022	0.016	4.000	4.000	5.000	0.000	5.000	4.000	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.782
Claude Sonnet 5	0.020	0.016	5.000	4.000	5.000	0.000	4.000	4.000	6.000	0.000	4.000	4.000	4.000	4.000	4.000	0.778
DeepSeek-chat	-0.017	0.016	6.000	6.000	8.000	0.000	7.000	5.000	8.000	0.000	7.000	7.000	8.000	7.000	7.000	0.768
GPT-5.4-mini	-0.022	0.016	7.000	6.000	8.000	0.000	8.000	6.000	8.000	0.000	8.000	8.000	7.000	8.000	8.000	0.773
DeepSeek-R1	-0.036	0.016	8.000	6.975	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	6.000	6.000	6.000	0.776
gpt-oss-120b	-0.086	0.016	9.000	9.000	9.000	0.000	9.000	9.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.755
Llama 3.3 70B	-0.161	0.016	10.000	10.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.720

Table 23: Full leaderboard under rule top5; weights: em 0.40, num_tol 0.10, num_decay 0.15, tok_prec 0.10, edit_sim 0.10, rouge_l 0.15.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.125	0.016	1.000	1.000	2.000	0.920	3.000	2.000	4.000	0.014	3.000	3.000	3.000	3.000	3.000	0.790
Claude Opus 5	0.109	0.000	2.000	1.000	2.000	0.080	1.500	1.000	3.000	0.597	1.000	1.000	1.000	1.000	1.000	0.792
GPT-5.5	0.059	0.016	3.000	3.000	3.000	0.000	1.500	1.000	3.000	0.381	2.000	2.000	2.000	2.000	2.000	0.788
Claude Sonnet 5	0.031	0.016	4.000	4.000	5.000	0.000	4.000	4.000	6.500	0.000	4.000	4.000	4.000	4.000	4.000	0.776
Grok 4	0.019	0.016	5.000	4.000	6.000	0.000	5.000	4.000	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.778
GPT-5.4-mini	-0.001	0.016	6.000	5.000	7.000	0.000	7.000	5.000	8.000	0.000	6.000	7.000	7.000	7.000	6.000	0.773
DeepSeek-chat	-0.022	0.016	7.000	6.000	7.000	0.000	8.000	5.000	8.000	0.000	7.000	7.000	8.000	8.000	8.000	0.763
DeepSeek-R1	-0.062	0.016	8.000	8.000	8.000	0.000	6.000	4.975	8.000	0.000	8.000	7.000	6.000	6.000	7.000	0.768
gpt-oss-120b	-0.111	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.750
Llama 3.3 70B	-0.147	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.713

Table 24: Full leaderboard under rule v1 rubric; weights: num_decay 0.35, tok_prec 0.15, tok_rec 0.15, tok_f1 0.35.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.121	0.016	1.000	1.000	1.000	0.992	3.000	1.000	3.000	0.036	3.000	3.000	3.000	3.000	3.000	0.812
Claude Opus 5	0.095	0.000	2.000	2.000	2.000	0.008	2.000	1.000	3.000	0.476	1.000	1.000	2.000	1.000	1.000	0.811
GPT-5.5	0.052	0.016	3.000	3.000	4.000	0.000	1.000	1.000	3.000	0.479	2.000	2.000	1.000	2.000	2.000	0.808
Claude Sonnet 5	0.029	0.016	4.000	3.000	5.000	0.000	4.000	4.000	6.000	0.000	4.000	4.000	4.000	4.000	4.000	0.795
Grok 4	0.027	0.016	5.000	4.000	5.000	0.000	5.000	4.000	6.000	0.000	5.000	5.000	5.000	5.000	5.000	0.797
GPT-5.4-mini	-0.010	0.016	6.000	6.000	8.000	0.000	8.000	5.988	8.000	0.000	7.000	6.500	7.000	8.000	6.000	0.790
DeepSeek-chat	-0.018	0.016	7.000	6.000	8.000	0.000	7.000	5.000	8.000	0.000	8.000	8.000	8.000	7.000	8.000	0.784
DeepSeek-R1	-0.025	0.016	8.000	6.000	8.000	0.000	6.000	5.000	8.000	0.000	6.000	6.500	6.000	6.000	7.000	0.788
gpt-oss-120b	-0.100	0.016	9.000	9.000	9.000	0.000	9.000	9.000	10.000	0.000	9.000	9.000	9.000	9.000	9.000	0.767
Llama 3.3 70B	-0.170	0.016	10.000	10.000	10.000	0.000	10.000	9.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.738

Table 25: Leave-one-dataset-out, every dataset.

kind	held_out	weights_from	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J_rest	tau_heldout_mean	n_heldout
dataset	abt_buy	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.430	0.809	1
dataset	abt_buy	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.430	0.809	1
dataset	abt_buy	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.391	0.809	1
dataset	alaska_camera	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.471	-0.442	1
dataset	alaska_camera	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.471	-0.442	1
dataset	alaska_camera	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.436	-0.442	1
dataset	alaska_schema	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.453	0.024	1
dataset	alaska_schema	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.450	0.024	1
dataset	alaska_schema	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.424	-0.200	1
dataset	amazon_google	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.457	-0.024	1
dataset	amazon_google	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.457	-0.024	1
dataset	amazon_google	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.419	-0.024	1
dataset	bird	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.433	0.705	1
dataset	bird	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.433	0.705	1
dataset	bird	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.405	0.539	1
dataset	dblp_acm	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.433	0.581	1
dataset	dblp_acm	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.433	0.581	1
dataset	dblp_acm	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.397	0.637	1
dataset	dblp_scholar	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.457	0.000	1
dataset	dblp_scholar	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.457	0.000	1
dataset	dblp_scholar	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.421	0.000	1
dataset	fetaqa	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.509	1
dataset	fetaqa	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.509	1
dataset	fetaqa	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.407	0.422	1
dataset	fintagging	heldout_weights	0.200	0.300	0.000	0.100	0.250	0.000	0.050	0.050	0.050	0.431	0.511	1
dataset	fintagging	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.428	0.788	1
dataset	fintagging	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.402	0.467	1
dataset	hitab	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.458	-0.210	1
dataset	hitab	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.458	-0.210	1
dataset	hitab	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.433	-0.315	1
dataset	machamp	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.432	0.738	1
dataset	machamp	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.432	0.738	1
dataset	machamp	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.393	0.738	1
dataset	magneto_gdc	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.535	1
dataset	magneto_gdc	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.535	1
dataset	magneto_gdc	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.405	0.539	1
dataset	mmtu	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.556	1

dataset	mmtu	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.432	0.613	1
dataset	mmtu	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.401	0.556	1
dataset	officeqa	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.438	0.549	1
dataset	officeqa	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.432	0.614	1
dataset	officeqa	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.399	0.556	1
dataset	officeqa_pro_v2	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.436	0.629	1
dataset	officeqa_pro_v2	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.436	0.629	1
dataset	officeqa_pro_v2	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.399	0.556	1
dataset	opensanctions_pairs	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.451	0.090	1
dataset	opensanctions_pairs	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.446	0.045	1
dataset	opensanctions_pairs	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.416	0.045	1
dataset	papadakis_dn	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.452	0.026	1
dataset	papadakis_dn	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.449	0.026	1
dataset	papadakis_dn	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.417	0.078	1
dataset	realhitbench	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.431	0.764	1
dataset	realhitbench	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.431	0.764	1
dataset	realhitbench	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.397	0.600	1
dataset	smat	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.444	0.248	1
dataset	smat	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.441	0.248	1
dataset	smat	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.407	0.378	1
dataset	suc	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.690	1
dataset	suc	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.690	1
dataset	suc	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.398	0.584	1
dataset	tab_fact	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.455	-0.090	1
dataset	tab_fact	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.455	-0.090	1
dataset	tab_fact	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.423	-0.045	1
dataset	tabis	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.441	0.477	1
dataset	tabis	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.441	0.477	1
dataset	tabis	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.403	0.432	1
dataset	tablebench	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.477	1
dataset	tablebench	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.436	0.552	1
dataset	tablebench	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.403	0.511	1
dataset	tableeval	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.447	0.276	1
dataset	tableeval	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.447	0.276	1
dataset	tableeval	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.409	0.333	1
dataset	tat_qa	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.270	1
dataset	tat_qa	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.270	1
dataset	tat_qa	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.405	0.539	1
dataset	tpcdi_cells	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.438	0.545	1
dataset	tpcdi_cells	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.438	0.545	1
dataset	tpcdi_cells	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.403	0.535	1
dataset	valentine	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.597	1
dataset	valentine	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.597	1
dataset	valentine	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.402	0.477	1

dataset	walmart_amazon	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.431	0.768	1
dataset	walmart_amazon	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.431	0.768	1
dataset	walmart_amazon	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.393	0.722	1
dataset	wdc_lspc	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.441	0.484	1
dataset	wdc_lspc	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.441	0.484	1
dataset	wdc_lspc	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.405	0.435	1
dataset	wdc_products	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.444	0.369	1
dataset	wdc_products	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.444	0.369	1
dataset	wdc_products	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.405	0.369	1
dataset	wikitablequestions	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.428	0.782	1
dataset	wikitablequestions	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.426	0.920	1
dataset	wikitablequestions	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.392	0.778	1

Table 26: Leave-one-family-out.

kind	held_out	weights_from	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J_rest	tau_heldout_mean	n_heldout
family	EM	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.505	0.328	11
family	EM	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.505	0.328	11
family	EM	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.456	0.301	11
family	SM	heldout_weights	0.200	0.300	0.000	0.100	0.250	0.000	0.050	0.050	0.050	0.430	0.358	6
family	SM	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.427	0.425	6
family	SM	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.409	0.252	6
family	TQA	heldout_weights	0.100	0.300	0.100	0.200	0.050	0.050	0.000	0.100	0.100	0.427	0.403	11
family	TQA	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.416	0.448	11
family	TQA	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.403	0.379	11
family	FIN	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.429	0.458	3
family	FIN	full_best	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.429	0.458	3
family	FIN	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.381	0.511	3

Table 27: Every quantity saved by the nested item bootstrap.

quantity	value
B	1000
n_candidates	305
reference	v1_rubric
in_bag_J_of_selected_mean	0.387
out_of_bag_J_of_selected_mean	0.337
optimism_gap	0.050
out_of_bag_J_ref_mean	0.339
out_of_bag_J_full_best_mean	0.342
selected_vs_ref_out_of_bag	"mean_diff": -0.0019807294309139253, "ci95_lo": -0.03339549601078033, "ci95_hi": 0.023845080286264413, "p_boot_one_sided": 0.544, "share_a_ahead": 0.456
full_best_vs_ref_out_of_bag	"mean_diff": 0.0031523545384407044, "ci95_lo": -0.031030044704675675, "ci95_hi": 0.03641698732972145, "p_boot_one_sided": 0.425, "share_a_ahead": 0.575
full_best_vs_ref_in_bag	"mean_diff": 0.006695200175046921, "ci95_lo": -0.02151138260960579, "ci95_hi": 0.03264723569154739, "p_boot_one_sided": 0.31, "share_a_ahead": 0.69
share_resamples_ref_in_top_5pct	0.012
selected_weights_mean	"em": 0.181, "num_tol": 0.235, "num_decay": 0.056, "tok_prec": 0.084, "tok_rec": 0.099, "tok_f1": 0.076, "edit_sim": 0.054, "rouge_l": 0.131, "jaccard": 0.083
selected_weights_sd	"em": 0.258, "num_tol": 0.273, "num_decay": 0.092, "tok_prec": 0.091, "tok_rec": 0.122, "tok_f1": 0.103, "edit_sim": 0.086, "rouge_l": 0.139, "jaccard": 0.104
n_distinct_selected	204
plateau_size	305
plateau_share	1.000
ref_in_plateau	yes
ref_p_vs_best	0.310
practical_plateau_size	303
practical_margin	0.020
ref_in_practical_plateau	yes
practical_plateau_weight_ranges	"em": [0.0, 1.0], "num_tol": [0.0, 1.0], "num_decay": [0.0, 0.6], "tok_prec": [0.0, 0.55], "tok_rec": [0.0, 0.5], "tok_f1": [0.0, 1.0], "edit_sim": [0.0, 0.6], "rouge_l": [0.0, 0.55], "jaccard": [0.0, 0.65]

Table 28: Official-split estimates under the headline rule for every dataset and model.

dataset	model	suite_mean	official_ estimate	official_lo95	official_hi95	effective_n	suite_f1_pos	official_f1_ pos	official_f1_ lo95	official_f1_ hi95	official_ positive_ share
abt_buy	Claude Opus 5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	Claude Sonnet 5	0.970	0.993	0.985	1.000	63.075	0.968	0.968	0.925	1.000	0.108
abt_buy	DeepSeek-chat	0.960	0.976	0.933	0.998	63.075	0.958	0.893	0.735	0.990	0.108
abt_buy	DeepSeek-R1	0.970	0.978	0.931	1.000	63.075	0.969	0.904	0.755	1.000	0.108
abt_buy	Gemini 3.8 Flash	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.944	1.000	0.108
abt_buy	GPT-5.4-mini	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.750	0.990	0.108
abt_buy	GPT-5.5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	gpt-oss-120b	0.940	0.987	0.974	0.996	63.075	0.935	0.935	0.879	0.981	0.108
abt_buy	Grok 4	0.960	0.976	0.934	0.998	63.075	0.958	0.893	0.745	0.990	0.108
abt_buy	Llama 3.3 70B	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.735	0.992	0.108
alaska_camera	Claude Opus 5	0.820	0.820	0.740	0.890	100.000	0.795	0.795	0.687	0.881	0.500
alaska_camera	Claude Sonnet 5	0.980	0.980	0.950	1.000	100.000	0.981	0.981	0.951	1.000	0.500
alaska_camera	DeepSeek-chat	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	DeepSeek-R1	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.969	1.000	0.500
alaska_camera	Gemini 3.8 Flash	0.890	0.890	0.830	0.950	100.000	0.884	0.884	0.805	0.943	0.500
alaska_camera	GPT-5.4-mini	0.970	0.970	0.930	1.000	100.000	0.971	0.971	0.929	1.000	0.500
alaska_camera	GPT-5.5	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	gpt-oss-120b	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	Grok 4	0.940	0.940	0.890	0.980	100.000	0.940	0.940	0.886	0.982	0.500
alaska_camera	Llama 3.3 70B	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.967	1.000	0.500
alaska_schema	Claude Opus 5	0.945	0.945	0.900	0.981	93.000	–	–	–	–	–
alaska_schema	Claude Sonnet 5	0.929	0.929	0.878	0.973	93.000	–	–	–	–	–
alaska_schema	DeepSeek-chat	0.951	0.951	0.908	0.986	93.000	–	–	–	–	–
alaska_schema	DeepSeek-R1	0.948	0.948	0.907	0.982	93.000	–	–	–	–	–
alaska_schema	Gemini 3.8 Flash	0.918	0.918	0.866	0.965	93.000	–	–	–	–	–
alaska_schema	GPT-5.4-mini	0.958	0.958	0.916	0.988	93.000	–	–	–	–	–
alaska_schema	GPT-5.5	0.928	0.928	0.882	0.968	93.000	–	–	–	–	–
alaska_schema	gpt-oss-120b	0.934	0.934	0.888	0.976	93.000	–	–	–	–	–
alaska_schema	Grok 4	0.929	0.929	0.876	0.970	93.000	–	–	–	–	–
alaska_schema	Llama 3.3 70B	0.840	0.840	0.787	0.891	93.000	–	–	–	–	–
amazon_google	Claude Opus 5	0.940	0.916	0.841	0.981	67.152	0.936	0.704	0.526	0.907	0.102
amazon_google	Claude Sonnet 5	0.960	0.935	0.865	0.985	67.152	0.957	0.758	0.587	0.933	0.102
amazon_google	DeepSeek-chat	0.960	0.949	0.886	0.998	67.152	0.957	0.796	0.628	0.992	0.102
amazon_google	DeepSeek-R1	0.950	0.932	0.867	0.985	67.152	0.946	0.747	0.583	0.930	0.102
amazon_google	Gemini 3.8 Flash	0.930	0.928	0.863	0.979	67.152	0.923	0.725	0.561	0.926	0.102
amazon_google	GPT-5.4-mini	0.930	0.942	0.876	0.990	67.152	0.921	0.762	0.590	0.962	0.102

amazon_google	GPT-5.5	0.940	0.916	0.840	0.981	67.152	0.936	0.704	0.529	0.917	0.102
amazon_google	gpt-oss-120b	0.860	0.926	0.865	0.975	67.152	0.829	0.676	0.502	0.855	0.102
amazon_google	Grok 4	0.940	0.944	0.886	0.994	67.152	0.933	0.773	0.606	0.968	0.102
amazon_google	Llama 3.3 70B	0.950	0.932	0.867	0.985	67.152	0.946	0.747	0.583	0.932	0.102
bird	Claude Opus 5	0.293	0.323	0.234	0.415	74.082	–	–	–	–	–
bird	Claude Sonnet 5	0.232	0.258	0.176	0.347	74.082	–	–	–	–	–
bird	DeepSeek-chat	0.187	0.203	0.129	0.285	74.082	–	–	–	–	–
bird	DeepSeek-R1	0.178	0.214	0.136	0.297	74.082	–	–	–	–	–
bird	Gemini 3.8 Flash	0.293	0.313	0.224	0.408	74.082	–	–	–	–	–
bird	GPT-5.4-mini	0.178	0.218	0.134	0.310	74.082	–	–	–	–	–
bird	GPT-5.5	0.185	0.204	0.122	0.291	74.082	–	–	–	–	–
bird	gpt-oss-120b	0.084	0.109	0.047	0.186	74.082	–	–	–	–	–
bird	Grok 4	0.122	0.144	0.077	0.223	74.082	–	–	–	–	–
bird	Llama 3.3 70B	0.122	0.144	0.081	0.217	74.082	–	–	–	–	–
dblp_acm	Claude Opus 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Claude Sonnet 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-chat	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-R1	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.966	1.000	0.180
dblp_acm	Gemini 3.8 Flash	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	GPT-5.4-mini	0.990	0.996	0.987	1.000	72.166	0.990	0.990	0.964	1.000	0.180
dblp_acm	GPT-5.5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	gpt-oss-120b	0.910	0.967	0.943	0.987	72.166	0.899	0.899	0.819	0.958	0.180
dblp_acm	Grok 4	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Llama 3.3 70B	0.970	0.952	0.893	1.000	72.166	0.970	0.882	0.752	1.000	0.180
dblp_scholar	Claude Opus 5	0.950	0.956	0.906	0.993	70.463	0.950	0.888	0.777	0.983	0.186
dblp_scholar	Claude Sonnet 5	0.970	0.976	0.941	1.000	70.463	0.970	0.937	0.836	1.000	0.186
dblp_scholar	DeepSeek-chat	0.960	0.959	0.911	0.996	70.463	0.961	0.898	0.793	0.991	0.186
dblp_scholar	DeepSeek-R1	0.950	0.969	0.930	0.993	70.463	0.949	0.917	0.811	0.990	0.186
dblp_scholar	Gemini 3.8 Flash	0.930	0.961	0.920	0.990	70.463	0.928	0.895	0.795	0.972	0.186
dblp_scholar	GPT-5.4-mini	0.940	0.965	0.922	0.992	70.463	0.939	0.906	0.788	0.980	0.186
dblp_scholar	GPT-5.5	0.960	0.959	0.907	0.996	70.463	0.961	0.898	0.778	0.991	0.186
dblp_scholar	gpt-oss-120b	0.910	0.967	0.943	0.987	70.463	0.903	0.903	0.838	0.962	0.186
dblp_scholar	Grok 4	0.950	0.969	0.930	0.996	70.463	0.949	0.917	0.822	0.990	0.186
dblp_scholar	Llama 3.3 70B	0.970	0.976	0.938	1.000	70.463	0.970	0.937	0.851	1.000	0.186
fetaqa	Claude Opus 5	0.401	0.401	0.373	0.429	100.000	–	–	–	–	–
fetaqa	Claude Sonnet 5	0.373	0.373	0.345	0.400	100.000	–	–	–	–	–
fetaqa	DeepSeek-chat	0.434	0.434	0.407	0.462	100.000	–	–	–	–	–
fetaqa	DeepSeek-R1	0.413	0.413	0.383	0.443	100.000	–	–	–	–	–
fetaqa	Gemini 3.8 Flash	0.445	0.445	0.421	0.469	100.000	–	–	–	–	–
fetaqa	GPT-5.4-mini	0.312	0.312	0.289	0.336	100.000	–	–	–	–	–
fetaqa	GPT-5.5	0.438	0.438	0.412	0.464	100.000	–	–	–	–	–
fetaqa	gpt-oss-120b	0.322	0.322	0.304	0.342	100.000	–	–	–	–	–

fetaqa	Grok 4	0.428	0.428	0.403	0.453	100.000	–	–	–	–	–
fetaqa	Llama 3.3 70B	0.253	0.253	0.225	0.279	100.000	–	–	–	–	–
fintagging	Claude Opus 5	0.909	0.932	0.879	0.976	51.769	–	–	–	–	–
fintagging	Claude Sonnet 5	0.898	0.897	0.838	0.952	51.769	–	–	–	–	–
fintagging	DeepSeek-chat	0.858	0.847	0.779	0.917	51.769	–	–	–	–	–
fintagging	DeepSeek-R1	0.840	0.855	0.768	0.926	51.769	–	–	–	–	–
fintagging	Gemini 3.8 Flash	0.897	0.908	0.848	0.958	51.769	–	–	–	–	–
fintagging	GPT-5.4-mini	0.892	0.863	0.792	0.928	51.769	–	–	–	–	–
fintagging	GPT-5.5	0.893	0.899	0.830	0.957	51.769	–	–	–	–	–
fintagging	gpt-oss-120b	0.863	0.847	0.780	0.913	51.769	–	–	–	–	–
fintagging	Grok 4	0.887	0.876	0.805	0.939	51.769	–	–	–	–	–
fintagging	Llama 3.3 70B	0.858	0.836	0.770	0.902	51.769	–	–	–	–	–
hitab	Claude Opus 5	0.781	0.790	0.719	0.863	94.537	–	–	–	–	–
hitab	Claude Sonnet 5	0.763	0.758	0.678	0.839	94.537	–	–	–	–	–
hitab	DeepSeek-chat	0.814	0.822	0.754	0.888	94.537	–	–	–	–	–
hitab	DeepSeek-R1	0.771	0.771	0.698	0.849	94.537	–	–	–	–	–
hitab	Gemini 3.8 Flash	0.748	0.757	0.678	0.831	94.537	–	–	–	–	–
hitab	GPT-5.4-mini	0.784	0.783	0.707	0.862	94.537	–	–	–	–	–
hitab	GPT-5.5	0.757	0.764	0.684	0.838	94.537	–	–	–	–	–
hitab	gpt-oss-120b	0.750	0.752	0.674	0.834	94.537	–	–	–	–	–
hitab	Grok 4	0.746	0.749	0.662	0.828	94.537	–	–	–	–	–
hitab	Llama 3.3 70B	0.784	0.783	0.706	0.853	94.537	–	–	–	–	–
machamp	Claude Opus 5	0.900	0.912	0.833	0.970	32.491	0.904	0.774	0.623	0.920	0.175
machamp	Claude Sonnet 5	0.900	0.923	0.844	0.971	32.491	0.902	0.793	0.645	0.922	0.175
machamp	DeepSeek-chat	0.890	0.892	0.801	0.959	32.491	0.897	0.742	0.572	0.897	0.175
machamp	DeepSeek-R1	0.900	0.897	0.807	0.959	32.491	0.906	0.751	0.593	0.897	0.175
machamp	Gemini 3.8 Flash	0.900	0.912	0.827	0.967	32.491	0.904	0.774	0.614	0.914	0.175
machamp	GPT-5.4-mini	0.890	0.893	0.805	0.956	32.491	0.895	0.739	0.574	0.897	0.175
machamp	GPT-5.5	0.900	0.897	0.805	0.963	32.491	0.906	0.751	0.593	0.904	0.175
machamp	gpt-oss-120b	0.830	0.794	0.642	0.902	32.491	0.835	0.558	0.372	0.774	0.175
machamp	Grok 4	0.890	0.896	0.812	0.960	32.491	0.895	0.747	0.594	0.887	0.175
machamp	Llama 3.3 70B	0.870	0.880	0.783	0.951	32.491	0.874	0.698	0.541	0.852	0.175
magneto_gdc	Claude Opus 5	0.971	0.974	0.946	0.995	93.027	–	–	–	–	–
magneto_gdc	Claude Sonnet 5	0.955	0.955	0.916	0.986	93.027	–	–	–	–	–
magneto_gdc	DeepSeek-chat	0.953	0.956	0.920	0.986	93.027	–	–	–	–	–
magneto_gdc	DeepSeek-R1	0.970	0.972	0.941	0.994	93.027	–	–	–	–	–
magneto_gdc	Gemini 3.8 Flash	0.963	0.967	0.938	0.992	93.027	–	–	–	–	–
magneto_gdc	GPT-5.4-mini	0.912	0.914	0.862	0.961	93.027	–	–	–	–	–
magneto_gdc	GPT-5.5	0.967	0.972	0.940	0.994	93.027	–	–	–	–	–
magneto_gdc	gpt-oss-120b	0.961	0.965	0.928	0.993	93.027	–	–	–	–	–
magneto_gdc	Grok 4	0.939	0.944	0.905	0.978	93.027	–	–	–	–	–
magneto_gdc	Llama 3.3 70B	0.753	0.759	0.691	0.815	93.027	–	–	–	–	–

mmtu	Claude Opus 5	0.882	0.846	0.759	0.927	86.538	–	–	–	–	–
mmtu	Claude Sonnet 5	0.872	0.838	0.756	0.916	86.538	–	–	–	–	–
mmtu	DeepSeek-chat	0.865	0.831	0.748	0.911	86.538	–	–	–	–	–
mmtu	DeepSeek-R1	0.822	0.770	0.669	0.865	86.538	–	–	–	–	–
mmtu	Gemini 3.8 Flash	0.894	0.856	0.763	0.929	86.538	–	–	–	–	–
mmtu	GPT-5.4-mini	0.882	0.851	0.768	0.923	86.538	–	–	–	–	–
mmtu	GPT-5.5	0.917	0.882	0.800	0.957	86.538	–	–	–	–	–
mmtu	gpt-oss-120b	0.830	0.774	0.680	0.862	86.538	–	–	–	–	–
mmtu	Grok 4	0.901	0.869	0.786	0.941	86.538	–	–	–	–	–
mmtu	Llama 3.3 70B	0.807	0.773	0.684	0.854	86.538	–	–	–	–	–
officeqa	Claude Opus 5	0.187	0.187	0.146	0.235	100.000	–	–	–	–	–
officeqa	Claude Sonnet 5	0.170	0.170	0.128	0.217	100.000	–	–	–	–	–
officeqa	DeepSeek-chat	0.075	0.075	0.040	0.118	100.000	–	–	–	–	–
officeqa	DeepSeek-R1	0.083	0.083	0.053	0.119	100.000	–	–	–	–	–
officeqa	Gemini 3.8 Flash	0.194	0.194	0.160	0.235	100.000	–	–	–	–	–
officeqa	GPT-5.4-mini	0.107	0.107	0.082	0.137	100.000	–	–	–	–	–
officeqa	GPT-5.5	0.168	0.168	0.134	0.206	100.000	–	–	–	–	–
officeqa	gpt-oss-120b	0.079	0.079	0.052	0.114	100.000	–	–	–	–	–
officeqa	Grok 4	0.129	0.129	0.095	0.173	100.000	–	–	–	–	–
officeqa	Llama 3.3 70B	0.097	0.097	0.073	0.127	100.000	–	–	–	–	–
officeqa_pro_v2	Claude Opus 5	0.250	0.250	0.188	0.314	90.000	–	–	–	–	–
officeqa_pro_v2	Claude Sonnet 5	0.133	0.133	0.089	0.182	90.000	–	–	–	–	–
officeqa_pro_v2	DeepSeek-chat	0.121	0.121	0.090	0.161	90.000	–	–	–	–	–
officeqa_pro_v2	DeepSeek-R1	0.036	0.036	0.021	0.054	90.000	–	–	–	–	–
officeqa_pro_v2	Gemini 3.8 Flash	0.198	0.198	0.152	0.246	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.4-mini	0.123	0.123	0.094	0.155	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.5	0.188	0.188	0.127	0.254	90.000	–	–	–	–	–
officeqa_pro_v2	gpt-oss-120b	0.104	0.104	0.068	0.148	90.000	–	–	–	–	–
officeqa_pro_v2	Grok 4	0.119	0.119	0.083	0.164	90.000	–	–	–	–	–
officeqa_pro_v2	Llama 3.3 70B	0.115	0.115	0.082	0.152	90.000	–	–	–	–	–
opensanctions_pairs	Claude Opus 5	0.940	0.906	0.827	0.969	76.249	0.935	0.935	0.874	0.979	0.769
opensanctions_pairs	Claude Sonnet 5	0.730	0.587	0.478	0.704	76.249	0.630	0.636	0.496	0.761	0.769
opensanctions_pairs	DeepSeek-chat	0.760	0.635	0.521	0.753	76.249	0.684	0.691	0.556	0.792	0.769
opensanctions_pairs	DeepSeek-R1	0.850	0.765	0.655	0.860	76.249	0.819	0.819	0.716	0.905	0.769
opensanctions_pairs	Gemini 3.8 Flash	0.860	0.780	0.669	0.881	76.249	0.833	0.833	0.734	0.909	0.769
opensanctions_pairs	GPT-5.4-mini	0.900	0.865	0.782	0.942	76.249	0.891	0.905	0.837	0.963	0.769
opensanctions_pairs	GPT-5.5	0.750	0.608	0.497	0.724	76.249	0.658	0.658	0.520	0.776	0.769
opensanctions_pairs	gpt-oss-120b	0.780	0.666	0.561	0.770	76.249	0.718	0.725	0.594	0.831	0.769
opensanctions_pairs	Grok 4	0.850	0.776	0.677	0.868	76.249	0.824	0.830	0.733	0.914	0.769
opensanctions_pairs	Llama 3.3 70B	0.790	0.682	0.571	0.789	76.249	0.734	0.741	0.618	0.834	0.769
papadakis_dn	Claude Opus 5	0.920	0.955	0.892	0.996	32.601	0.789	0.322	0.130	0.840	0.018

papadakis_dn	Claude Sonnet 5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.569	0.947	0.018
papadakis_dn	DeepSeek-chat	0.940	0.993	0.985	0.998	32.601	0.833	0.757	0.520	0.929	0.018
papadakis_dn	DeepSeek-R1	0.940	0.993	0.985	0.998	32.601	0.833	0.757	0.533	0.927	0.018
papadakis_dn	Gemini 3.8 Flash	0.940	0.992	0.984	0.998	32.601	0.824	0.702	0.459	0.889	0.018
papadakis_dn	GPT-5.4-mini	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.535	0.936	0.018
papadakis_dn	GPT-5.5	0.950	0.994	0.986	0.999	32.601	0.857	0.786	0.573	0.942	0.018
papadakis_dn	gpt-oss-120b	0.930	0.992	0.983	0.997	32.601	0.788	0.682	0.425	0.881	0.018
papadakis_dn	Grok 4	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.547	0.939	0.018
papadakis_dn	Llama 3.3 70B	0.940	0.975	0.925	0.998	32.601	0.833	0.463	0.180	0.916	0.018
realhitbench	Claude Opus 5	0.743	0.745	0.668	0.818	98.080	–	–	–	–	–
realhitbench	Claude Sonnet 5	0.681	0.684	0.606	0.766	98.080	–	–	–	–	–
realhitbench	DeepSeek-chat	0.662	0.656	0.569	0.735	98.080	–	–	–	–	–
realhitbench	DeepSeek-R1	0.659	0.662	0.581	0.745	98.080	–	–	–	–	–
realhitbench	Gemini 3.8 Flash	0.746	0.747	0.672	0.818	98.080	–	–	–	–	–
realhitbench	GPT-5.4-mini	0.661	0.658	0.570	0.746	98.080	–	–	–	–	–
realhitbench	GPT-5.5	0.724	0.721	0.642	0.797	98.080	–	–	–	–	–
realhitbench	gpt-oss-120b	0.603	0.602	0.514	0.686	98.080	–	–	–	–	–
realhitbench	Grok 4	0.640	0.638	0.552	0.725	98.080	–	–	–	–	–
realhitbench	Llama 3.3 70B	0.611	0.616	0.538	0.694	98.080	–	–	–	–	–
smat	Claude Opus 5	0.933	0.931	0.883	0.968	98.100	–	–	–	–	–
smat	Claude Sonnet 5	0.938	0.936	0.893	0.973	98.100	–	–	–	–	–
smat	DeepSeek-chat	0.918	0.916	0.872	0.956	98.100	–	–	–	–	–
smat	DeepSeek-R1	0.913	0.909	0.860	0.952	98.100	–	–	–	–	–
smat	Gemini 3.8 Flash	0.921	0.919	0.872	0.961	98.100	–	–	–	–	–
smat	GPT-5.4-mini	0.956	0.958	0.923	0.987	98.100	–	–	–	–	–
smat	GPT-5.5	0.940	0.939	0.894	0.973	98.100	–	–	–	–	–
smat	gpt-oss-120b	0.916	0.913	0.862	0.957	98.100	–	–	–	–	–
smat	Grok 4	0.936	0.933	0.889	0.971	98.100	–	–	–	–	–
smat	Llama 3.3 70B	0.806	0.803	0.743	0.860	98.100	–	–	–	–	–
suc	Claude Opus 5	0.986	0.988	0.963	1.000	98.920	–	–	–	–	–
suc	Claude Sonnet 5	0.911	0.908	0.851	0.955	98.920	–	–	–	–	–
suc	DeepSeek-chat	0.858	0.858	0.796	0.914	98.920	–	–	–	–	–
suc	DeepSeek-R1	0.935	0.932	0.887	0.970	98.920	–	–	–	–	–
suc	Gemini 3.8 Flash	0.976	0.978	0.945	1.000	98.920	–	–	–	–	–
suc	GPT-5.4-mini	0.914	0.910	0.856	0.956	98.920	–	–	–	–	–
suc	GPT-5.5	0.943	0.938	0.895	0.976	98.920	–	–	–	–	–
suc	gpt-oss-120b	0.932	0.930	0.881	0.970	98.920	–	–	–	–	–
suc	Grok 4	0.943	0.938	0.894	0.977	98.920	–	–	–	–	–
suc	Llama 3.3 70B	0.740	0.738	0.662	0.808	98.920	–	–	–	–	–
tab_fact	Claude Opus 5	0.660	0.667	0.571	0.760	99.360	0.673	0.696	0.589	0.785	0.500
tab_fact	Claude Sonnet 5	0.740	0.733	0.646	0.818	99.360	0.698	0.710	0.585	0.809	0.500

tab_fact	DeepSeek-chat	0.840	0.826	0.749	0.902	99.360	0.789	0.789	0.676	0.874	0.500
tab_fact	DeepSeek-R1	0.760	0.754	0.669	0.842	99.360	0.721	0.732	0.625	0.825	0.500
tab_fact	Gemini 3.8 Flash	0.860	0.848	0.781	0.924	99.360	0.821	0.821	0.709	0.909	0.500
tab_fact	GPT-5.4-mini	0.820	0.808	0.723	0.883	99.360	0.769	0.772	0.652	0.867	0.500
tab_fact	GPT-5.5	0.840	0.829	0.745	0.902	99.360	0.800	0.803	0.700	0.892	0.500
tab_fact	gpt-oss-120b	0.830	0.815	0.731	0.891	99.360	0.773	0.773	0.656	0.880	0.500
tab_fact	Grok 4	0.850	0.839	0.756	0.912	99.360	0.810	0.812	0.700	0.897	0.500
tab_fact	Llama 3.3 70B	0.780	0.769	0.690	0.849	99.360	0.725	0.732	0.605	0.835	0.500
tabis	Claude Opus 5	0.960	0.964	0.915	0.994	77.069	–	–	–	–	–
tabis	Claude Sonnet 5	0.920	0.917	0.853	0.970	77.069	–	–	–	–	–
tabis	DeepSeek-chat	0.770	0.710	0.609	0.817	77.069	–	–	–	–	–
tabis	DeepSeek-R1	0.970	0.982	0.958	1.000	77.069	–	–	–	–	–
tabis	Gemini 3.8 Flash	0.950	0.970	0.938	0.994	77.069	–	–	–	–	–
tabis	GPT-5.4-mini	0.900	0.905	0.831	0.960	77.069	–	–	–	–	–
tabis	GPT-5.5	0.960	0.976	0.947	0.994	77.069	–	–	–	–	–
tabis	gpt-oss-120b	0.920	0.951	0.914	0.982	77.069	–	–	–	–	–
tabis	Grok 4	0.980	0.988	0.970	1.000	77.069	–	–	–	–	–
tabis	Llama 3.3 70B	0.820	0.821	0.735	0.908	77.069	–	–	–	–	–
tablebench	Claude Opus 5	0.846	0.846	0.776	0.911	89.187	–	–	–	–	–
tablebench	Claude Sonnet 5	0.877	0.875	0.811	0.928	89.187	–	–	–	–	–
tablebench	DeepSeek-chat	0.833	0.844	0.776	0.905	89.187	–	–	–	–	–
tablebench	DeepSeek-R1	0.783	0.793	0.720	0.863	89.187	–	–	–	–	–
tablebench	Gemini 3.8 Flash	0.835	0.833	0.767	0.902	89.187	–	–	–	–	–
tablebench	GPT-5.4-mini	0.841	0.835	0.765	0.902	89.187	–	–	–	–	–
tablebench	GPT-5.5	0.852	0.850	0.778	0.912	89.187	–	–	–	–	–
tablebench	gpt-oss-120b	0.813	0.808	0.736	0.876	89.187	–	–	–	–	–
tablebench	Grok 4	0.839	0.837	0.764	0.908	89.187	–	–	–	–	–
tablebench	Llama 3.3 70B	0.660	0.641	0.558	0.720	89.187	–	–	–	–	–
tableeval	Claude Opus 5	0.765	0.784	0.684	0.873	69.689	–	–	–	–	–
tableeval	Claude Sonnet 5	0.733	0.764	0.668	0.851	69.689	–	–	–	–	–
tableeval	DeepSeek-chat	0.711	0.708	0.608	0.802	69.689	–	–	–	–	–
tableeval	DeepSeek-R1	0.756	0.757	0.662	0.852	69.689	–	–	–	–	–
tableeval	Gemini 3.8 Flash	0.791	0.805	0.714	0.889	69.689	–	–	–	–	–
tableeval	GPT-5.4-mini	0.742	0.751	0.648	0.838	69.689	–	–	–	–	–
tableeval	GPT-5.5	0.784	0.794	0.690	0.882	69.689	–	–	–	–	–
tableeval	gpt-oss-120b	0.773	0.774	0.683	0.864	69.689	–	–	–	–	–
tableeval	Grok 4	0.802	0.785	0.677	0.881	69.689	–	–	–	–	–
tableeval	Llama 3.3 70B	0.636	0.618	0.514	0.718	69.689	–	–	–	–	–
tat_qa	Claude Opus 5	0.861	0.846	0.773	0.915	60.725	–	–	–	–	–
tat_qa	Claude Sonnet 5	0.845	0.845	0.774	0.909	60.725	–	–	–	–	–
tat_qa	DeepSeek-chat	0.806	0.774	0.684	0.859	60.725	–	–	–	–	–

tat_qa	DeepSeek-R1	0.806	0.763	0.673	0.852	60.725	–	–	–	–	–
tat_qa	Gemini 3.8 Flash	0.829	0.815	0.735	0.889	60.725	–	–	–	–	–
tat_qa	GPT-5.4-mini	0.789	0.719	0.627	0.811	60.725	–	–	–	–	–
tat_qa	GPT-5.5	0.789	0.748	0.669	0.827	60.725	–	–	–	–	–
tat_qa	gpt-oss-120b	0.765	0.684	0.588	0.775	60.725	–	–	–	–	–
tat_qa	Grok 4	0.791	0.710	0.612	0.804	60.725	–	–	–	–	–
tat_qa	Llama 3.3 70B	0.752	0.679	0.573	0.780	60.725	–	–	–	–	–
tpcdi_cells	Claude Opus 5	0.462	0.465	0.377	0.557	95.091	–	–	–	–	–
tpcdi_cells	Claude Sonnet 5	0.462	0.465	0.378	0.563	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-chat	0.462	0.465	0.377	0.558	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-R1	0.462	0.466	0.377	0.557	95.091	–	–	–	–	–
tpcdi_cells	Gemini 3.8 Flash	0.462	0.465	0.376	0.555	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.4-mini	0.455	0.458	0.362	0.537	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.5	0.462	0.466	0.370	0.552	95.091	–	–	–	–	–
tpcdi_cells	gpt-oss-120b	0.442	0.453	0.360	0.550	95.091	–	–	–	–	–
tpcdi_cells	Grok 4	0.447	0.456	0.370	0.545	95.091	–	–	–	–	–
tpcdi_cells	Llama 3.3 70B	0.442	0.453	0.368	0.542	95.091	–	–	–	–	–
valentine	Claude Opus 5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Claude Sonnet 5	0.994	0.989	0.960	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-chat	0.979	0.967	0.928	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-R1	0.990	0.986	0.956	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.8 Flash	0.970	0.988	0.970	1.000	81.736	–	–	–	–	–
valentine	GPT-5.4-mini	0.985	0.978	0.943	1.000	81.736	–	–	–	–	–
valentine	GPT-5.5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	gpt-oss-120b	0.979	0.971	0.928	1.000	81.736	–	–	–	–	–
valentine	Grok 4	0.973	0.957	0.912	0.992	81.736	–	–	–	–	–
valentine	Llama 3.3 70B	0.970	0.958	0.913	0.991	81.736	–	–	–	–	–
walmart_amazon	Claude Opus 5	0.980	0.982	0.948	1.000	70.807	0.976	0.912	0.767	1.000	0.094
walmart_amazon	Claude Sonnet 5	0.950	0.962	0.914	0.997	70.807	0.938	0.823	0.645	0.980	0.094
walmart_amazon	DeepSeek-chat	0.950	0.962	0.912	0.998	70.807	0.938	0.823	0.657	0.977	0.094
walmart_amazon	DeepSeek-R1	0.960	0.978	0.939	0.998	70.807	0.950	0.887	0.737	0.989	0.094
walmart_amazon	Gemini 3.8 Flash	0.990	0.998	0.993	1.000	70.807	0.988	0.988	0.959	1.000	0.094
walmart_amazon	GPT-5.4-mini	0.950	0.975	0.938	0.998	70.807	0.937	0.874	0.722	0.987	0.094
walmart_amazon	GPT-5.5	0.970	0.967	0.917	1.000	70.807	0.964	0.848	0.671	1.000	0.094
walmart_amazon	gpt-oss-120b	0.940	0.973	0.933	0.995	70.807	0.923	0.860	0.698	0.974	0.094
walmart_amazon	Grok 4	0.960	0.978	0.942	0.998	70.807	0.950	0.887	0.720	0.989	0.094
walmart_amazon	Llama 3.3 70B	0.930	0.971	0.932	0.994	70.807	0.909	0.847	0.681	0.965	0.094
wdc_lspc	Claude Opus 5	0.990	0.996	0.987	1.000	75.082	0.991	0.992	0.973	1.000	0.264
wdc_lspc	Claude Sonnet 5	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.958	1.000	0.264
wdc_lspc	DeepSeek-chat	0.960	0.940	0.868	0.994	75.082	0.964	0.895	0.777	0.990	0.264
wdc_lspc	DeepSeek-R1	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Gemini 3.8 Flash	0.980	0.992	0.979	1.000	75.082	0.981	0.984	0.956	1.000	0.264

wdc_lspc	GPT-5.4-mini	0.970	0.986	0.967	1.000	75.082	0.971	0.973	0.937	1.000	0.264
wdc_lspc	GPT-5.5	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	gpt-oss-120b	0.980	0.990	0.974	1.000	75.082	0.981	0.981	0.951	1.000	0.264
wdc_lspc	Grok 4	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Llama 3.3 70B	0.960	0.940	0.872	0.994	75.082	0.964	0.895	0.781	0.990	0.264
wdc_products	Claude Opus 5	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Claude Sonnet 5	0.970	0.986	0.967	1.000	77.046	0.971	0.974	0.936	1.000	0.280
wdc_products	DeepSeek-chat	0.970	0.977	0.944	1.000	77.046	0.972	0.958	0.895	1.000	0.280
wdc_products	DeepSeek-R1	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.8 Flash	0.990	0.995	0.983	1.000	77.046	0.991	0.991	0.968	1.000	0.280
wdc_products	GPT-5.4-mini	0.970	0.986	0.966	1.000	77.046	0.971	0.974	0.940	1.000	0.280
wdc_products	GPT-5.5	0.980	0.979	0.940	1.000	77.046	0.981	0.964	0.902	1.000	0.280
wdc_products	gpt-oss-120b	0.980	0.979	0.945	1.000	77.046	0.981	0.964	0.898	1.000	0.280
wdc_products	Grok 4	0.990	0.995	0.983	1.000	77.046	0.991	0.991	0.969	1.000	0.280
wdc_products	Llama 3.3 70B	0.970	0.977	0.940	1.000	77.046	0.972	0.958	0.901	1.000	0.280
wikitablequestions	Claude Opus 5	0.879	0.878	0.815	0.933	97.686	–	–	–	–	–
wikitablequestions	Claude Sonnet 5	0.863	0.863	0.795	0.922	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-chat	0.843	0.842	0.767	0.906	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-R1	0.846	0.844	0.778	0.908	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.8 Flash	0.881	0.881	0.820	0.934	97.686	–	–	–	–	–
wikitablequestions	GPT-5.4-mini	0.842	0.839	0.770	0.901	97.686	–	–	–	–	–
wikitablequestions	GPT-5.5	0.865	0.862	0.799	0.922	97.686	–	–	–	–	–
wikitablequestions	gpt-oss-120b	0.824	0.821	0.753	0.887	97.686	–	–	–	–	–
wikitablequestions	Grok 4	0.867	0.864	0.803	0.923	97.686	–	–	–	–	–
wikitablequestions	Llama 3.3 70B	0.749	0.747	0.672	0.823	97.686	–	–	–	–	–

Table 29: Official-split estimates under the headline rule averaged over the ten models.

dataset	suite_mean	official_ estimate	suite_f1_ pos	official_precision_ pos	official_recall_ pos	official_f1_ pos	official_positive_ share
abt_buy	0.970	0.986	0.968	0.926	0.949	0.936	0.108
alaska_camera	0.958	0.958	0.955	1.000	0.921	0.955	0.500
alaska_schema	0.928	0.928	–	–	–	–	–
amazon_google	0.936	0.932	0.929	0.611	0.942	0.739	0.102
bird	0.187	0.213	–	–	–	–	–
dblp_acm	0.986	0.991	0.985	0.979	0.978	0.976	0.180
dblp_scholar	0.949	0.966	0.948	0.900	0.924	0.910	0.186
fetaqa	0.382	0.382	–	–	–	–	–
fintagging	0.880	0.876	–	–	–	–	–
hitab	0.770	0.773	–	–	–	–	–
machamp	0.887	0.890	0.892	0.644	0.853	0.733	0.175
magneto_gdc	0.934	0.938	–	–	–	–	–
mmtu	0.867	0.829	–	–	–	–	–
officeqa	0.129	0.129	–	–	–	–	–
officeqa_pro_v2	0.139	0.139	–	–	–	–	–
opensanctions_ pairs	0.821	0.727	0.773	0.993	0.649	0.777	0.769
papadakis_dn	0.941	0.988	0.833	0.550	0.963	0.683	0.018
realhitbench	0.673	0.673	–	–	–	–	–
smat	0.918	0.916	–	–	–	–	–
suc	0.914	0.912	–	–	–	–	–
tab_fact	0.798	0.789	0.758	0.896	0.674	0.764	0.500
tabis	0.915	0.918	–	–	–	–	–
tablebench	0.818	0.816	–	–	–	–	–
tableeval	0.749	0.754	–	–	–	–	–
tat_qa	0.803	0.758	–	–	–	–	–
tpcdi_cells	0.456	0.461	–	–	–	–	–
valentine	0.984	0.979	–	–	–	–	–
walmart_amazon	0.958	0.975	0.947	0.832	0.927	0.875	0.094
wdc_lspc	0.982	0.983	0.983	0.965	0.979	0.971	0.264
wdc_products	0.978	0.985	0.979	0.980	0.968	0.974	0.280
wikitablequestions	0.846	0.844	–	–	–	–	–

Table 30: Cross-dataset pairwise tests under rule best-rho: Friedman $\chi^2 = 78.3$ ($p = 3.6\text{e-}13$), Iman–Davenport $F = 11.7$, Nemenyi CD = 2.43, Kendall's $W = 0.272$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2581	22	7	2	0.0261	0.5485	no	no
Claude Opus 5	DeepSeek-chat	-2.5000	22	8	1	0.0143	0.3572	yes	no
Claude Opus 5	DeepSeek-R1	-2.0161	19	9	3	0.0202	0.4440	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5645	18	11	2	0.2242	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.3387	25	6	0	0.0051	0.1468	no	no
Claude Opus 5	GPT-5.5	0.0968	13	12	6	0.5497	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0806	26	4	1	0.0004	0.0130	yes	yes
Claude Opus 5	Grok 4	-1.6129	18	10	3	0.0745	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.5968	26	5	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2419	20	7	4	0.0424	0.7345	no	no
Claude Sonnet 5	DeepSeek-R1	-0.7581	15	14	2	0.4443	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.6935	11	17	3	0.1102	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.0806	20	8	3	0.0598	0.9566	no	no
Claude Sonnet 5	GPT-5.5	1.3548	9	19	3	0.0334	0.6676	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8226	22	8	1	0.0044	0.1306	yes	no
Claude Sonnet 5	Grok 4	-0.3548	19	10	2	0.5369	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.3387	25	4	2	0.0001	0.0041	yes	yes
DeepSeek-chat	DeepSeek-R1	0.4839	17	13	1	0.7609	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.9355	6	22	3	0.0061	0.1697	no	no
DeepSeek-chat	GPT-5.4-mini	0.1613	11	16	4	0.3777	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.5968	5	22	4	0.0024	0.0736	yes	no
DeepSeek-chat	gpt-oss-120b	-1.5806	21	9	1	0.0387	0.7345	no	no
DeepSeek-chat	Grok 4	0.8871	12	16	3	0.2898	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0968	21	6	4	0.0006	0.0219	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.4516	10	19	2	0.0396	0.7345	no	no
DeepSeek-R1	GPT-5.4-mini	-0.3226	17	13	1	0.7388	1.0000	no	no
DeepSeek-R1	GPT-5.5	2.1129	7	20	4	0.0083	0.2248	no	no
DeepSeek-R1	gpt-oss-120b	-2.0645	21	9	1	0.0159	0.3823	no	no
DeepSeek-R1	Grok 4	0.4032	12	15	4	0.4156	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.5806	22	6	3	0.0006	0.0220	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7742	21	9	1	0.0177	0.4069	no	no
Gemini 3.8 Flash	GPT-5.5	0.6613	13	16	2	0.9297	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.5161	26	4	1	0.0001	0.0028	yes	yes
Gemini 3.8 Flash	Grok 4	-1.0484	17	12	2	0.0863	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.0323	26	4	1	0.0001	0.0032	yes	yes
GPT-5.4-mini	GPT-5.5	2.4355	5	25	1	0.0022	0.0691	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.7419	23	8	0	0.0018	0.0584	no	no
GPT-5.4-mini	Grok 4	0.7258	12	16	3	0.4927	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2581	26	3	2	0.0000	0.0021	no	yes
GPT-5.5	gpt-oss-120b	-4.1774	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7097	20	7	4	0.0108	0.2806	no	no
GPT-5.5	Llama 3.3 70B	-4.6935	27	4	0	0.0000	0.0006	yes	yes
gpt-oss-120b	Grok 4	2.4677	5	26	0	0.0001	0.0037	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5161	19	12	0	0.2475	1.0000	no	no
Grok 4	Llama 3.3 70B	-2.9839	25	5	1	0.0001	0.0046	yes	yes

Table 31: Cross-dataset pairwise tests under rule exact match: Friedman $\chi^2 = 82.5$ ($p = 5.3\text{e-}14$), Iman–Davenport $F = 12.6$, Nemenyi CD = 2.43, Kendall's $W = 0.279$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3387	21	7	3	0.0266	0.5590	no	no
Claude Opus 5	DeepSeek-chat	-2.6290	22	6	3	0.0049	0.1564	yes	no
Claude Opus 5	DeepSeek-R1	-1.9032	18	6	7	0.0116	0.3133	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0161	19	8	4	0.0504	0.9126	no	no
Claude Opus 5	GPT-5.4-mini	-2.8065	25	5	1	0.0057	0.1703	yes	no
Claude Opus 5	GPT-5.5	-0.3065	13	9	9	0.3064	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.1935	26	3	2	0.0004	0.0127	yes	yes
Claude Opus 5	Grok 4	-1.7903	18	9	4	0.0559	0.9126	no	no
Claude Opus 5	Llama 3.3 70B	-4.9839	26	5	0	0.0001	0.0049	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2903	19	5	7	0.0161	0.4034	no	no
Claude Sonnet 5	DeepSeek-R1	-0.5645	14	13	4	0.6305	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.3226	11	14	6	0.3217	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4677	19	7	5	0.1052	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.0323	7	15	9	0.0397	0.7936	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8548	22	7	2	0.0094	0.2627	yes	no
Claude Sonnet 5	Grok 4	-0.4516	18	9	4	0.4924	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6452	25	4	2	0.0001	0.0038	yes	yes
DeepSeek-chat	DeepSeek-R1	0.7258	11	13	7	0.6016	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.6129	6	18	7	0.0166	0.4034	no	no
DeepSeek-chat	GPT-5.4-mini	-0.1774	12	14	5	0.4859	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3226	6	19	6	0.0052	0.1604	no	no
DeepSeek-chat	gpt-oss-120b	-1.5645	19	9	3	0.0827	1.0000	no	no
DeepSeek-chat	Grok 4	0.8387	12	15	4	0.2936	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.3548	22	3	6	0.0001	0.0053	no	yes
DeepSeek-R1	Gemini 3.8 Flash	0.8871	10	17	4	0.0897	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-0.9032	17	12	2	0.9373	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.5968	7	19	5	0.0203	0.4672	no	no
DeepSeek-R1	gpt-oss-120b	-2.2903	21	5	5	0.0038	0.1258	no	no
DeepSeek-R1	Grok 4	0.1129	13	14	4	0.7011	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.0806	25	3	3	0.0000	0.0021	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7903	21	9	1	0.0498	0.9126	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	10	15	6	0.5755	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.1774	24	2	5	0.0001	0.0033	yes	yes
Gemini 3.8 Flash	Grok 4	-0.7742	15	10	6	0.2506	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.9677	26	4	1	0.0001	0.0038	yes	yes
GPT-5.4-mini	GPT-5.5	2.5000	5	24	2	0.0064	0.1852	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.3871	20	10	1	0.0134	0.3494	no	no
GPT-5.4-mini	Grok 4	1.0161	9	16	6	0.3613	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1774	23	4	4	0.0001	0.0044	no	yes
GPT-5.5	gpt-oss-120b	-3.8871	26	1	4	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.4839	16	6	9	0.0480	0.9126	no	no
GPT-5.5	Llama 3.3 70B	-4.6774	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.4032	3	24	4	0.0002	0.0054	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.7903	20	8	3	0.0217	0.4779	no	no
Grok 4	Llama 3.3 70B	-3.1935	24	4	3	0.0001	0.0038	yes	yes

Table 32: Cross-dataset pairwise tests under rule knee: Friedman $\chi^2 = 80.9$ ($p = 1.1\text{e-}13$), Iman–Davenport $F = 12.3$, Nemenyi CD = 2.43, Kendall's $W = 0.282$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3871	22	7	2	0.0224	0.4932	no	no
Claude Opus 5	DeepSeek-chat	-2.6613	22	8	1	0.0086	0.2331	yes	no
Claude Opus 5	DeepSeek-R1	-2.1774	20	8	3	0.0139	0.3194	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.7258	20	9	2	0.1168	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.4677	25	6	0	0.0042	0.1264	yes	no
Claude Opus 5	GPT-5.5	0.0000	14	11	6	0.4045	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.2097	26	4	1	0.0004	0.0136	yes	yes
Claude Opus 5	Grok 4	-1.7742	18	10	3	0.0548	0.8746	no	no
Claude Opus 5	Llama 3.3 70B	-4.7581	26	5	0	0.0002	0.0070	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2742	20	7	4	0.0318	0.6042	no	no
Claude Sonnet 5	DeepSeek-R1	-0.7903	15	14	2	0.4560	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.6613	11	17	3	0.1389	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.0806	20	8	3	0.0547	0.8746	no	no
Claude Sonnet 5	GPT-5.5	1.3871	9	19	3	0.0288	0.5758	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8226	22	8	1	0.0056	0.1611	yes	no
Claude Sonnet 5	Grok 4	-0.3871	19	10	2	0.5499	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.3710	25	4	2	0.0001	0.0034	yes	yes
DeepSeek-chat	DeepSeek-R1	0.4839	17	13	1	0.9296	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.9355	6	22	3	0.0068	0.1911	no	no
DeepSeek-chat	GPT-5.4-mini	0.1935	10	17	4	0.3270	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.6613	5	22	4	0.0018	0.0602	yes	no
DeepSeek-chat	gpt-oss-120b	-1.5484	21	9	1	0.0425	0.7649	no	no
DeepSeek-chat	Grok 4	0.8871	12	16	3	0.2809	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0968	21	6	4	0.0004	0.0137	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.4516	10	19	2	0.0435	0.7649	no	no
DeepSeek-R1	GPT-5.4-mini	-0.2903	16	14	1	0.7836	1.0000	no	no
DeepSeek-R1	GPT-5.5	2.1774	7	20	4	0.0088	0.2331	no	no
DeepSeek-R1	gpt-oss-120b	-2.0323	21	9	1	0.0121	0.3031	no	no
DeepSeek-R1	Grok 4	0.4032	13	14	4	0.4269	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.5806	22	6	3	0.0003	0.0104	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7419	21	9	1	0.0230	0.4932	no	no
Gemini 3.8 Flash	GPT-5.5	0.7258	12	17	2	0.8216	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.4839	27	3	1	0.0001	0.0024	yes	yes
Gemini 3.8 Flash	Grok 4	-1.0484	18	11	2	0.0976	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.0323	26	4	1	0.0001	0.0030	yes	yes
GPT-5.4-mini	GPT-5.5	2.4677	5	25	1	0.0025	0.0787	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.7419	23	8	0	0.0032	0.0985	no	no
GPT-5.4-mini	Grok 4	0.6935	11	17	3	0.3884	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2903	25	4	2	0.0001	0.0027	no	yes
GPT-5.5	gpt-oss-120b	-4.2097	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7742	21	6	4	0.0128	0.3061	no	no
GPT-5.5	Llama 3.3 70B	-4.7581	27	4	0	0.0000	0.0006	yes	yes
gpt-oss-120b	Grok 4	2.4355	5	26	0	0.0001	0.0035	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5484	20	11	0	0.0957	1.0000	no	no
Grok 4	Llama 3.3 70B	-2.9839	26	4	1	0.0001	0.0033	yes	yes

Table 33: Cross-dataset pairwise tests under rule top1: Friedman $\chi^2 = 81.8$ ($p = 7.2\text{e-}14$), Iman–Davenport $F = 12.4$, Nemenyi CD = 2.43, Kendall's $W = 0.276$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2903	20	7	4	0.0294	0.5878	no	no
Claude Opus 5	DeepSeek-chat	-2.7097	22	6	3	0.0049	0.1565	yes	no
Claude Opus 5	DeepSeek-R1	-1.7419	18	6	7	0.0126	0.2890	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.8548	16	9	6	0.1750	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.6290	24	5	2	0.0066	0.1911	yes	no
Claude Opus 5	GPT-5.5	-0.0968	13	9	9	0.3813	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0645	26	4	1	0.0006	0.0191	yes	yes
Claude Opus 5	Grok 4	-1.6290	19	9	3	0.0413	0.7430	no	no
Claude Opus 5	Llama 3.3 70B	-4.8226	25	5	1	0.0002	0.0066	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.4194	20	5	6	0.0104	0.2640	no	no
Claude Sonnet 5	DeepSeek-R1	-0.4516	13	13	5	0.7530	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.4355	11	13	7	0.3822	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.3387	20	7	4	0.0856	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.1935	7	17	7	0.0097	0.2614	no	no
Claude Sonnet 5	gpt-oss-120b	-2.7742	22	7	2	0.0066	0.1911	yes	no
Claude Sonnet 5	Grok 4	-0.3387	15	11	5	0.5827	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.5323	25	4	2	0.0001	0.0049	yes	yes
DeepSeek-chat	DeepSeek-R1	0.9677	10	17	4	0.2210	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.8548	5	19	7	0.0107	0.2640	no	no
DeepSeek-chat	GPT-5.4-mini	0.0806	11	13	7	0.3767	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.6129	4	20	7	0.0008	0.0251	yes	yes
DeepSeek-chat	gpt-oss-120b	-1.3548	18	8	5	0.1014	1.0000	no	no
DeepSeek-chat	Grok 4	1.0806	11	16	4	0.1724	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1129	23	3	5	0.0001	0.0052	no	yes
DeepSeek-R1	Gemini 3.8 Flash	0.8871	11	16	4	0.1098	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-0.8871	17	11	3	0.7905	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.6452	7	19	5	0.0128	0.2890	no	no
DeepSeek-R1	gpt-oss-120b	-2.3226	20	6	5	0.0060	0.1810	no	no
DeepSeek-R1	Grok 4	0.1129	12	15	4	0.4427	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.0806	24	4	3	0.0001	0.0049	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7742	20	9	2	0.0679	1.0000	no	no
Gemini 3.8 Flash	GPT-5.5	0.7581	10	15	6	0.4730	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.2097	24	2	5	0.0001	0.0033	yes	yes
Gemini 3.8 Flash	Grok 4	-0.7742	17	11	3	0.2191	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.9677	26	4	1	0.0001	0.0048	yes	yes
GPT-5.4-mini	GPT-5.5	2.5323	5	25	1	0.0053	0.1642	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.4355	21	9	1	0.0102	0.2640	no	no
GPT-5.4-mini	Grok 4	1.0000	10	15	6	0.4437	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1935	23	3	5	0.0001	0.0035	no	yes
GPT-5.5	gpt-oss-120b	-3.9677	25	1	5	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.5323	17	6	8	0.0270	0.5669	no	no
GPT-5.5	Llama 3.3 70B	-4.7258	27	4	0	0.0000	0.0010	yes	yes
gpt-oss-120b	Grok 4	2.4355	3	24	4	0.0001	0.0053	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.7581	19	8	4	0.0302	0.5878	no	no
Grok 4	Llama 3.3 70B	-3.1935	23	4	4	0.0001	0.0049	yes	yes

Table 34: Cross-dataset pairwise tests under rule top2: Friedman $\chi^2 = 82.5$ ($p = 5.3\text{e-}14$), Iman–Davenport $F = 12.6$, Nemenyi CD = 2.43, Kendall's $W = 0.279$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3387	21	7	3	0.0266	0.5590	no	no
Claude Opus 5	DeepSeek-chat	-2.6290	22	6	3	0.0049	0.1564	yes	no
Claude Opus 5	DeepSeek-R1	-1.9032	18	6	7	0.0116	0.3133	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0161	19	8	4	0.0504	0.9126	no	no
Claude Opus 5	GPT-5.4-mini	-2.8065	25	5	1	0.0057	0.1703	yes	no
Claude Opus 5	GPT-5.5	-0.3065	13	9	9	0.3064	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.1935	26	3	2	0.0004	0.0127	yes	yes
Claude Opus 5	Grok 4	-1.7903	18	9	4	0.0559	0.9126	no	no
Claude Opus 5	Llama 3.3 70B	-4.9839	26	5	0	0.0001	0.0049	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2903	19	5	7	0.0161	0.4034	no	no
Claude Sonnet 5	DeepSeek-R1	-0.5645	14	13	4	0.6305	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.3226	11	14	6	0.3217	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4677	19	7	5	0.1052	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.0323	7	15	9	0.0397	0.7936	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8548	22	7	2	0.0094	0.2627	yes	no
Claude Sonnet 5	Grok 4	-0.4516	18	9	4	0.4924	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.6452	25	4	2	0.0001	0.0038	yes	yes
DeepSeek-chat	DeepSeek-R1	0.7258	11	13	7	0.6016	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.6129	6	18	7	0.0166	0.4034	no	no
DeepSeek-chat	GPT-5.4-mini	-0.1774	12	14	5	0.4859	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3226	6	19	6	0.0052	0.1604	no	no
DeepSeek-chat	gpt-oss-120b	-1.5645	19	9	3	0.0827	1.0000	no	no
DeepSeek-chat	Grok 4	0.8387	12	15	4	0.2936	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.3548	22	3	6	0.0001	0.0053	no	yes
DeepSeek-R1	Gemini 3.8 Flash	0.8871	10	17	4	0.0897	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-0.9032	17	12	2	0.9373	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.5968	7	19	5	0.0203	0.4672	no	no
DeepSeek-R1	gpt-oss-120b	-2.2903	21	5	5	0.0038	0.1258	no	no
DeepSeek-R1	Grok 4	0.1129	13	14	4	0.7011	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-3.0806	25	3	3	0.0000	0.0021	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7903	21	9	1	0.0498	0.9126	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	10	15	6	0.5755	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.1774	24	2	5	0.0001	0.0033	yes	yes
Gemini 3.8 Flash	Grok 4	-0.7742	15	10	6	0.2506	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.9677	26	4	1	0.0001	0.0038	yes	yes
GPT-5.4-mini	GPT-5.5	2.5000	5	24	2	0.0064	0.1852	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.3871	20	10	1	0.0134	0.3494	no	no
GPT-5.4-mini	Grok 4	1.0161	9	16	6	0.3613	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1774	23	4	4	0.0001	0.0044	no	yes
GPT-5.5	gpt-oss-120b	-3.8871	26	1	4	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.4839	16	6	9	0.0480	0.9126	no	no
GPT-5.5	Llama 3.3 70B	-4.6774	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.4032	3	24	4	0.0002	0.0054	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.7903	20	8	3	0.0217	0.4779	no	no
Grok 4	Llama 3.3 70B	-3.1935	24	4	3	0.0001	0.0038	yes	yes

Table 35: Cross-dataset pairwise tests under rule top3: Friedman $\chi^2 = 81.2$ ($p = 9.2\text{e-}14$), Iman–Davenport $F = 12.3$, Nemenyi CD = 2.43, Kendall's $W = 0.283$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4839	22	7	2	0.0213	0.4471	no	no
Claude Opus 5	DeepSeek-chat	-2.8226	23	7	1	0.0054	0.1618	yes	no
Claude Opus 5	DeepSeek-R1	-2.0484	20	8	3	0.0139	0.3472	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0161	21	8	2	0.0654	0.9154	no	no
Claude Opus 5	GPT-5.4-mini	-2.5645	25	6	0	0.0045	0.1434	yes	no
Claude Opus 5	GPT-5.5	-0.1290	14	11	6	0.3414	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.2742	26	4	1	0.0004	0.0147	yes	yes
Claude Opus 5	Grok 4	-1.8710	19	9	3	0.0396	0.7126	no	no
Claude Opus 5	Llama 3.3 70B	-4.9194	26	5	0	0.0001	0.0054	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.3387	20	7	4	0.0172	0.3786	no	no
Claude Sonnet 5	DeepSeek-R1	-0.5645	15	14	2	0.4680	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.4677	11	17	3	0.2133	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.0806	20	8	3	0.0880	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.3548	9	19	3	0.0318	0.6357	no	no
Claude Sonnet 5	gpt-oss-120b	-2.7903	22	8	1	0.0063	0.1812	yes	no
Claude Sonnet 5	Grok 4	-0.3871	19	10	2	0.4159	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.4355	25	4	2	0.0001	0.0033	yes	yes
DeepSeek-chat	DeepSeek-R1	0.7742	14	16	1	0.6445	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.8065	7	21	3	0.0102	0.2761	no	no
DeepSeek-chat	GPT-5.4-mini	0.2581	10	17	4	0.2555	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.6935	5	22	4	0.0015	0.0492	yes	yes
DeepSeek-chat	gpt-oss-120b	-1.4516	22	8	1	0.0467	0.7931	no	no
DeepSeek-chat	Grok 4	0.9516	12	16	3	0.2168	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0968	21	6	4	0.0004	0.0147	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.0323	11	18	2	0.0598	0.8977	no	no
DeepSeek-R1	GPT-5.4-mini	-0.5161	16	14	1	0.8139	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.9194	8	19	4	0.0154	0.3696	no	no
DeepSeek-R1	gpt-oss-120b	-2.2258	21	9	1	0.0103	0.2761	no	no
DeepSeek-R1	Grok 4	0.1774	13	14	4	0.4269	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.8710	24	4	3	0.0001	0.0049	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.5484	21	9	1	0.0326	0.6357	no	no
Gemini 3.8 Flash	GPT-5.5	0.8871	11	18	2	0.5898	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.2581	26	4	1	0.0001	0.0029	yes	yes
Gemini 3.8 Flash	Grok 4	-0.8548	17	12	2	0.1794	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-3.9032	26	4	1	0.0001	0.0032	yes	yes
GPT-5.4-mini	GPT-5.5	2.4355	6	24	1	0.0052	0.1618	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.7097	22	9	0	0.0062	0.1812	no	no
GPT-5.4-mini	Grok 4	0.6935	11	17	3	0.3777	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3548	25	4	2	0.0001	0.0032	no	yes
GPT-5.5	gpt-oss-120b	-4.1452	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7419	21	6	4	0.0159	0.3696	no	no
GPT-5.5	Llama 3.3 70B	-4.7903	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.4032	5	26	0	0.0001	0.0041	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.6452	20	11	0	0.0500	0.8000	no	no
Grok 4	Llama 3.3 70B	-3.0484	26	4	1	0.0001	0.0033	yes	yes

Table 36: Cross-dataset pairwise tests under rule top4: Friedman $\chi^2 = 81.4$ ($p = 8.5\text{e-}14$), Iman–Davenport $F = 12.4$, Nemenyi CD = 2.43, Kendall's $W = 0.282$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3548	22	7	2	0.0192	0.4220	no	no
Claude Opus 5	DeepSeek-chat	-2.4839	22	8	1	0.0081	0.2218	yes	no
Claude Opus 5	DeepSeek-R1	-1.9677	20	8	3	0.0202	0.4238	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.7097	19	10	2	0.1524	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.8871	26	5	0	0.0040	0.1188	yes	no
Claude Opus 5	GPT-5.5	-0.1129	14	11	6	0.3935	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0484	26	4	1	0.0004	0.0132	yes	yes
Claude Opus 5	Grok 4	-1.7097	18	10	3	0.0654	0.9158	no	no
Claude Opus 5	Llama 3.3 70B	-4.8871	26	5	0	0.0002	0.0066	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.1290	20	6	5	0.0476	0.7143	no	no
Claude Sonnet 5	DeepSeek-R1	-0.6129	15	13	3	0.6448	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.6452	11	16	4	0.1442	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.5323	20	8	3	0.0377	0.6404	no	no
Claude Sonnet 5	GPT-5.5	1.2419	9	18	4	0.0235	0.4701	no	no
Claude Sonnet 5	gpt-oss-120b	-2.6935	21	9	1	0.0079	0.2218	yes	no
Claude Sonnet 5	Grok 4	-0.3548	19	10	2	0.6450	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.5323	25	4	2	0.0001	0.0043	yes	yes
DeepSeek-chat	DeepSeek-R1	0.5161	14	15	2	0.7164	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.7742	6	22	3	0.0068	0.1979	no	no
DeepSeek-chat	GPT-5.4-mini	-0.4032	14	13	4	0.9844	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3710	5	21	5	0.0020	0.0643	no	no
DeepSeek-chat	gpt-oss-120b	-1.5645	22	8	1	0.0335	0.6021	no	no
DeepSeek-chat	Grok 4	0.7742	13	15	3	0.3672	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.4032	23	4	4	0.0001	0.0046	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.2581	9	19	3	0.0425	0.6792	no	no
DeepSeek-R1	GPT-5.4-mini	-0.9194	20	10	1	0.3775	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.8548	7	20	4	0.0099	0.2571	no	no
DeepSeek-R1	gpt-oss-120b	-2.0806	22	8	1	0.0029	0.0897	no	no
DeepSeek-R1	Grok 4	0.2581	12	15	4	0.5761	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.9194	23	5	3	0.0001	0.0039	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.1774	21	9	1	0.0143	0.3567	no	no
Gemini 3.8 Flash	GPT-5.5	0.5968	13	15	3	0.7988	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.3387	26	4	1	0.0001	0.0029	yes	yes
Gemini 3.8 Flash	Grok 4	-1.0000	17	12	2	0.1058	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.1774	26	4	1	0.0001	0.0030	yes	yes
GPT-5.4-mini	GPT-5.5	2.7742	5	25	1	0.0013	0.0417	yes	yes
GPT-5.4-mini	gpt-oss-120b	-1.1613	20	11	0	0.0182	0.4181	no	no
GPT-5.4-mini	Grok 4	1.1774	11	17	3	0.2395	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.0000	24	5	2	0.0001	0.0039	no	yes
GPT-5.5	gpt-oss-120b	-3.9355	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.5968	20	7	4	0.0167	0.4016	no	no
GPT-5.5	Llama 3.3 70B	-4.7742	27	4	0	0.0000	0.0006	yes	yes
gpt-oss-120b	Grok 4	2.3387	5	26	0	0.0001	0.0043	no	yes
gpt-oss-120b	Llama 3.3 70B	-0.8387	22	9	0	0.0282	0.5349	no	no
Grok 4	Llama 3.3 70B	-3.1774	26	4	1	0.0001	0.0039	yes	yes

Table 37: Cross-dataset pairwise tests under rule top5: Friedman $\chi^2 = 80.9$ ($p = 1.1\text{e-}13$), Iman–Davenport $F = 12.3$, Nemenyi CD = 2.43, Kendall's $W = 0.282$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3871	22	7	2	0.0224	0.4932	no	no
Claude Opus 5	DeepSeek-chat	-2.6613	22	8	1	0.0086	0.2331	yes	no
Claude Opus 5	DeepSeek-R1	-2.1774	20	8	3	0.0139	0.3194	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.7258	20	9	2	0.1168	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.4677	25	6	0	0.0042	0.1264	yes	no
Claude Opus 5	GPT-5.5	0.0000	14	11	6	0.4045	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.2097	26	4	1	0.0004	0.0136	yes	yes
Claude Opus 5	Grok 4	-1.7742	18	10	3	0.0548	0.8746	no	no
Claude Opus 5	Llama 3.3 70B	-4.7581	26	5	0	0.0002	0.0070	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.2742	20	7	4	0.0318	0.6042	no	no
Claude Sonnet 5	DeepSeek-R1	-0.7903	15	14	2	0.4560	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.6613	11	17	3	0.1389	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.0806	20	8	3	0.0547	0.8746	no	no
Claude Sonnet 5	GPT-5.5	1.3871	9	19	3	0.0288	0.5758	no	no
Claude Sonnet 5	gpt-oss-120b	-2.8226	22	8	1	0.0056	0.1611	yes	no
Claude Sonnet 5	Grok 4	-0.3871	19	10	2	0.5499	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.3710	25	4	2	0.0001	0.0034	yes	yes
DeepSeek-chat	DeepSeek-R1	0.4839	17	13	1	0.9296	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.9355	6	22	3	0.0068	0.1911	no	no
DeepSeek-chat	GPT-5.4-mini	0.1935	10	17	4	0.3270	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.6613	5	22	4	0.0018	0.0602	yes	no
DeepSeek-chat	gpt-oss-120b	-1.5484	21	9	1	0.0425	0.7649	no	no
DeepSeek-chat	Grok 4	0.8871	12	16	3	0.2809	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.0968	21	6	4	0.0004	0.0137	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.4516	10	19	2	0.0435	0.7649	no	no
DeepSeek-R1	GPT-5.4-mini	-0.2903	16	14	1	0.7836	1.0000	no	no
DeepSeek-R1	GPT-5.5	2.1774	7	20	4	0.0088	0.2331	no	no
DeepSeek-R1	gpt-oss-120b	-2.0323	21	9	1	0.0121	0.3031	no	no
DeepSeek-R1	Grok 4	0.4032	13	14	4	0.4269	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.5806	22	6	3	0.0003	0.0104	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.7419	21	9	1	0.0230	0.4932	no	no
Gemini 3.8 Flash	GPT-5.5	0.7258	12	17	2	0.8216	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.4839	27	3	1	0.0001	0.0024	yes	yes
Gemini 3.8 Flash	Grok 4	-1.0484	18	11	2	0.0976	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.0323	26	4	1	0.0001	0.0030	yes	yes
GPT-5.4-mini	GPT-5.5	2.4677	5	25	1	0.0025	0.0787	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.7419	23	8	0	0.0032	0.0985	no	no
GPT-5.4-mini	Grok 4	0.6935	11	17	3	0.3884	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2903	25	4	2	0.0001	0.0027	no	yes
GPT-5.5	gpt-oss-120b	-4.2097	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7742	21	6	4	0.0128	0.3061	no	no
GPT-5.5	Llama 3.3 70B	-4.7581	27	4	0	0.0000	0.0006	yes	yes
gpt-oss-120b	Grok 4	2.4355	5	26	0	0.0001	0.0035	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5484	20	11	0	0.0957	1.0000	no	no
Grok 4	Llama 3.3 70B	-2.9839	26	4	1	0.0001	0.0033	yes	yes

Table 38: Cross-dataset pairwise tests under rule v1 rubric: Friedman $\chi^2 = 76.0$ ($p = 1.0\text{e-}12$), Iman–Davenport $F = 11.2$, Nemenyi CD = 2.43, Kendall's $W = 0.264$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.2581	22	7	2	0.0303	0.6068	no	no
Claude Opus 5	DeepSeek-chat	-2.3548	21	8	2	0.0151	0.3620	no	no
Claude Opus 5	DeepSeek-R1	-1.8710	19	9	3	0.0207	0.4550	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.4677	17	11	3	0.2595	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.4194	25	6	0	0.0048	0.1382	no	no
Claude Opus 5	GPT-5.5	0.0161	13	11	7	0.5300	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.0000	26	4	1	0.0003	0.0121	yes	yes
Claude Opus 5	Grok 4	-1.4839	18	10	3	0.0900	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-4.5484	25	6	0	0.0002	0.0082	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.0968	20	7	4	0.0559	0.8948	no	no
Claude Sonnet 5	DeepSeek-R1	-0.6129	15	14	2	0.4680	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.7903	11	17	3	0.0957	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.1613	20	8	3	0.0499	0.8484	no	no
Claude Sonnet 5	GPT-5.5	1.2742	9	19	3	0.0334	0.6342	no	no
Claude Sonnet 5	gpt-oss-120b	-2.7419	21	8	2	0.0044	0.1306	yes	no
Claude Sonnet 5	Grok 4	-0.2258	18	11	2	0.6592	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.2903	25	4	2	0.0001	0.0037	yes	yes
DeepSeek-chat	DeepSeek-R1	0.4839	15	15	1	0.9608	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.8871	6	22	3	0.0068	0.1911	no	no
DeepSeek-chat	GPT-5.4-mini	-0.0645	13	14	4	0.6241	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.3710	6	21	4	0.0025	0.0810	no	no
DeepSeek-chat	gpt-oss-120b	-1.6452	20	9	2	0.0351	0.6342	no	no
DeepSeek-chat	Grok 4	0.8710	13	15	3	0.3270	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1935	22	5	4	0.0004	0.0141	no	yes
DeepSeek-R1	Gemini 3.8 Flash	1.4032	10	19	2	0.0197	0.4520	no	no
DeepSeek-R1	GPT-5.4-mini	-0.5484	17	13	1	0.9063	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.8871	7	20	4	0.0090	0.2339	no	no
DeepSeek-R1	gpt-oss-120b	-2.1290	22	8	1	0.0082	0.2201	no	no
DeepSeek-R1	Grok 4	0.3871	12	15	4	0.4324	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-2.6774	22	6	3	0.0005	0.0153	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-1.9516	21	9	1	0.0135	0.3377	no	no
Gemini 3.8 Flash	GPT-5.5	0.4839	14	15	2	0.8909	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.5323	26	4	1	0.0001	0.0024	yes	yes
Gemini 3.8 Flash	Grok 4	-1.0161	17	12	2	0.1145	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.0806	26	4	1	0.0001	0.0034	yes	yes
GPT-5.4-mini	GPT-5.5	2.4355	5	25	1	0.0013	0.0417	yes	yes
GPT-5.4-mini	gpt-oss-120b	-1.5806	22	9	0	0.0042	0.1306	no	no
GPT-5.4-mini	Grok 4	0.9355	12	16	3	0.3468	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1290	25	4	2	0.0001	0.0025	no	yes
GPT-5.5	gpt-oss-120b	-4.0161	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.5000	18	8	5	0.0212	0.4550	no	no
GPT-5.5	Llama 3.3 70B	-4.5645	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.5161	5	26	0	0.0001	0.0032	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5484	19	12	0	0.1169	1.0000	no	no
Grok 4	Llama 3.3 70B	-3.0645	26	4	1	0.0001	0.0043	yes	yes

Table 39: The (J, ρ) Pareto front.

w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J	rho	auc
0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.442	0.796	0.989
1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.439	0.828	0.989
0.650	0.150	0.100	0.000	0.000	0.000	0.100	0.000	0.000	0.434	0.864	0.994
0.400	0.100	0.150	0.100	0.000	0.000	0.100	0.150	0.000	0.432	0.884	0.994
0.200	0.000	0.250	0.250	0.050	0.050	0.150	0.050	0.000	0.415	0.884	0.994

Table 40: Mean score per dataset and model under the headline rule (v1 rubric).

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.960	0.960	0.960	0.970	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.980	0.940	0.970	1.000	0.990	1.000	0.990
alaska_schema	0.918	0.945	0.928	0.929	0.929	0.958	0.951	0.948	0.934	0.840
amazon_google	0.930	0.940	0.940	0.960	0.940	0.930	0.960	0.950	0.860	0.950
bird	0.293	0.293	0.185	0.232	0.122	0.178	0.187	0.178	0.084	0.122
dblp_acm	1.000	1.000	1.000	1.000	1.000	0.990	1.000	0.990	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.970	0.950	0.940	0.960	0.950	0.910	0.970
fetaqa	0.445	0.401	0.438	0.373	0.428	0.312	0.434	0.413	0.322	0.253
fintagging	0.897	0.909	0.893	0.898	0.887	0.892	0.858	0.840	0.863	0.858
hitab	0.748	0.781	0.757	0.763	0.746	0.784	0.814	0.771	0.750	0.784
machamp	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.900	0.830	0.870
magneto_gdc	0.963	0.971	0.967	0.955	0.939	0.912	0.953	0.970	0.961	0.753
mmtu	0.894	0.882	0.917	0.872	0.901	0.882	0.865	0.822	0.830	0.807
officeqa	0.194	0.187	0.168	0.170	0.129	0.107	0.075	0.083	0.079	0.097
officeqa_pro_v2	0.198	0.250	0.188	0.133	0.119	0.123	0.121	0.036	0.104	0.115
opensanctions_pairs	0.860	0.940	0.750	0.730	0.850	0.900	0.760	0.850	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.950	0.950	0.950	0.940	0.940	0.930	0.940
realhitbench	0.746	0.743	0.724	0.681	0.640	0.661	0.662	0.659	0.603	0.611
smat	0.921	0.933	0.940	0.938	0.936	0.956	0.918	0.913	0.916	0.806
suc	0.976	0.986	0.943	0.911	0.943	0.914	0.858	0.935	0.932	0.740
tab_fact	0.860	0.660	0.840	0.740	0.850	0.820	0.840	0.760	0.830	0.780
tabis	0.950	0.960	0.960	0.920	0.980	0.900	0.770	0.970	0.920	0.820
tablebench	0.835	0.846	0.852	0.877	0.839	0.841	0.833	0.783	0.813	0.660
tableeval	0.791	0.765	0.784	0.733	0.802	0.742	0.711	0.756	0.773	0.636
tat_qa	0.829	0.861	0.789	0.845	0.791	0.789	0.806	0.806	0.765	0.752
tpcdi_cells	0.462	0.462	0.462	0.462	0.447	0.455	0.462	0.462	0.442	0.442
valentine	0.970	1.000	1.000	0.994	0.973	0.985	0.979	0.990	0.979	0.970
walmart_amazon	0.990	0.980	0.970	0.950	0.960	0.950	0.950	0.960	0.940	0.930
wdc_lspc	0.980	0.990	1.000	0.980	1.000	0.970	0.960	1.000	0.980	0.960
wdc_products	0.990	0.980	0.980	0.970	0.990	0.970	0.970	0.980	0.980	0.970
wikitablequestions	0.881	0.879	0.865	0.863	0.867	0.842	0.843	0.846	0.824	0.749

Table 41: Per-family optimum.

family	best weights	J own best	J global best
EM	numeric tolerance 1.00	0.361	0.361
SM	numeric tolerance 1.00	0.580	0.580
TQA	exact match 0.50, numeric decay 0.05, token precision 0.15, token recall 0.05, token F1 0.10, edit similarity 0.05, ROUGE-L 0.10	0.513	0.512
FIN	exact match 0.10, numeric tolerance 0.10, token precision 0.20, token recall 0.35, token F1 0.10, ROUGE-L 0.15	0.834	0.619

Table 42: The 30 best candidates with bootstrap statistics.

rank_full	w_em	w_ num_ tol	w_ num_ decay	w_ tok_ prec	w_ tok_rec	w_ tok_f1	w_ edit_ sim	w_ rouge_l	w_ jaccard	J_ boot_ mean	J_ boot_ sd	J_ boot_lo	J_ boot_ hi	p_vs_ best	in_ plateau	in_ practical_ plateau	times_ selected	J_ oob_ mean
10	0.000	0.450	0.000	0.100	0.250	0.000	0.000	0.050	0.150	0.375	0.027	0.323	0.426	0.681	yes	yes	54	0.347
4	0.200	0.300	0.000	0.100	0.250	0.000	0.050	0.050	0.050	0.374	0.026	0.322	0.425	0.627	yes	yes	25	0.343
6	0.000	0.350	0.000	0.250	0.200	0.050	0.050	0.050	0.050	0.374	0.026	0.321	0.426	0.602	yes	yes	16	0.345
21	0.200	0.250	0.000	0.100	0.100	0.000	0.050	0.100	0.200	0.374	0.026	0.320	0.423	0.619	yes	yes	11	0.343
11	0.000	0.400	0.000	0.050	0.150	0.100	0.100	0.200	0.000	0.374	0.026	0.320	0.424	0.601	yes	yes	14	0.346
44	0.300	0.200	0.000	0.050	0.050	0.250	0.100	0.050	0.000	0.373	0.026	0.319	0.424	0.606	yes	yes	12	0.344
47	0.050	0.150	0.000	0.200	0.000	0.150	0.050	0.400	0.000	0.373	0.026	0.320	0.425	0.572	yes	yes	36	0.343
13	0.200	0.250	0.100	0.050	0.150	0.100	0.000	0.150	0.000	0.373	0.027	0.321	0.424	0.617	yes	yes	11	0.345
16	0.100	0.450	0.100	0.050	0.050	0.000	0.000	0.200	0.050	0.373	0.027	0.320	0.425	0.602	yes	yes	36	0.345
12	0.400	0.200	0.100	0.000	0.000	0.050	0.000	0.200	0.050	0.373	0.027	0.319	0.423	0.591	yes	yes	24	0.344
46	0.100	0.250	0.000	0.000	0.100	0.150	0.150	0.250	0.000	0.373	0.026	0.319	0.425	0.556	yes	yes	9	0.345
97	0.350	0.100	0.000	0.050	0.050	0.000	0.100	0.300	0.050	0.372	0.027	0.319	0.424	0.560	yes	yes	6	0.342
165	0.050	0.200	0.000	0.050	0.000	0.150	0.150	0.250	0.150	0.372	0.026	0.320	0.422	0.544	yes	yes	7	0.344
83	0.200	0.200	0.050	0.050	0.050	0.150	0.050	0.150	0.100	0.372	0.026	0.320	0.422	0.549	yes	yes	5	0.343
74	0.050	0.300	0.000	0.100	0.200	0.050	0.150	0.150	0.000	0.372	0.026	0.319	0.422	0.546	yes	yes	5	0.344
45	0.100	0.250	0.000	0.050	0.350	0.050	0.100	0.000	0.100	0.372	0.027	0.318	0.425	0.553	yes	yes	9	0.342
29	0.050	0.300	0.100	0.050	0.050	0.200	0.000	0.050	0.200	0.372	0.026	0.319	0.425	0.550	yes	yes	2	0.344
65	0.050	0.250	0.000	0.200	0.100	0.000	0.150	0.100	0.150	0.372	0.026	0.319	0.421	0.527	yes	yes	3	0.344
175	0.100	0.100	0.050	0.100	0.000	0.200	0.050	0.400	0.000	0.371	0.026	0.321	0.423	0.542	yes	yes	8	0.342
93	0.100	0.200	0.100	0.000	0.050	0.200	0.000	0.050	0.300	0.371	0.027	0.321	0.423	0.545	yes	yes	4	0.344
32	0.100	0.250	0.000	0.200	0.150	0.100	0.150	0.000	0.050	0.371	0.026	0.319	0.421	0.524	yes	yes	1	0.344
26	0.150	0.350	0.100	0.050	0.100	0.050	0.050	0.150	0.000	0.371	0.027	0.318	0.423	0.542	yes	yes	8	0.343
75	0.250	0.050	0.000	0.200	0.000	0.050	0.100	0.300	0.050	0.371	0.027	0.318	0.422	0.535	yes	yes	7	0.342
56	0.250	0.150	0.050	0.000	0.000	0.050	0.050	0.150	0.300	0.371	0.026	0.320	0.422	0.544	yes	yes	4	0.341
19	0.050	0.300	0.050	0.000	0.100	0.100	0.050	0.050	0.300	0.371	0.026	0.320	0.421	0.525	yes	yes	4	0.342
155	0.150	0.100	0.050	0.200	0.100	0.050	0.050	0.300	0.000	0.371	0.026	0.318	0.422	0.528	yes	yes	5	0.341
120	0.300	0.050	0.000	0.000	0.050	0.000	0.100	0.350	0.150	0.371	0.027	0.317	0.423	0.532	yes	yes	6	0.341
64	0.300	0.100	0.000	0.200	0.050	0.050	0.150	0.100	0.050	0.371	0.026	0.319	0.422	0.532	yes	yes	3	0.342
36	0.250	0.100	0.100	0.050	0.100	0.300	0.000	0.050	0.050	0.371	0.026	0.320	0.423	0.532	yes	yes	5	0.343
15	0.100	0.300	0.100	0.200	0.050	0.050	0.000	0.100	0.100	0.371	0.026	0.319	0.420	0.550	yes	yes	8	0.343

Table 43: Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	Claude Sonnet 5	Grok 4	GPT-5.4-mini	DeepSeek-chat	DeepSeek-R1	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
exact match	3 (3)	1 (1)	2 (2)	5 (4)	4 (5)	6 (8)	8 (7)	7 (6)	9 (9)	10 (10)	5.3e-14	0.28
v1 rubric	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (8)	7 (7)	8 (6)	9 (9)	10 (10)	1.0e-12	0.26
top1	3 (3)	1 (1)	2 (2)	5 (4)	4 (5)	6 (7)	8 (8)	7 (6)	9 (9)	10 (10)	7.2e-14	0.28
top2	3 (3)	1 (1)	2 (2)	5 (4)	4 (5)	6 (8)	8 (7)	7 (6)	9 (9)	10 (10)	5.3e-14	0.28
top3	1 (3)	2 (1)	3 (2)	4 (4)	5 (5)	6 (7)	7 (8)	8 (6)	9 (9)	10 (10)	9.2e-14	0.28
top4	1 (3)	2 (1)	3 (2)	5 (4)	4 (5)	7 (8)	6 (7)	8 (6)	9 (9)	10 (10)	8.5e-14	0.28
top5	1 (3)	2 (1)	3 (1)	4 (4)	5 (5)	6 (7)	7 (8)	8 (6)	9 (9)	10 (10)	1.1e-13	0.28
best-rho	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (7)	7 (8)	8 (6)	9 (9)	10 (10)	3.6e-13	0.27
knee	1 (3)	2 (1)	3 (1)	4 (4)	5 (5)	6 (7)	7 (8)	8 (6)	9 (9)	10 (10)	1.1e-13	0.28

Table 44: Anchor correctness per dataset.

dataset	v1	bestJ	bestrho
abt_buy	0.860	0.860	0.827
alaska_camera	0.857	0.857	0.827
alaska_schema	0.877	0.836	0.913
amazon_google	0.863	0.863	0.827
bird	0.889	0.737	0.910
dblp_acm	0.860	0.860	0.825
dblp_scholar	0.858	0.858	0.828
fetaqa	0.913	0.811	0.931
fintagging	0.874	0.848	0.879
hitab	0.906	0.736	0.921
machamp	0.857	0.857	0.825
magneto_gdc	0.872	0.838	0.910
mmtu	0.868	0.815	0.891
officeqa	0.894	0.728	0.900
officeqa_pro_v2	0.885	0.716	0.898
opensanctions_pairs	0.860	0.860	0.828
papadakis_dn	0.891	0.891	0.838
realhitbench	0.899	0.749	0.918
smat	0.914	0.839	0.926
suc	0.857	0.799	0.881
tab_fact	0.862	0.862	0.826
tabis	0.639	0.649	0.595
tablebench	0.898	0.740	0.917
tableeval	0.866	0.748	0.887
tat_qa	0.885	0.758	0.887
tpcdi_cells	0.905	0.756	0.908
valentine	0.846	0.844	0.891
walmart_amazon	0.867	0.867	0.828
wdc_lspc	0.856	0.856	0.825
wdc_products	0.856	0.856	0.825
wikitablequestions	0.877	0.763	0.911

Table 45: Kendall τ between the Bradley–Terry rankings of the rules.

rule	exact match	v1 rubric	top1	top2	top3	top4	top5	best-rho	knee
exact match	1.000	0.822	1.000	1.000	0.822	0.822	0.822	0.822	0.822
v1 rubric	0.822	1.000	0.822	0.822	1.000	0.911	1.000	1.000	1.000
top1	1.000	0.822	1.000	1.000	0.822	0.822	0.822	0.822	0.822
top2	1.000	0.822	1.000	1.000	0.822	0.822	0.822	0.822	0.822
top3	0.822	1.000	0.822	0.822	1.000	0.911	1.000	1.000	1.000
top4	0.822	0.911	0.822	0.822	0.911	1.000	0.911	0.911	0.911
top5	0.822	1.000	0.822	0.822	1.000	0.911	1.000	1.000	1.000
best-rho	0.822	1.000	0.822	0.822	1.000	0.911	1.000	1.000	1.000
knee	0.822	1.000	0.822	0.822	1.000	0.911	1.000	1.000	1.000

Table 46: One-step sensitivity.

point	n_neighbours	J_centre	J_min	J_max	J_mean	flatness_range	share_neighbours_better
v1_rubric	32.0000	0.4040	0.4011	0.4095	0.4055	0.0084	0.7188
best	8.0000	0.4419	0.4246	0.4456	0.4304	0.0210	0.1250